

Chapter 4

The Medium Access Control Sublayer

介质访问（接入）控制

(about 8 hours)

Data Link Types

■ Point to point links(点-点链路)

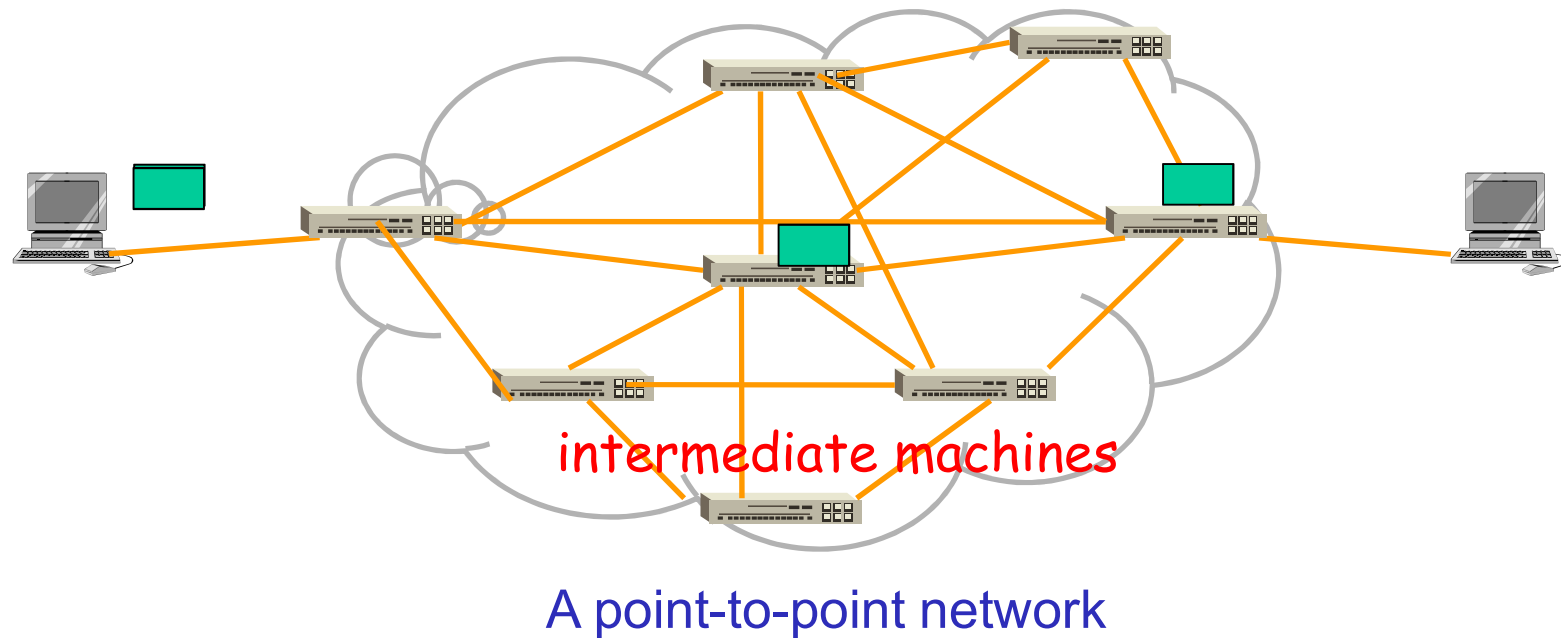
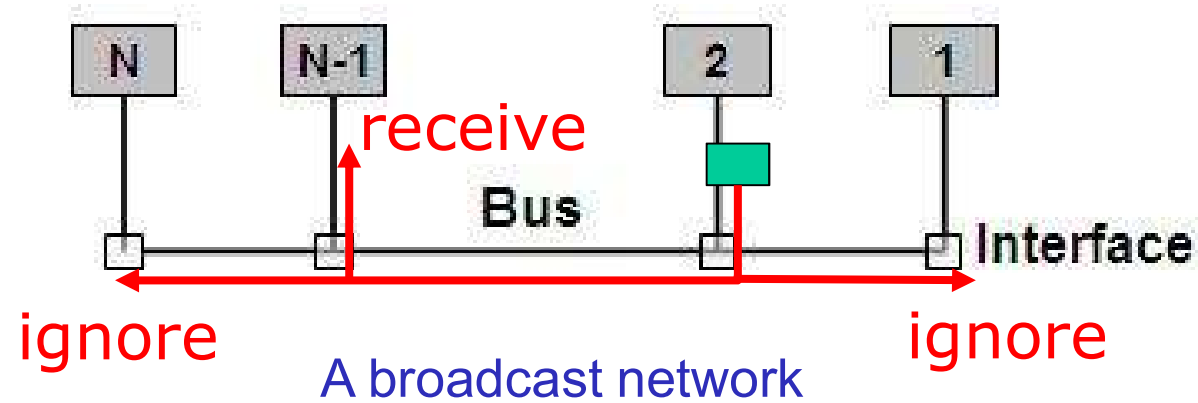
- ◆ One sender, one receiver
- ◆ Address may not be necessary
- ◆ Reliable communication
- ◆ Sample protocols: HDLC, PPP...

■ Broadcast links(广播链路)

- ◆ Many senders and potential receivers.
- ◆ Accessing a shared medium
 - Multiple access problem must be solved
- ◆ Addressing is necessary
- ◆ Sample protocols: ALOHA, CSMA/CD, MACA...

Medium Access Control: a means of controlling access to the medium to promote orderly and efficient use.

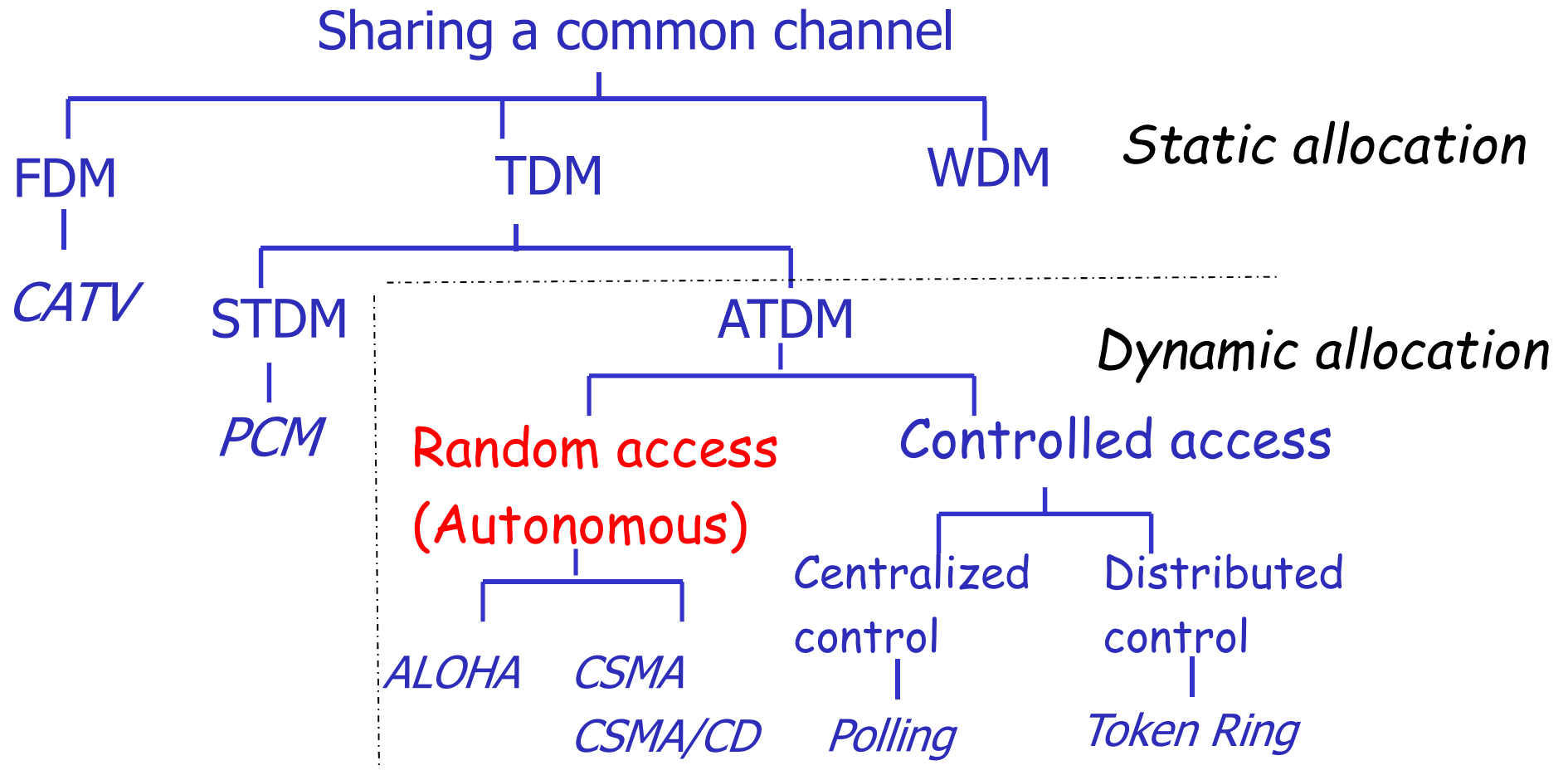
Broadcast vs. Point-to-point



Outline

- Channel Allocation Problem
- Multiple Access Protocols
- IEEE 802 Reference Model
- Ethernet
- Wireless LANs
- Data Link Layer Switching

Multiple Access Schemes



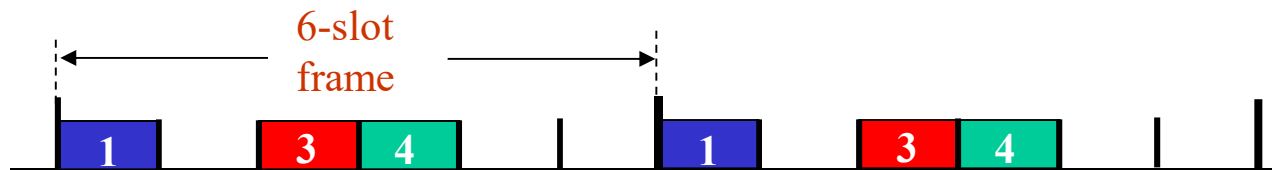
STDM: Synchronous TDM 同步时分复用

ATDM: Asynchronous TDM 异步时分复用(统计复用)

STDM Access

TDMA: time division multiple access

- access to channel in "rounds"
- each station gets fixed length **slot**(时隙/时槽) (length = frame transmission time) in each round
- unused slots go idle
- example: 6-station LAN, 1,3,4 have data, slots 2,5,6 idle



[Kurose]

Static Channel Allocation (静态信道分配)

- From Little's Law and M/M/1 queuing system

$$T = \frac{1}{\mu - \lambda}$$

μ : AverageServiceRate

λ : AverageArrivalRate

T : AverageWaitingTime(QueuingTime + ServiceTime)

- Channel is divided into N static sub-channels

- ◆ FDM: user number is small and constant, with heavy load of traffic of each user

$$T_{FDM} = \frac{1}{\mu C / N - \lambda / N} = NT$$

C: Channel capacity(data rate), bps

T: Mean delay(waiting time + service time)

λ : Frame arrival rate, frames/sec

$1/\mu$: mean frame length, bits

$$\begin{aligned} \text{AverageServiceRate} &= 1 / T_{\text{trans}} \\ &= 1 / ((1/\mu) / C) = \mu C \end{aligned}$$

Static Channel Allocation: Example

$$T = \frac{1}{\mu - \lambda}$$

■ Single channel

- ◆ data transmission rate: 100 Mbps, mean frame length 10,000 bits,

Mean service time $\mu = 10000$ frames/sec

Frame arrival rate $\lambda = 5000$ frames/sec

Mean delay = $1/(10000-5000) = 0.2$ ms

■ Divide the channel into N

- ◆ Replace the 100-Mbps network with 10 networks of 10 Mbps each and statically allocate each user to one of them

- ◆ For each sub-channel, $\mu = 1000$ frames/sec, $\lambda = 500$ frames/sec

Mean delay = $1/(1000-500) = 2$ ms

Static Channel Allocation: Poor performance

- Channel is divided into N static sub-channels
 - ◆ FDM: Much bigger delay than single channel
 - ◆ TDM: same problem. Poor performance.
- None of the static channel allocation methods work well with bursty traffic
- Dynamic Channel Allocation are adopted

Classification of Dynamic Channel Allocation

- Random multiple access
 - ◆ ALOHA
 - ◆ CSMA, CSMA/CD, CSMA/CA
- Controlled multiple access
 - ◆ Centralized: polling
 - ◆ Decentralized: token

Assumptions of Dynamic Allocation (动态信道分配)

1. Station model: independent and single user system
2. Single channel assumption
3. Collision assumption
4. Time assumption { Continuous time(连续时间)
Slotted time(分时隙)
5. Carrier sense assumption { Carrier sense
载波监听：发送之前
能否知道信道是否空闲
No carrier sense

Outline

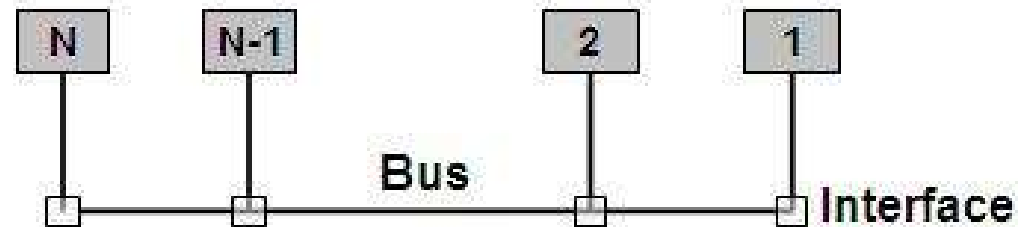
- Channel Allocation Problem
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- Wireless LANs
- Data Link Layer Switching

Multiple Access Protocols

- *ALOHA: Pure ALOHA & Slotted ALOHA*
- Carrier Sense Multiple Access Protocols
- Collision-Free Protocols
- Wireless LAN Protocols

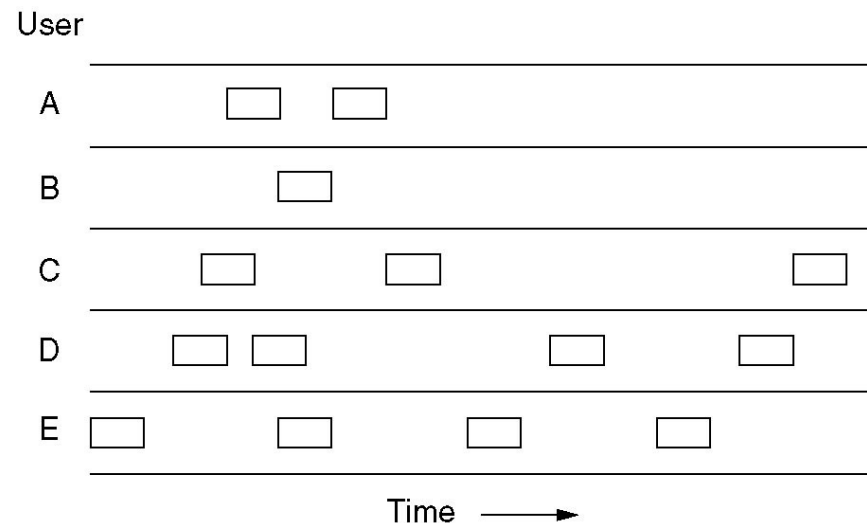
Pure ALOHA

- First protocol for multiple access channels; first analysis of such protocols (Norm Abramson, Univ. of Hawaii, 1970)

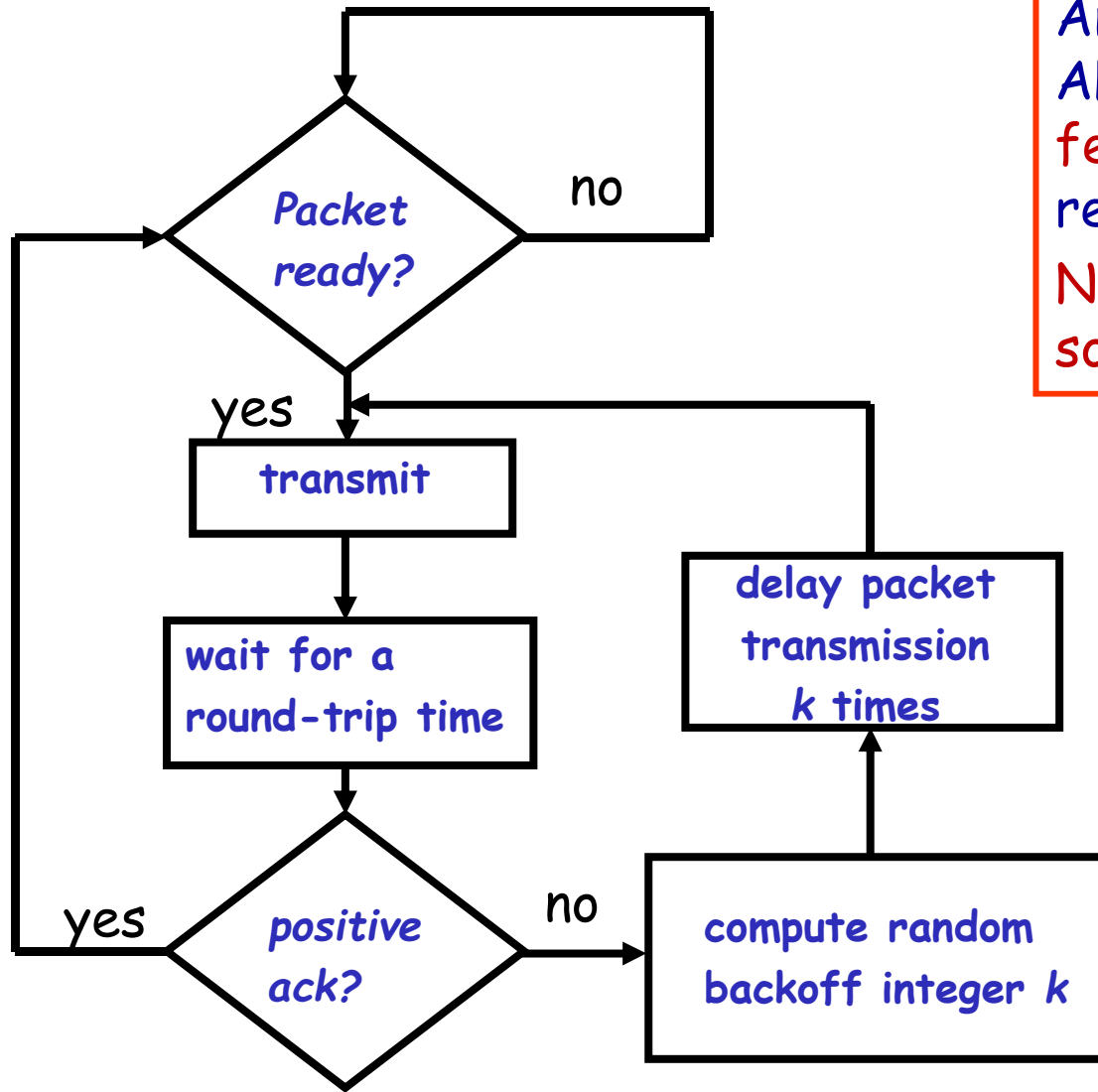


- Basic idea

- stations transmit a frame whenever they have one **without coordination** with others
- If the **acknowledge** not received (maybe collision), then do **retransmission**



Flow of Pure ALOHA



An integral part of the ALOHA protocol is **feedback** from the receiver.

No coordination among sources.

Pure ALOHA: Performance

- Performance parameters: Assumed each frame has same duration T_0

- ◆ **Throughput(吞吐量) S** --- Frames successfully transmitted in T_0

- ◆ **Network Load(网络负载) G** --- Frames arrived in T_0

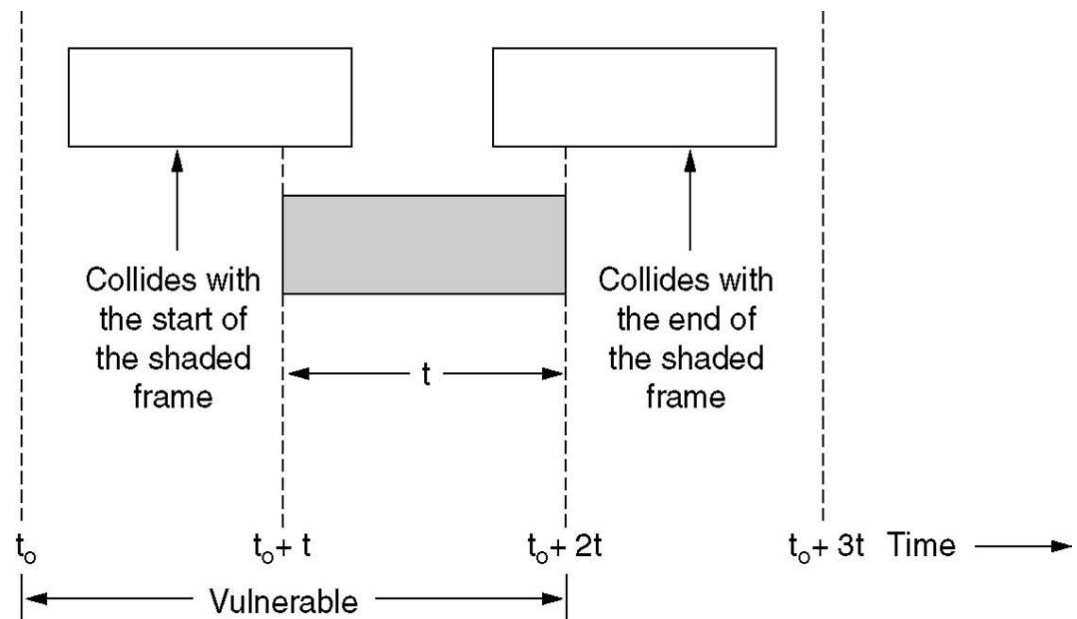
$$S = G \times P \text{ [successful transmission]}$$

- ◆ A frame is successful if

- ◆ no packets arrive within T_0 seconds before it starts or while it is being transmitted

Vulnerable Period: $2T_0$

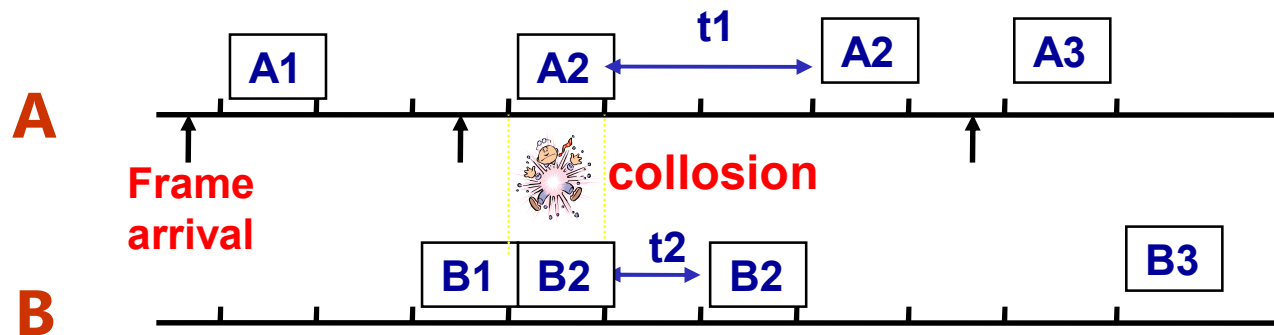
脆弱期（易冲突期）



Max throughput/utilization: 18.4%

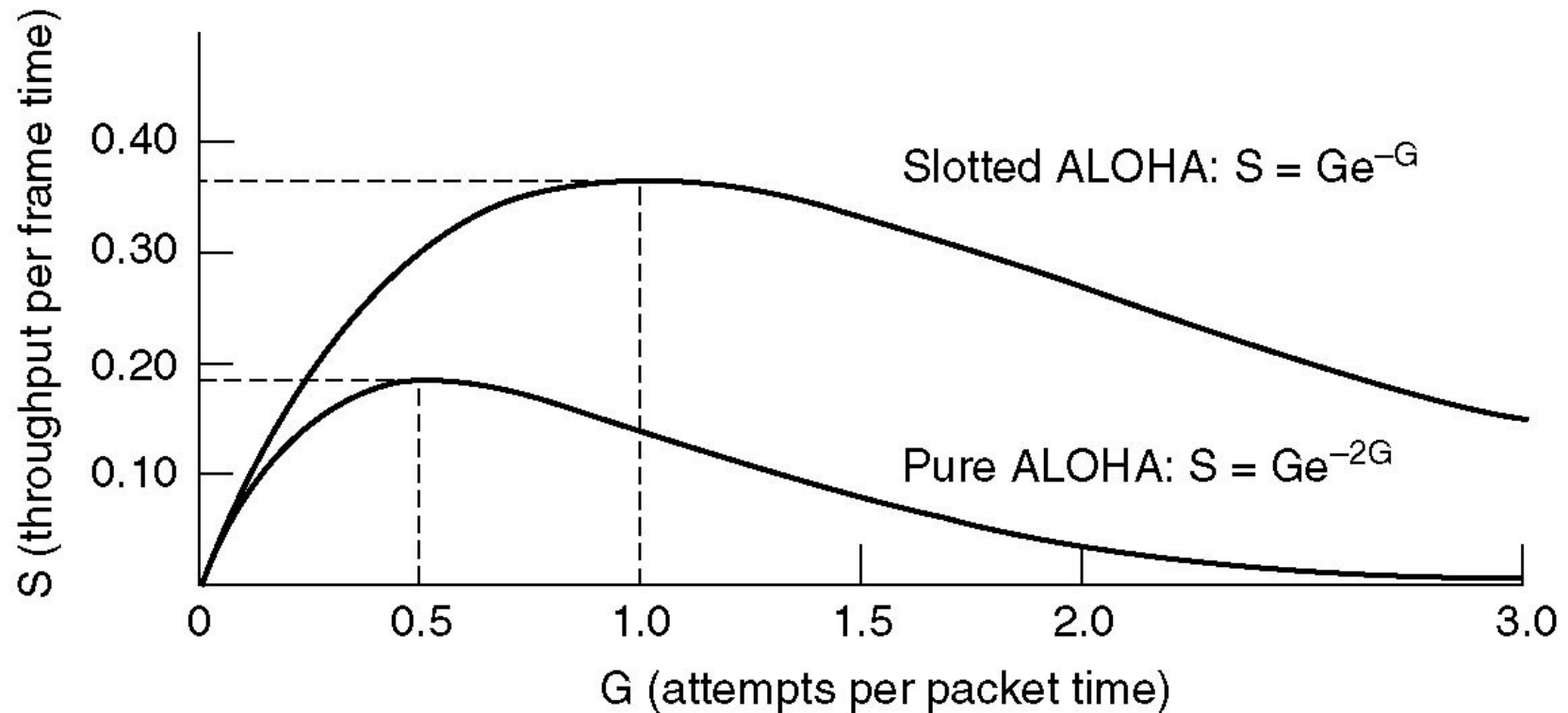
Slotted ALOHA(时隙/分槽ALOHA)

- Time in uniform slots equal T_0
- Need central clock (or other sync mechanism)
- Transmission begins at **slot boundary**



- Frames either miss or overlap totally
- Vulnerable Period: T_0
- Max utilization(S): 36.8%

Summary: Performance of ALOHA



Channel Utilization(信道利用率)

Multiple Access Protocols

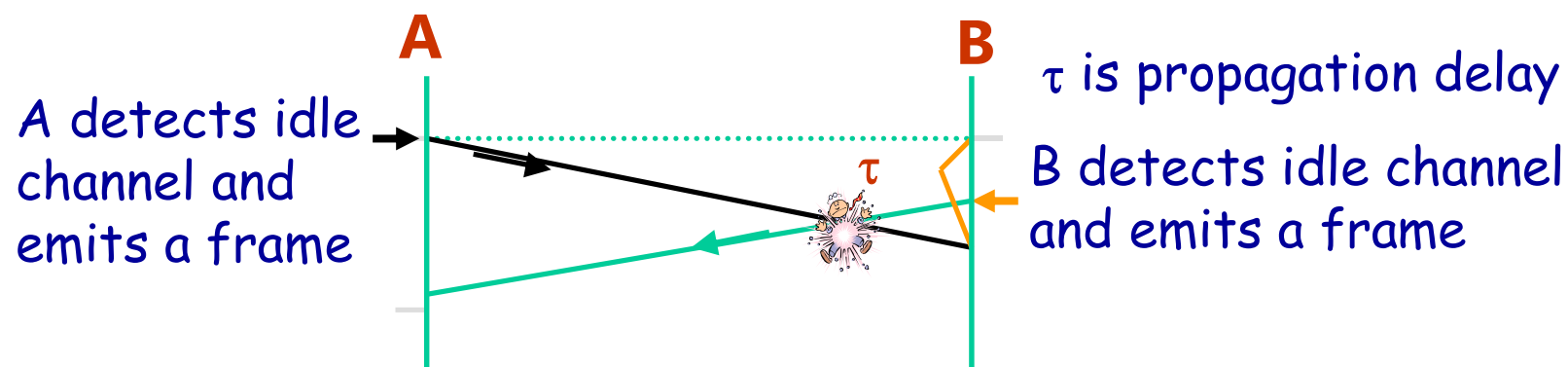
- ALOHA
- Carrier Sense Multiple Access Protocols
 - ◆ CSMA
 - ◆ CSMA/CD
- Collision-Free Protocols
- Wireless LAN Protocols

CSMA(载波监听多点接入/多路访问)

- Basic idea: By listening to the channel before sending (*Carrier Sense*), stations try to reduce the Vulnerability Period to *one propagation delay*
 - ◆ Propagation time is much less than transmission time
- CSMA (Kleinrock and Tobagi, UCLA, 1975)
 - ◆ If channel is idle, transmit with a *probability*.
 - ◆ If the channel is busy, the station waits until it becomes idle
 - ◆ If two stations start at the same instant, *collision*
 - ◆ Wait reasonable time (round trip plus ACK contention)
 - ◆ No ACK then retransmit

Vulnerable Period of CSMA

假设站A和站B分别位于电缆的两端，A和B之间的传播时延就是电缆的传播时延 τ



Vulnerable Period = Propagation delay (τ)

Nonpersistent vs. Persistent (1)

- Nonpersistent CSMA(非坚持CSMA)

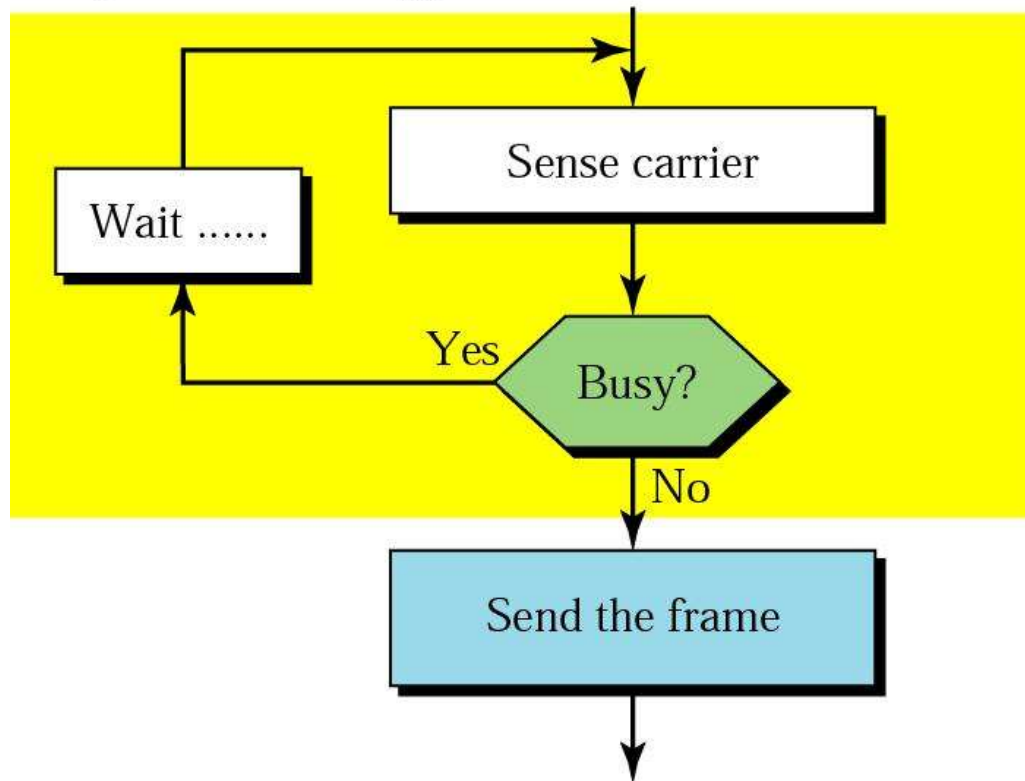
If the channel is busy, station waits for a random of time and try to sense again.

- Persistent-CSMA(坚持CSMA)

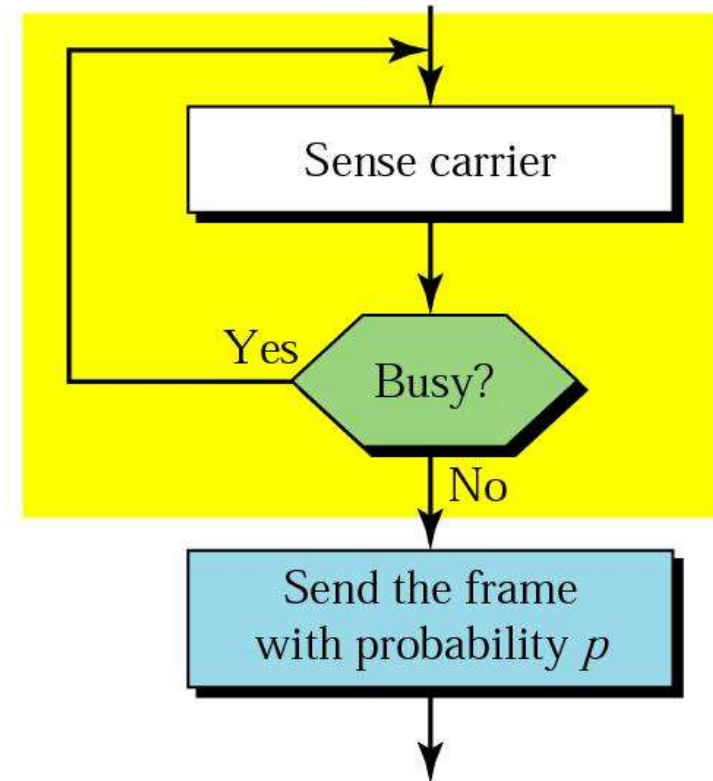
- If the channel is busy, waits until it becomes idle, then transmits frame immediately
- 1-persistent CSMA: Sending a frame when detecting that channel is idle (with probability 1)
- p-persistent CSMA: Sending a frame with probability p when detecting idle channel, i.e. delay transmission for a random time with probability $1-p$

Nonpersistent vs. Persistent(2)

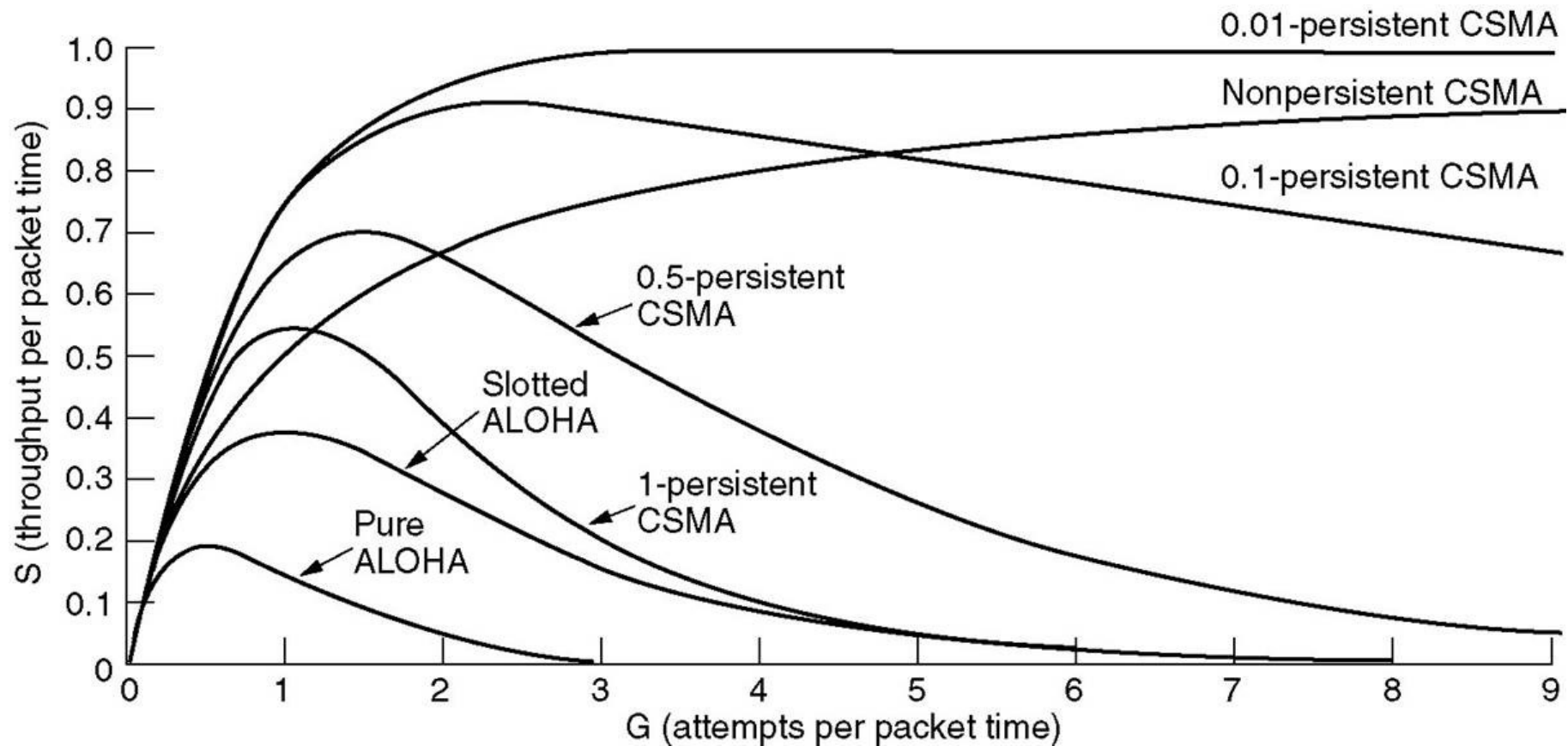
Nonpersistent strategy



Persistent strategy



Channel Utilization of ALOHA and CSMA



CSMA Comparison

■ 1-persistent CSMA

- High throughput and low delay when low load
- Low throughput when high load

■ Nonpersistent CSMA

- High throughput when high load

评价MAC协议的指标:

- ◆ 低负载（轻载）情况下的时延
- ◆ 高负载（重载）情况下的吞吐量

Summary of above random access protocols

- Pure ALOHA, Continuous time, no carrier sense
Vulnerable period: $2T_0$
- Slotted ALOHA, Slotted time, no carrier sense
Vulnerable period: T_0
- CSMA, Carrier sense
Vulnerable period: Propagation delay τ

- Similarity of above 3 protocols:
 - ◆ Only when timer expires, does the sender know that error (including collision) occurs and then start retransmission

CSMA/CD (CSMA with Collision Detection)

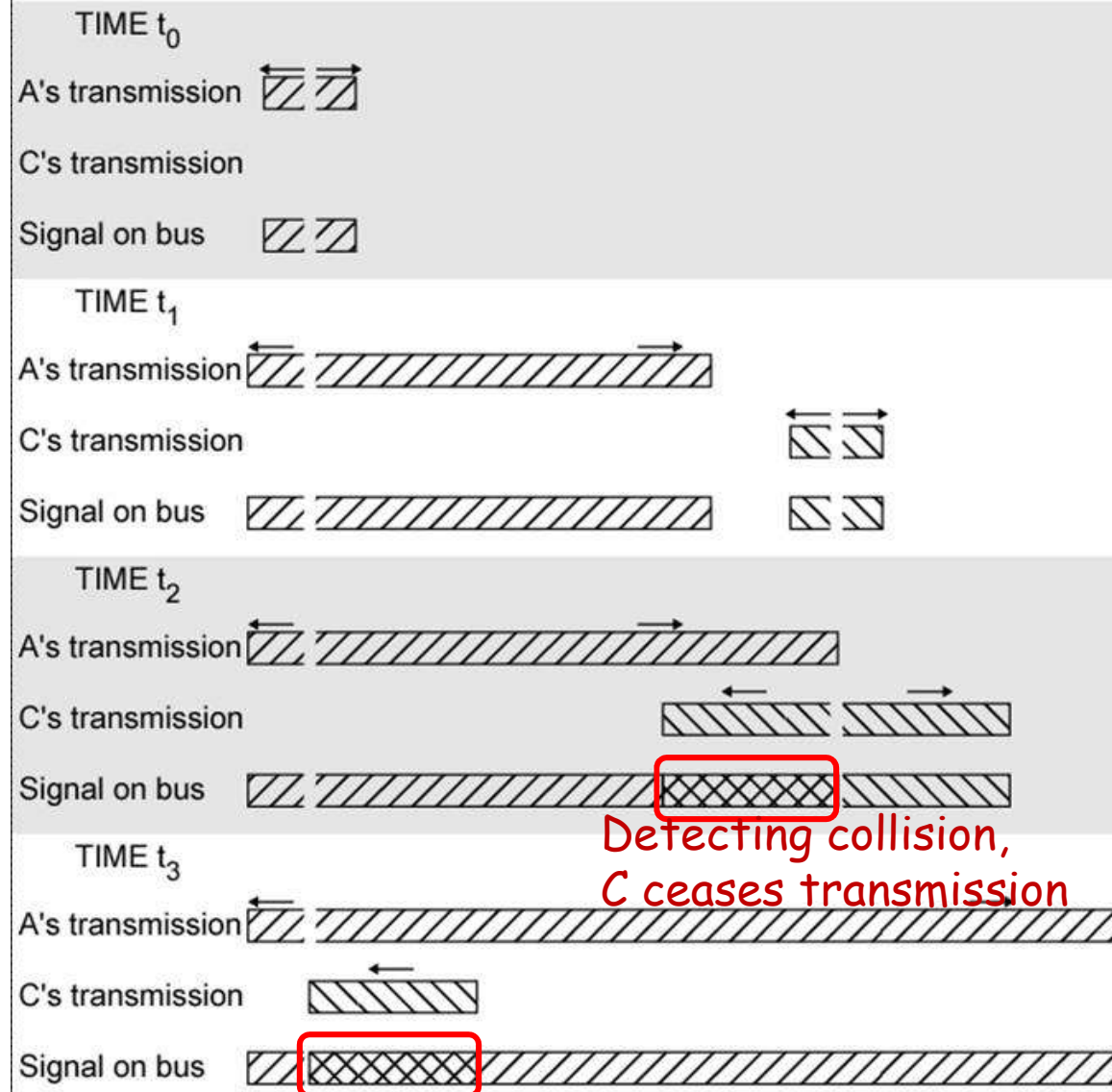
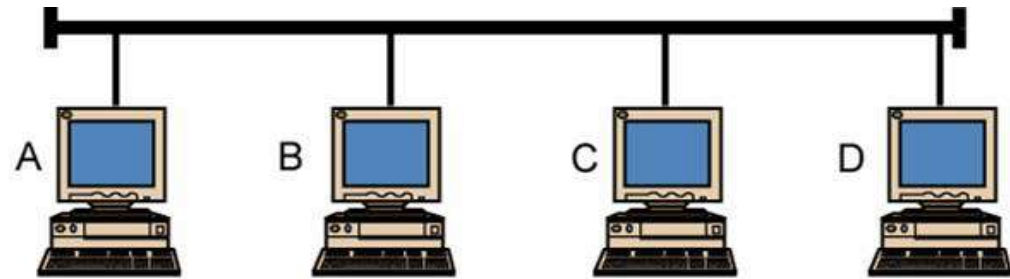
(带冲突检测的载波监听多路访问)

- Basic idea: listen to the channel while a packet is being sent. When collision is detected, transmission is aborted.

R.M Metcalfe and D.R. Boggs, "Ethernet: Distributed Packet Switching for Local Computer Networks," Comm. ACM, Vol. 19, 1976 (Xerox PARC).

- If collision detected, cease transmission (**backoff**, 退避) and transmit **jam** signal(强化信号) rather than finish transmitting the whole frame
- After jam, wait random time then start again
- **Half-duplex** system
- Saving time and bandwidth.
- Basis of Ethernet LAN.

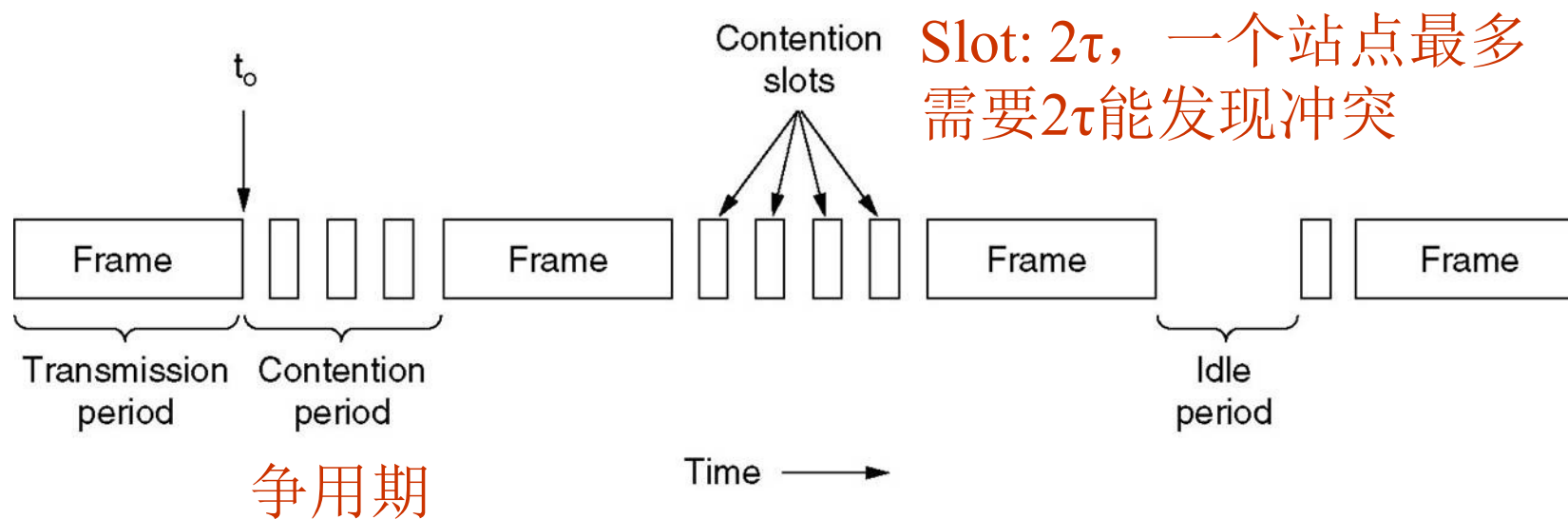
CSMA/CD Operations



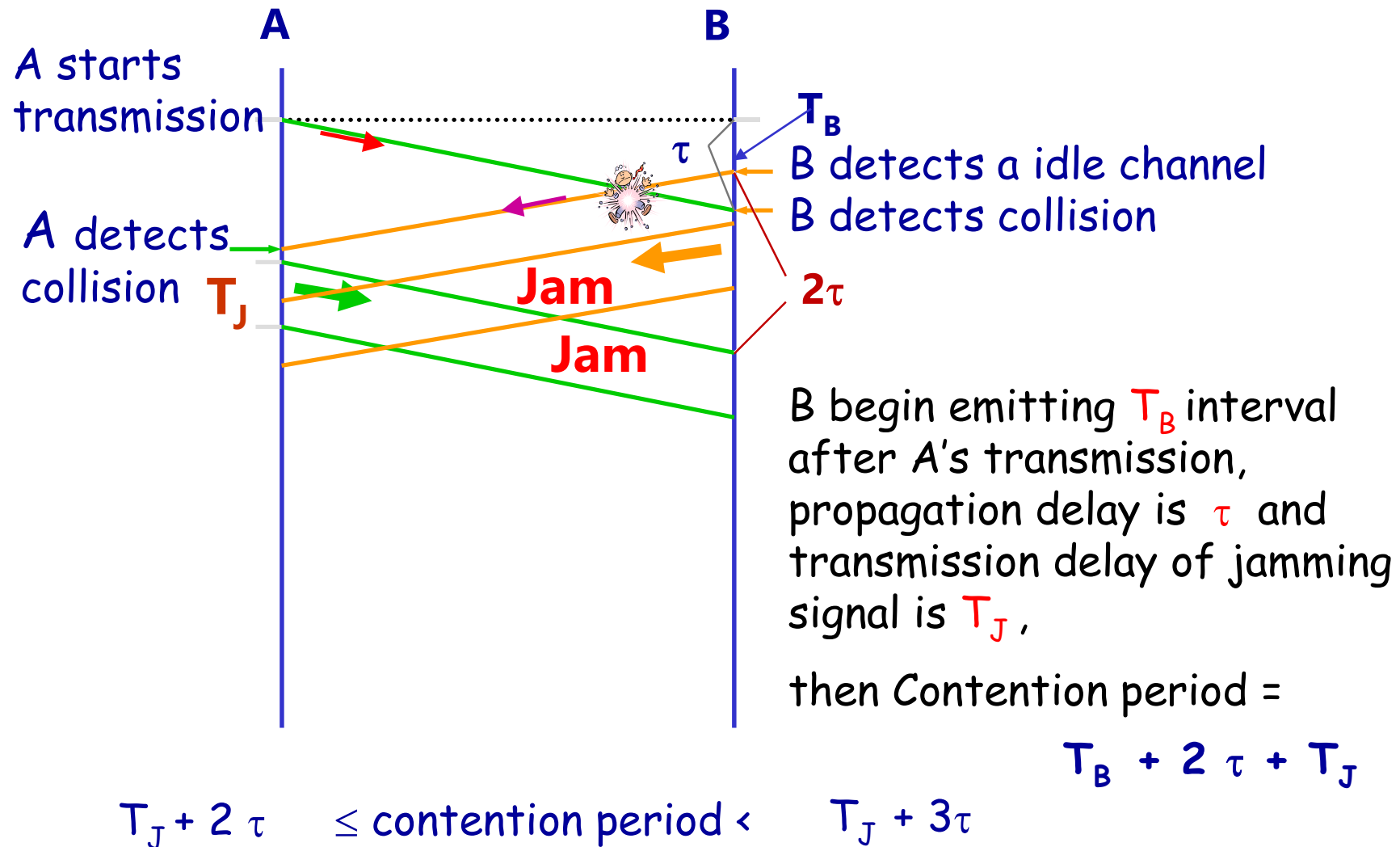
[Armstrong]

CSMA/CD: States

- CSMA/CD can be in one of three states: contention, transmission, or idle.



CSMA/CD: Contention Periods(争用期)



1-persistent CSMA/CD: Retransmission Waiting Time(1)

- Backoff Time(退避时间)
- Choose a random number k between 0 and d , and after a delay of k , check if the medium is clear.
 - ◆ If the medium is clear, transmit
 - ◆ If the medium is in use, wait until free.
- If a collision occurs on a subsequent attempt to send the same frame, repeat
 - ◆ Keep track of the number of attempts; if a preset maximum attempt number (often 16) is reached, abort
- Exponential backoff(指数退避): the maximum delay time is doubled between transmission retries

1-persistent CSMA/CD: Retransmission Waiting Time(2)

- Binary Exponential Backoff Algorithm (二进制指数退避算法)
 - ◆ Random pick a number r from $\{ 0, 1, 2, \dots, 2^k-1 \}$
 - ◆ Retransmission time = $r \times \text{basic retransmission_time}$
 - ◆ With $k = \min[\text{number of attempt}, 10]$

Binary Exponential Backoff Algorithm

Basic retransmission time: 51.2 μsec

1st collision, random number 0~1

2nd collision, random number 0~3

....

9th collision, random number 0~511

10th collision, random number 0~1023

11th collision, random number 0~1023

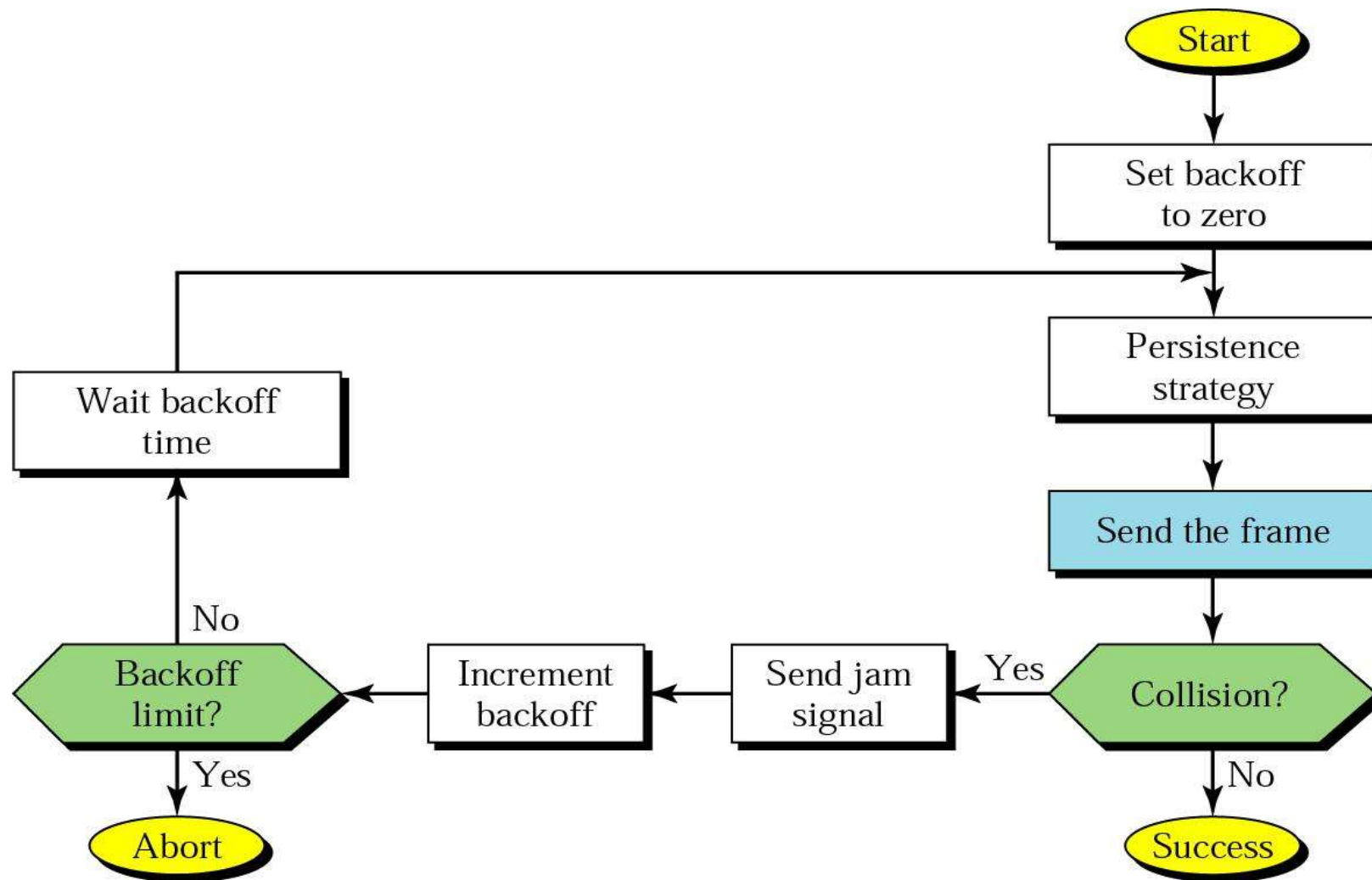
...

16th collision, random number 0~1023

After 16 collisions, reports failure

Further recovery is up to higher layers

Procedure of CSMA/CD



CSMA/CD

- 带冲突检测的载波监听多路访问协议
- 载波监听策略
 - ◆ 非坚持CSMA: 重载时, 吞吐量较高
 - ◆ 1坚持CSMA: 轻载时, 时延较低
 - ◆ p-坚持
- 冲突检测
 - ◆ 边发送边检测
 - ◆ 冲突: 收到了非自己发送的数据
 - ◆ 检测到冲突时, 停止发送, 并发送jam信号
- 退避时间: 二进制指数退避算法
 - ◆ 重发次数越多, 等待时间越长
- 性能: 帧发送时延远大于传播时延时, 信道效率高

Multiple Access Protocols

- ALOHA
- Carrier Sense Multiple Access Protocols
- *Collision-Free Protocols* 无冲突协议
- Wireless LAN Protocols

Polling(轮询)

- HDLC
- Primary station polling secondary stations one by one

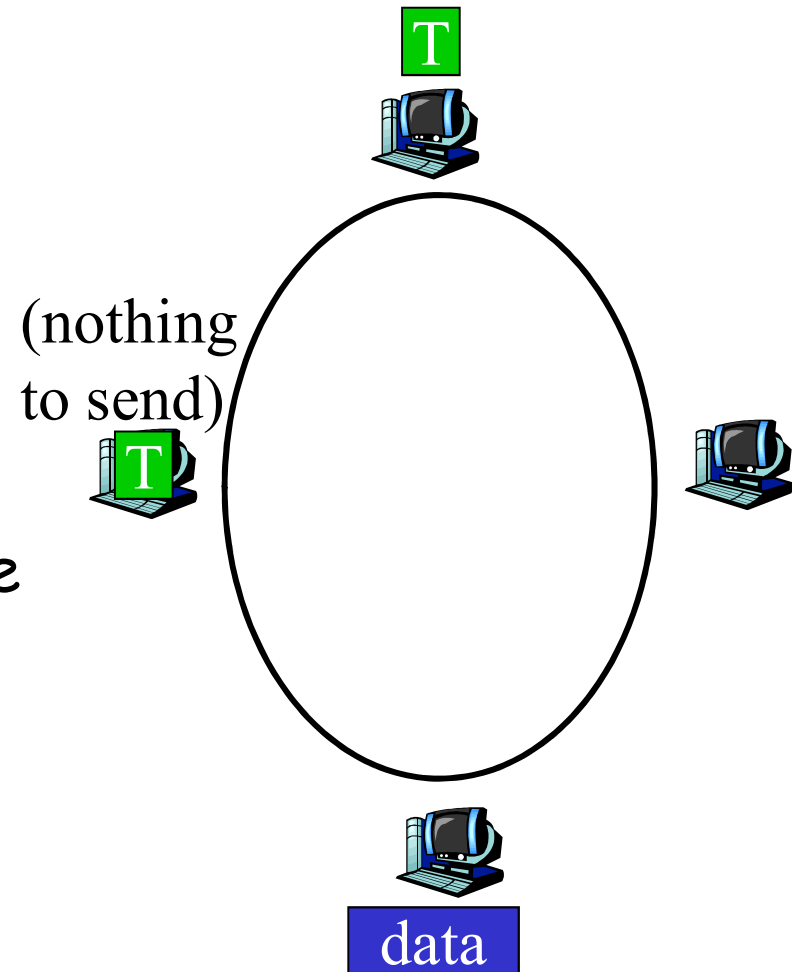


Token Passing (令牌传递)

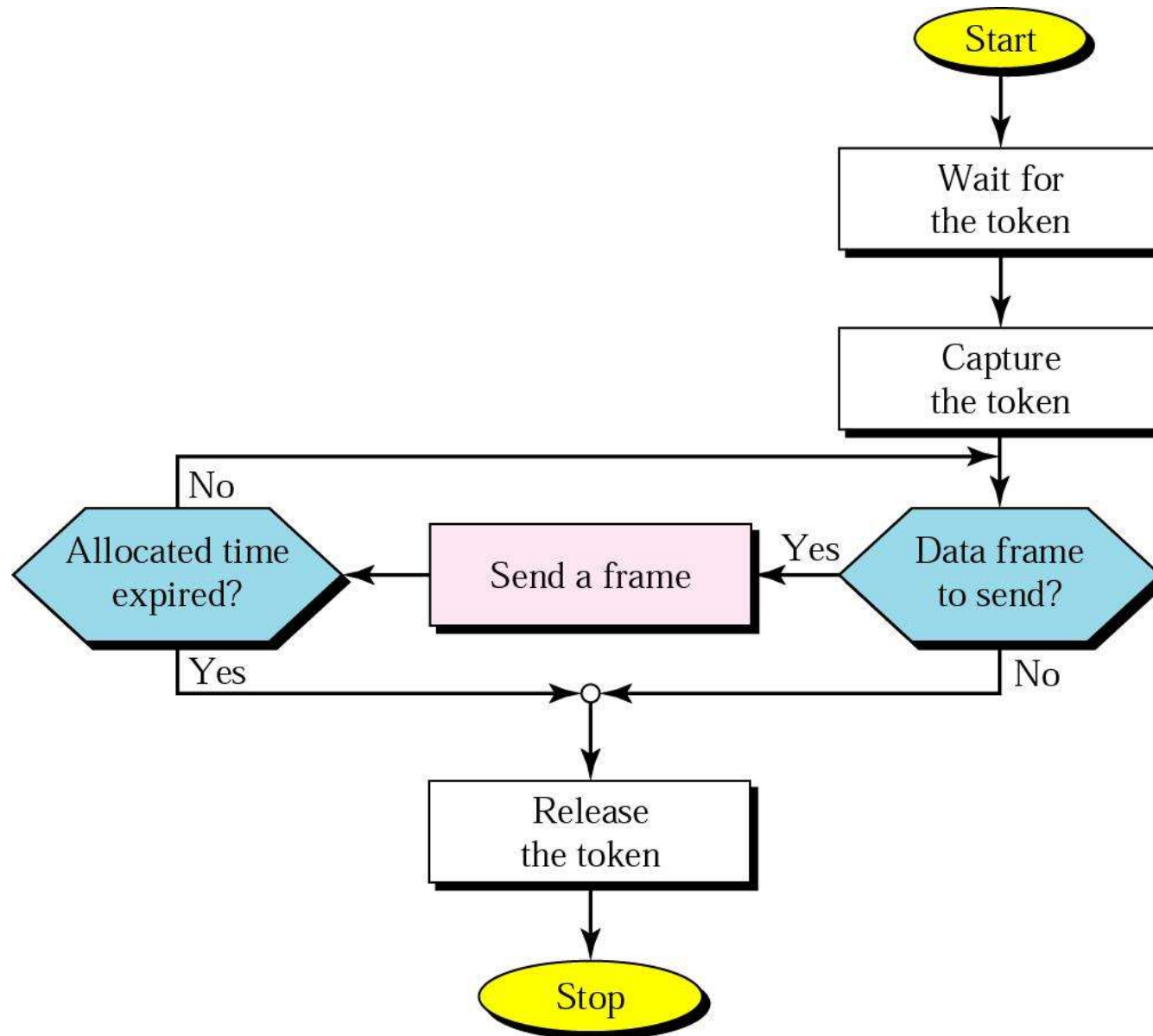
- Token: a small message representing permission to send.

- Token ring

- ◆ All stations connects to a ring
- ◆ Token passed from one node to next sequentially
- ◆ Station can transmit only when seizing token
- ◆ Collision free



Procedure of Token Passing

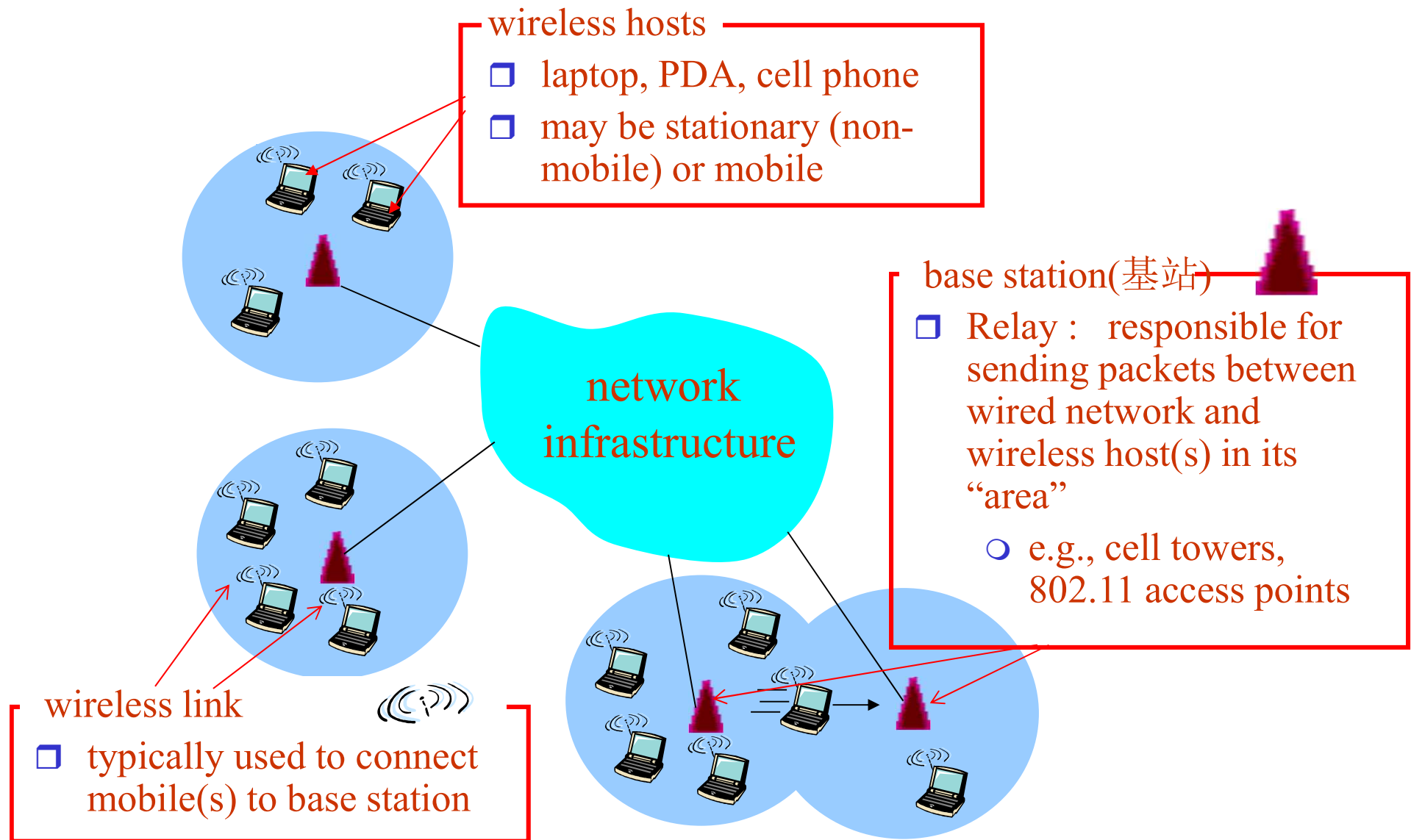


[Forouzan]

Multiple Access Protocols

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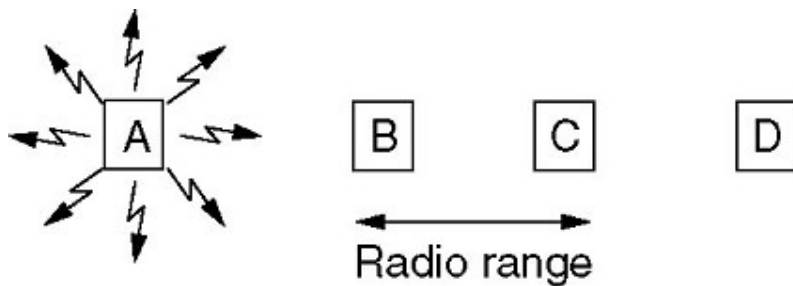
Elements of a wireless network



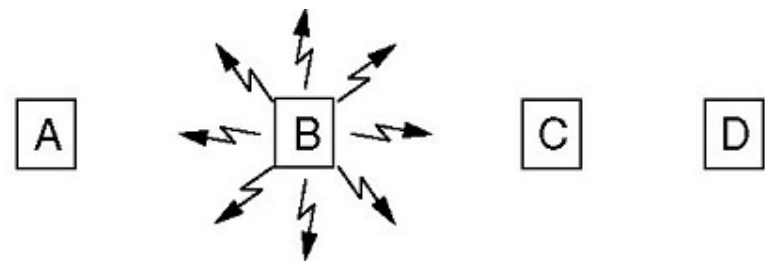
Problems

■ Radio Transmitter & Receiver

- ◆ Radio transmitters have some fixed range.
- ◆ In some wireless LANs, not all stations are within range of one another, thus collision may not be detected
- ◆ Interference at the **receiver**, not at the sender
- ◆ CSMA/CD is not appropriate



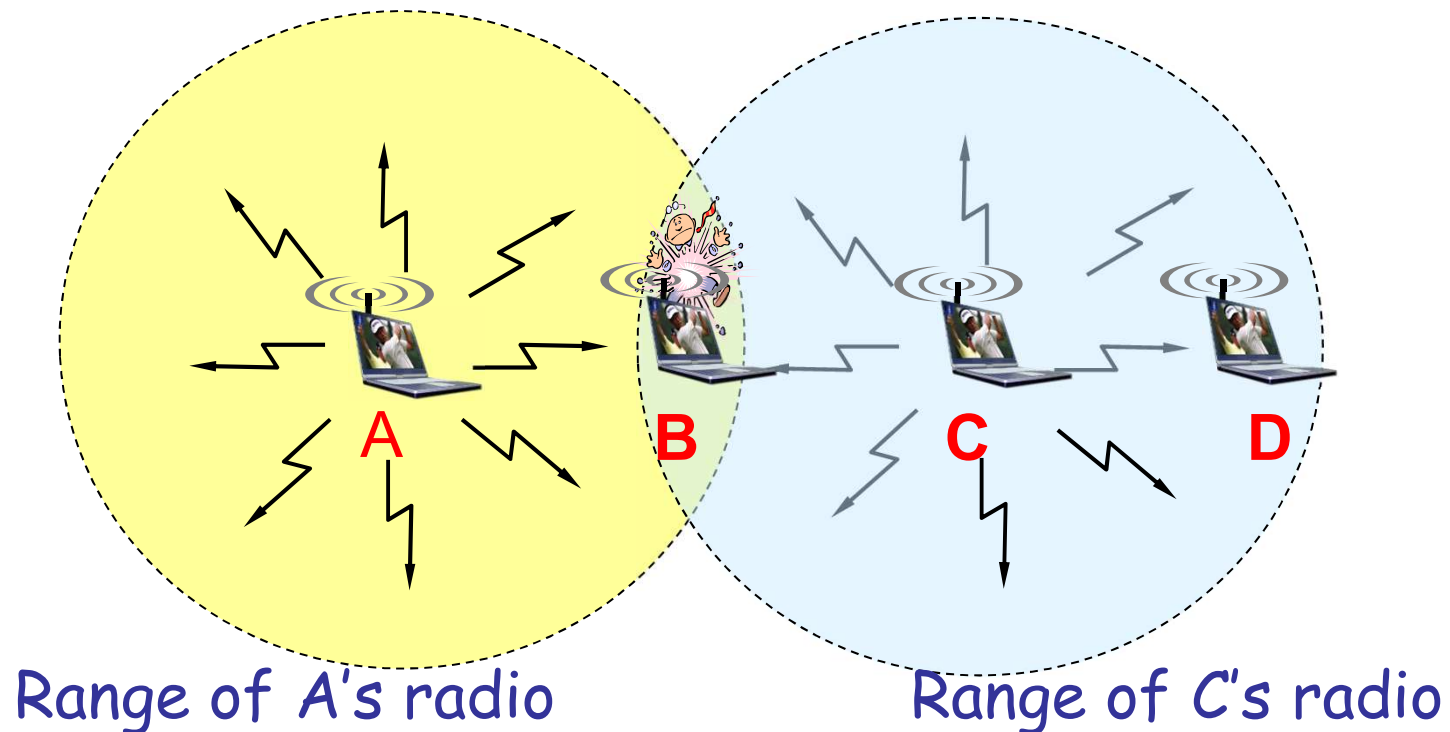
A is transmitting, C→B?



B is transmitting, C→D?

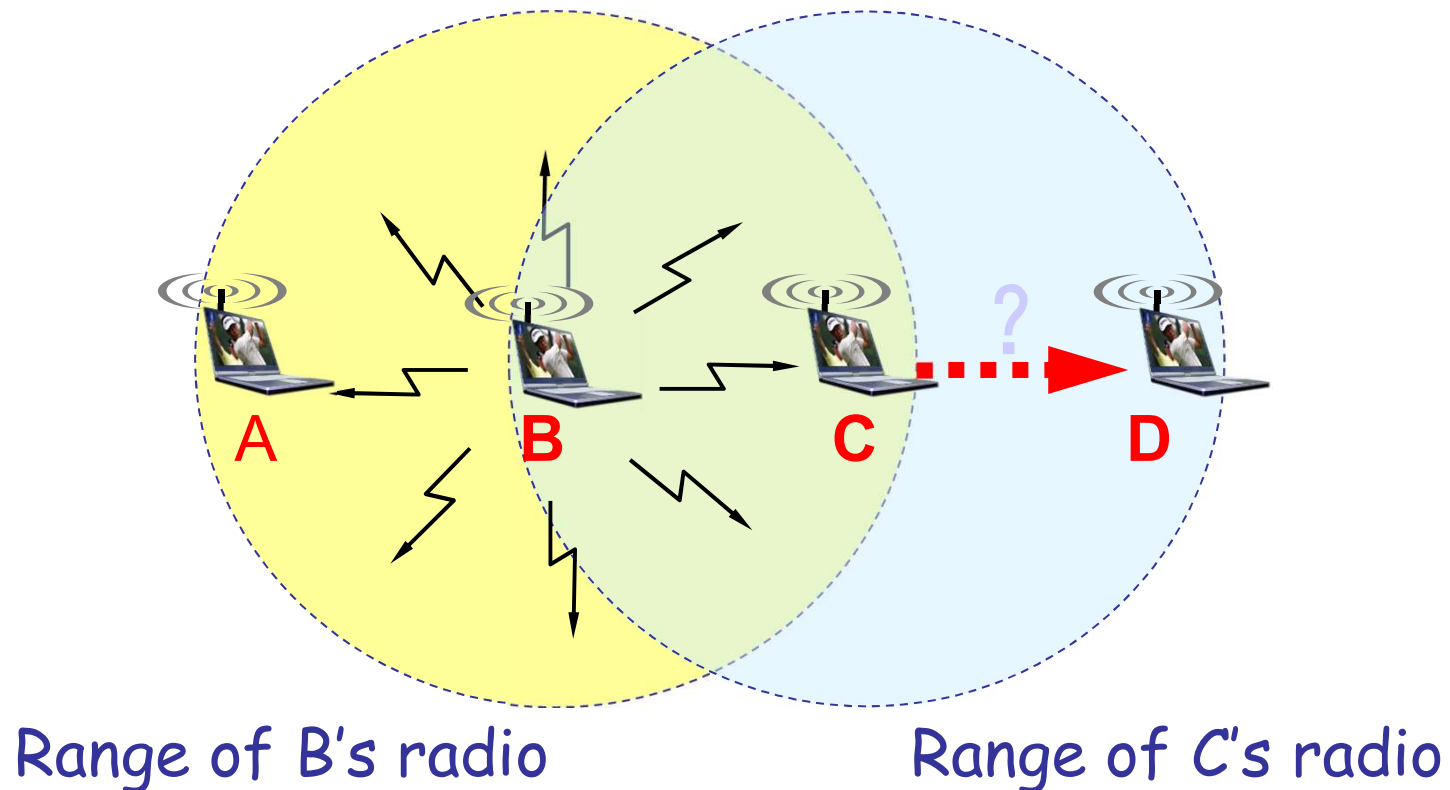
Hidden station (隐蔽站) problem

- A is transmitting to B
- C can not sense the signal, so C is sending to B at same time
- Collision at B (receiver)



Exposed station (暴露站/曝露站) problem

- B is sending to A
- C senses the signal, and **falsely** concluded that it can not send to D



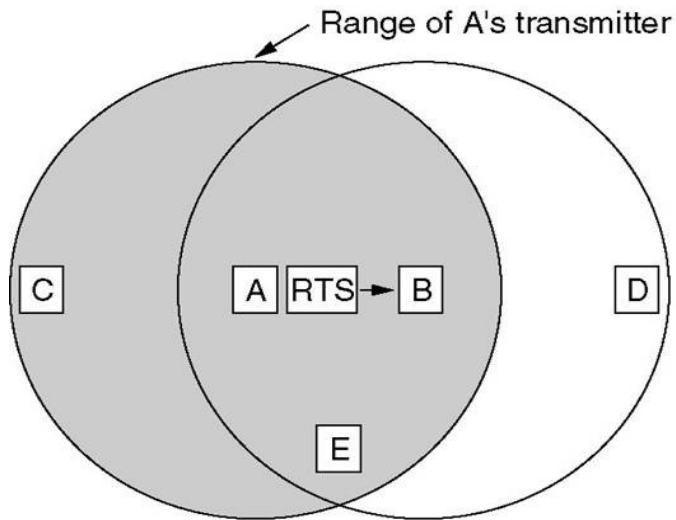
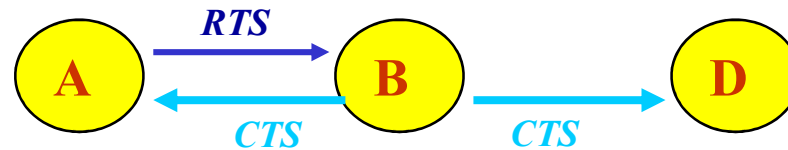
Solutions

- PCF (Point Coordination Function 点协调功能):
Polling(轮询) 集中控制
 - ◆ Uses **base station** to control all activity in its cell
 - ◆ Base station broadcasts a **beacon frame**(信标帧) periodically, inviting new stations to sign up for polling service
 - ◆ Once a station has signed up, it is effectively **guaranteed** a certain fraction of the **bandwidth**
- DCF (Distributed Coordination Function, 分布协调功能)
 - ◆ Stations use a **handshake** to determine which one can send
 - ◆ MACA: Multiple Access with Collision Avoidance, CSMA/CA: CSMA with Collision Avoidance

MACA (带冲突避免的多路访问)

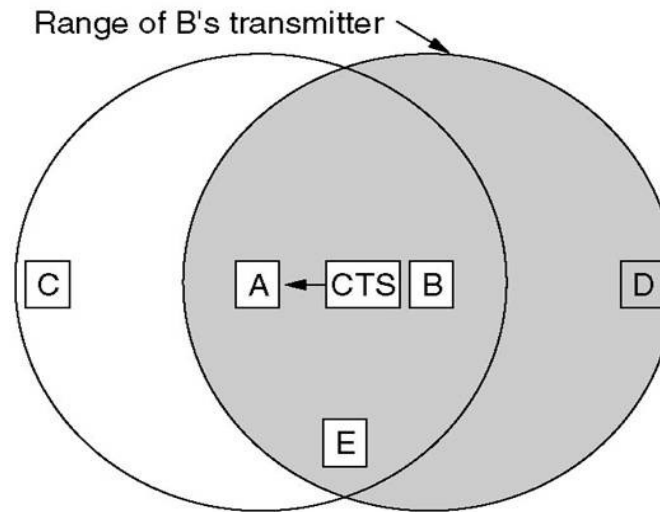
- Multiple Access Collision Avoidance
 - To further enhance reliability, four-frame exchange may be used
 - ◆ Source issues a Request to Send (RTS) frame to destination
 - ◆ Destination responds with Clear to Send (CTS)
 - ◆ After receiving CTS, source transmits data
 - ◆ Destination responds with ACK
- RTS* (S to R) -> *CTS* (R to S) -> *DATA* (S to R)
-> *ACK* (R to S)
- RTS and CTS are used to **reserve channel**
 - ◆ RTS alerts all stations within range of source
 - ◆ CTS alerts all stations within range of destination

MACA: Operation Example



Hearing **RTS**, **C** and **E** keep silent

RTS alerts all stations
within range of source

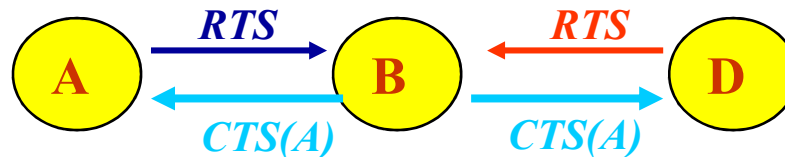


Hearing **CTS**, **D** and **E** keep silent
solving hidden station problem

CTS alerts all stations
within range of destination

MACA: Collisions

- 2 stations send **RTS** frames to one station at the same time
- Unsuccessful transmitter (does not hear a **CTS** within expected time interval) waits a random amount of time and tries again
- The algorithm used is **binary exponential backoff**(二进制指数退避)
 - Similar to **CSMA/CD**



Review of MAC Protocols

■ Contention-based

- ◆ ALOHA, sending at will
- ◆ CSMA/CD
 - listening before sending, checking collision while sending

■ Collision-Free

- ◆ token ring: token as right to send

■ Wireless LAN

- ◆ Hidden station problem and Exposed station problem
- ◆ PCF, polling by base station
- ◆ MACA, using RTS/CTS for channel reservation

Outline

- Channel Allocation Problem
- Multiple Access Protocols
- LAN Reference Model: IEEE 802 RM
- Ethernet
- Wireless LANs
- Data Link Layer Switching

Introductions to LANs

■ Features of LAN

- ◆ Limited number of stations and range
- ◆ Higher data rate
- ◆ Lower error rate
- ◆ Supporting broadcasting and multicasting
- ◆ Single administration

■ Topologies

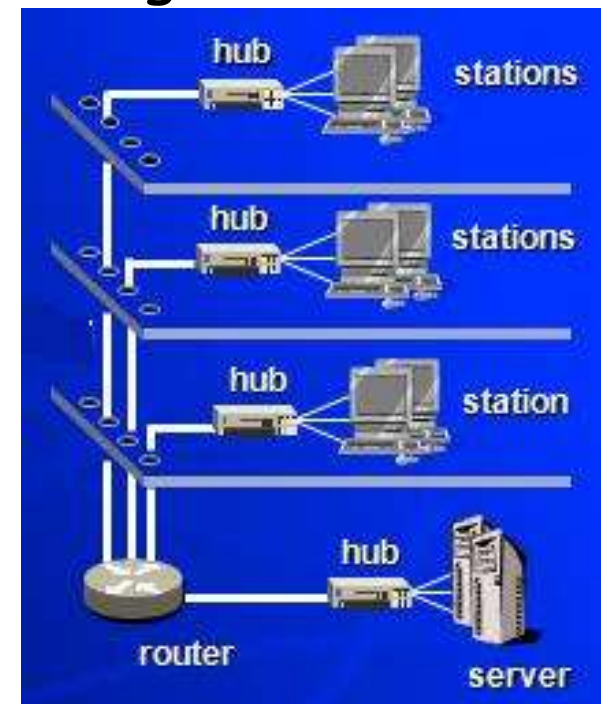
- ◆ Bus, Star, Ring, Tree, ...

■ Transmission Media

- ◆ Twisted pair, Coaxial Cable, Fiber, RF

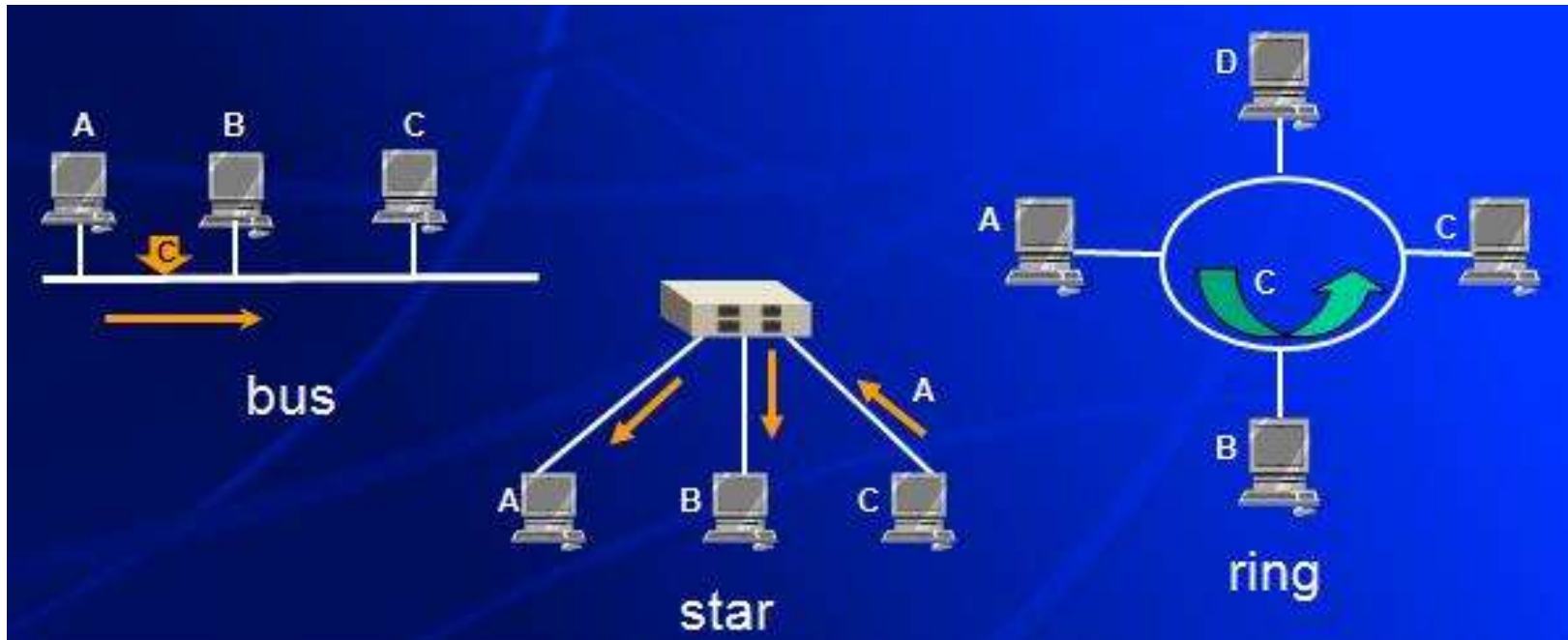
■ Media Access Method

- ◆ Following protocols to share a common channel

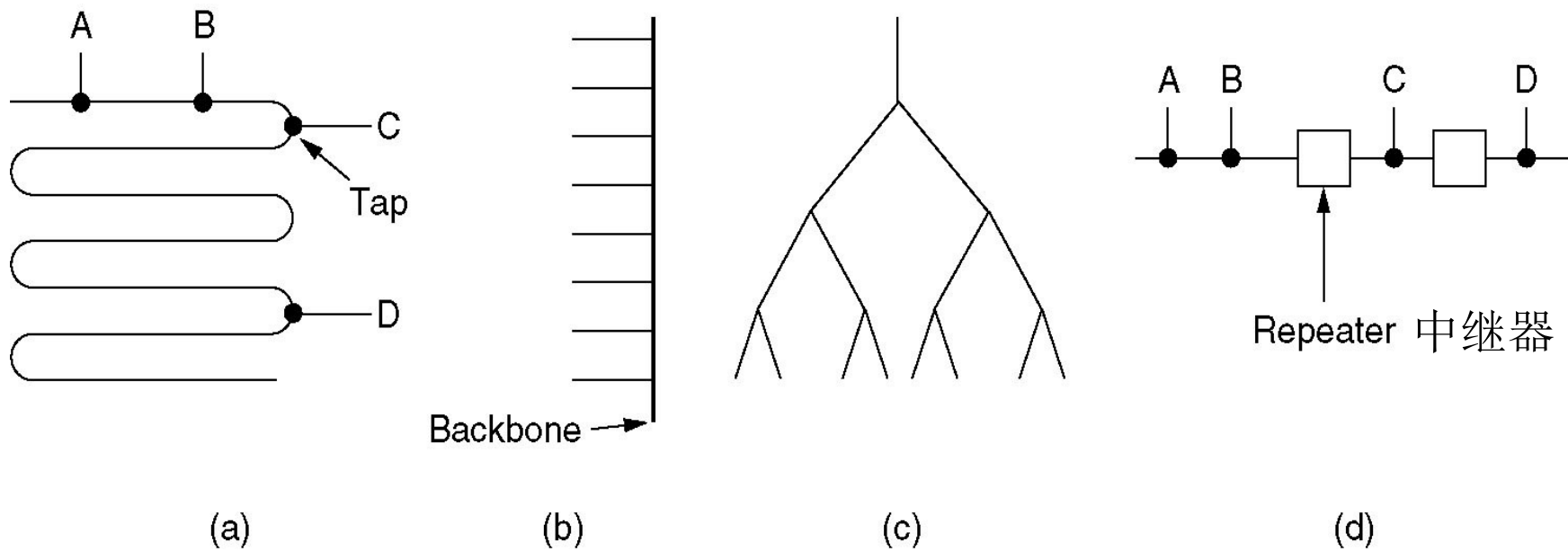


Typical topologies

- Bus/Tree: All stations connected to common channel
- Star: All stations connected to central node (eg. HUB)
- Ring: A station connects to neighbor with point-to-point link

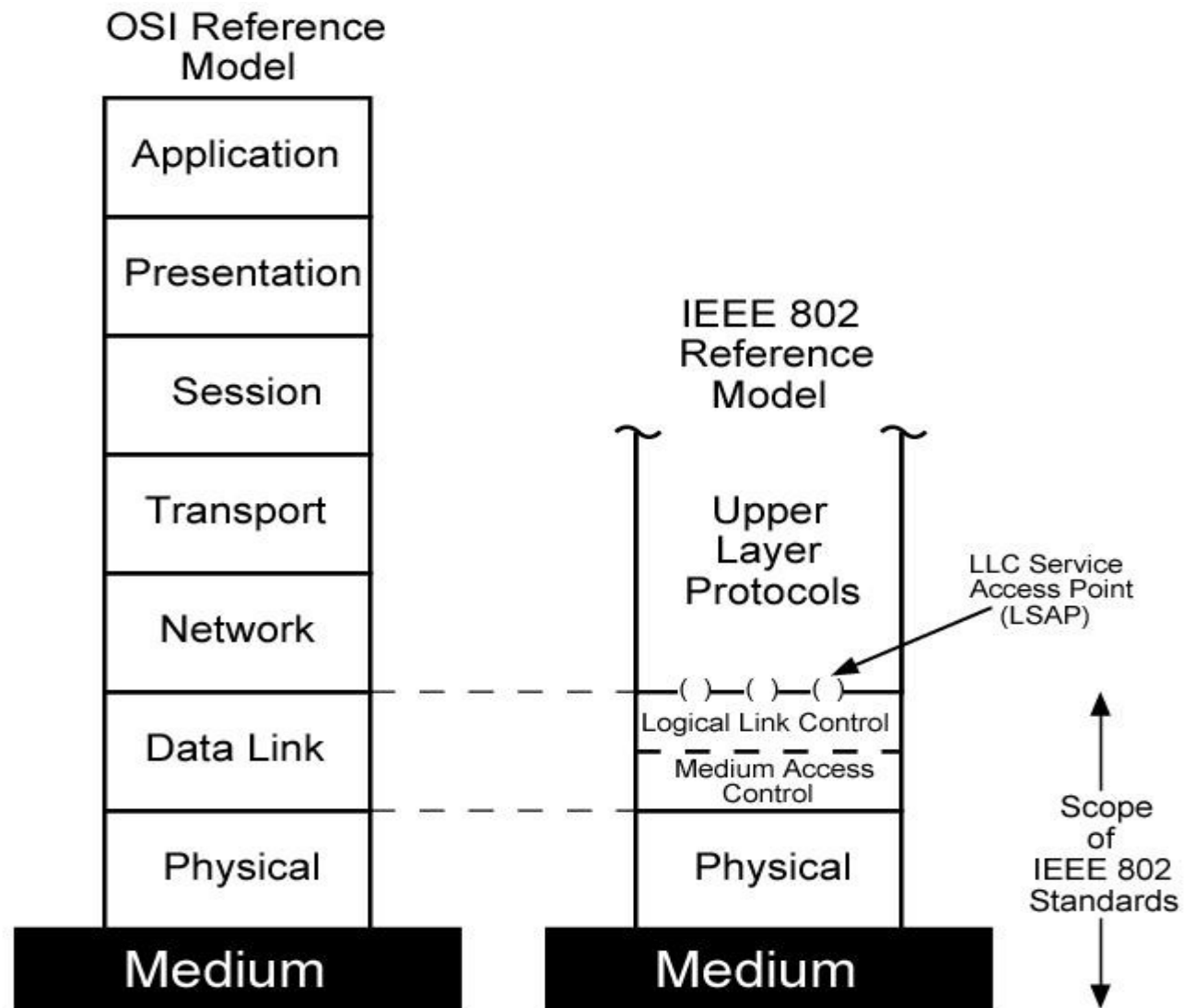


Other cabling topologies



(a) Linear, (b) Spine, (c) Tree, (d) Segmented(分段)

IEEE 802 Reference Model



[Stallings]

IEEE 802 Layers

- Physical Layer
 - ◆ Encoding/decoding
 - ◆ Bit transmission/reception
 - ◆ Transmission medium and topology
- MAC (Media Access Control, 介质访问控制)子层
 - ◆ Framing
 - ◆ Error checking
 - ◆ Physical Addresses: MAC addresses
 - ◆ Sharing of medium: Multiple Access Control
 - ◆ LAN switching, VLAN
- LLC (Logic Link Control, 逻辑链路控制)子层
 - ◆ Hides difference of MAC layers, providing a uniform interface to network layer
 - ◆ Flow control and error control
 - ◆ Inter-connecting LANs with bridge(网桥)

LLC Services and Protocol

■ LLC Services

- ◆ Unreliable datagram service, e.g. *Ethernet*
- ◆ Acknowledged datagram service, e.g. *MACA* (Multiple Access Collision Avoidance)
- ◆ Reliable connection-oriented service, e.g. Token Ring

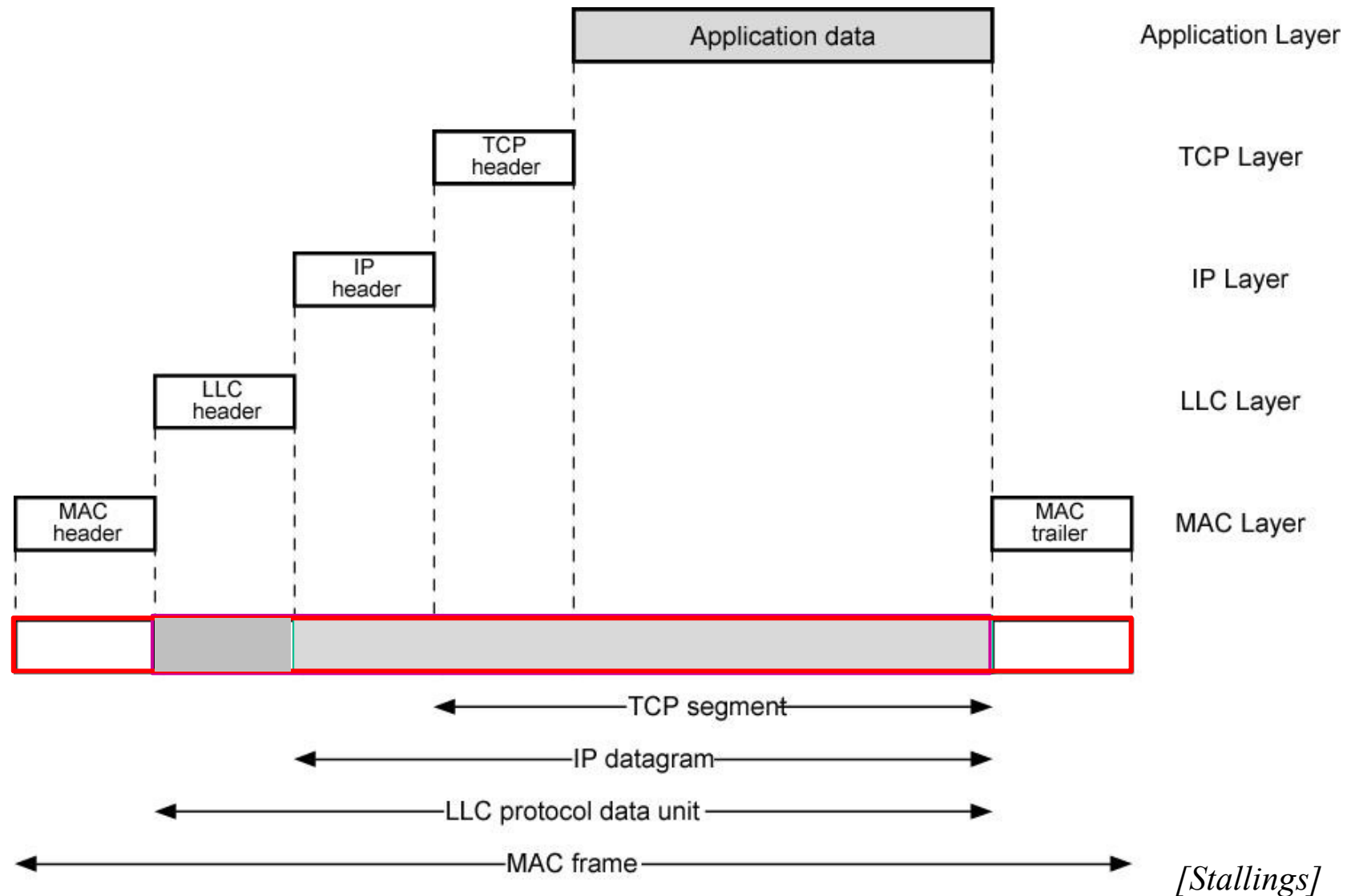
■ LLC protocol

- ◆ Based on HDLC, but not identical
- ◆ Multiplexing using LSAPs (LLC Service Access Points)

MAC Overview

- Assembly of data into frame with address and error detection fields at sender side
 - Disassembly of frame at receiver side
 - ◆ Address recognition
 - ◆ Error detection
 - Govern access to transmission medium
 - For the same LLC, several MAC options may be available
- (不同的LAN, **MAC**子层不同, **LLC**子层相同)

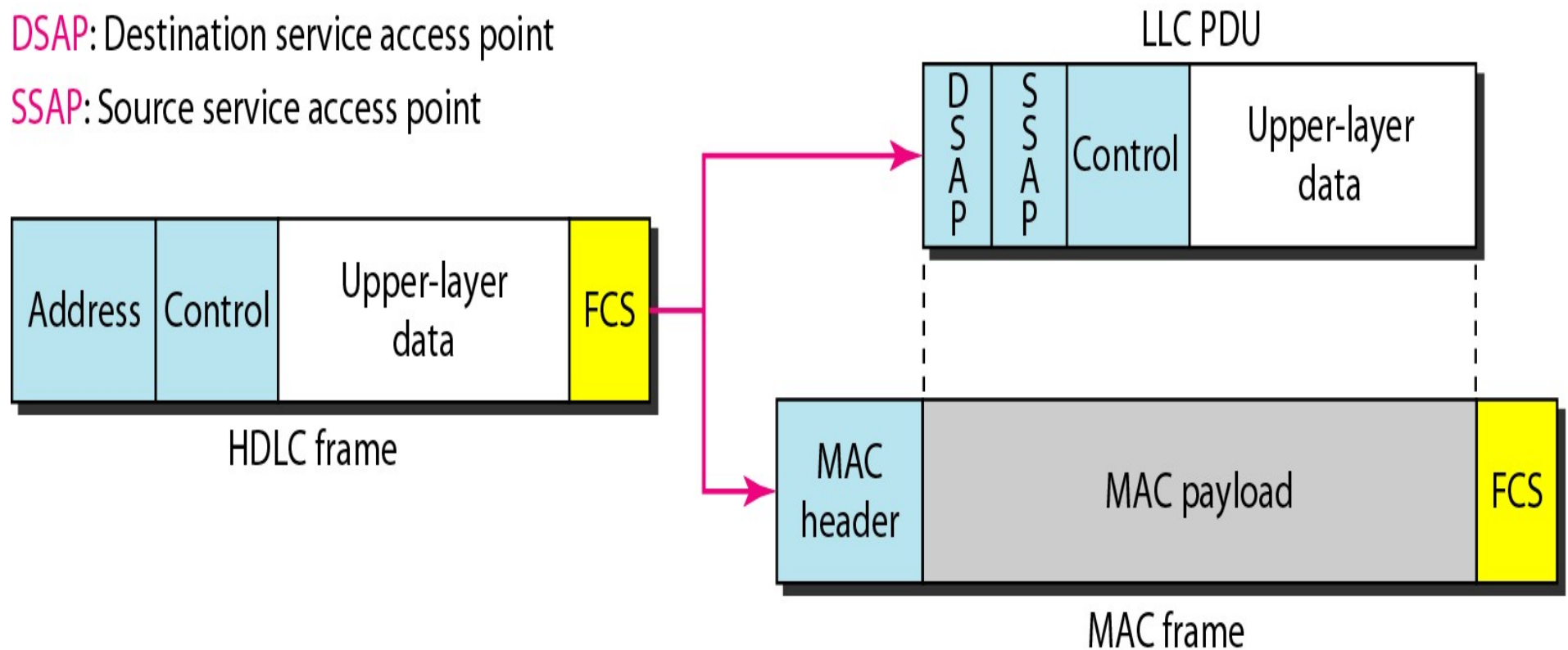
LAN Protocols in Context (封装)



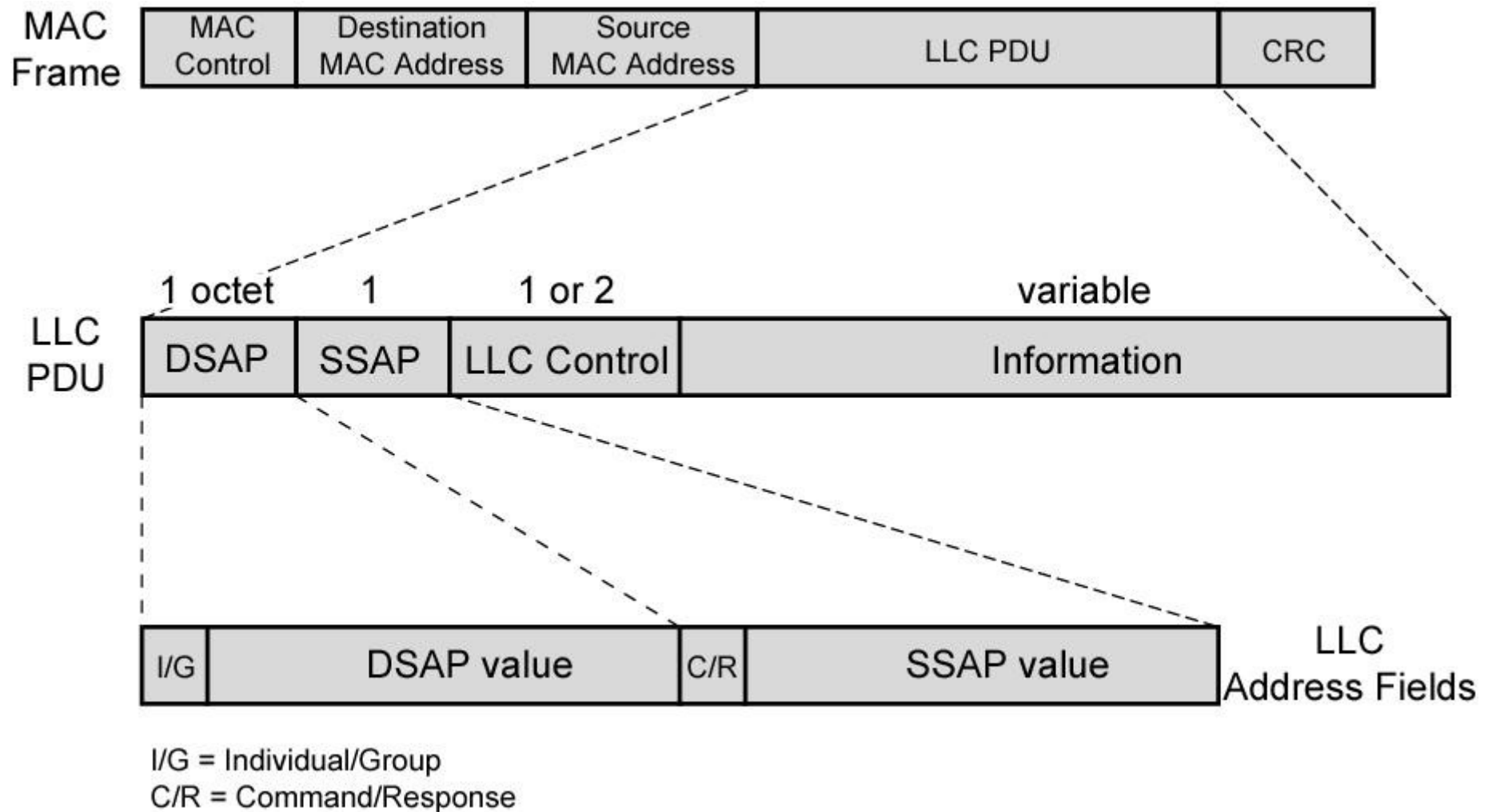
HDLC frame compared with LLC PDU and MAC frame

DSAP: Destination service access point

SSAP: Source service access point

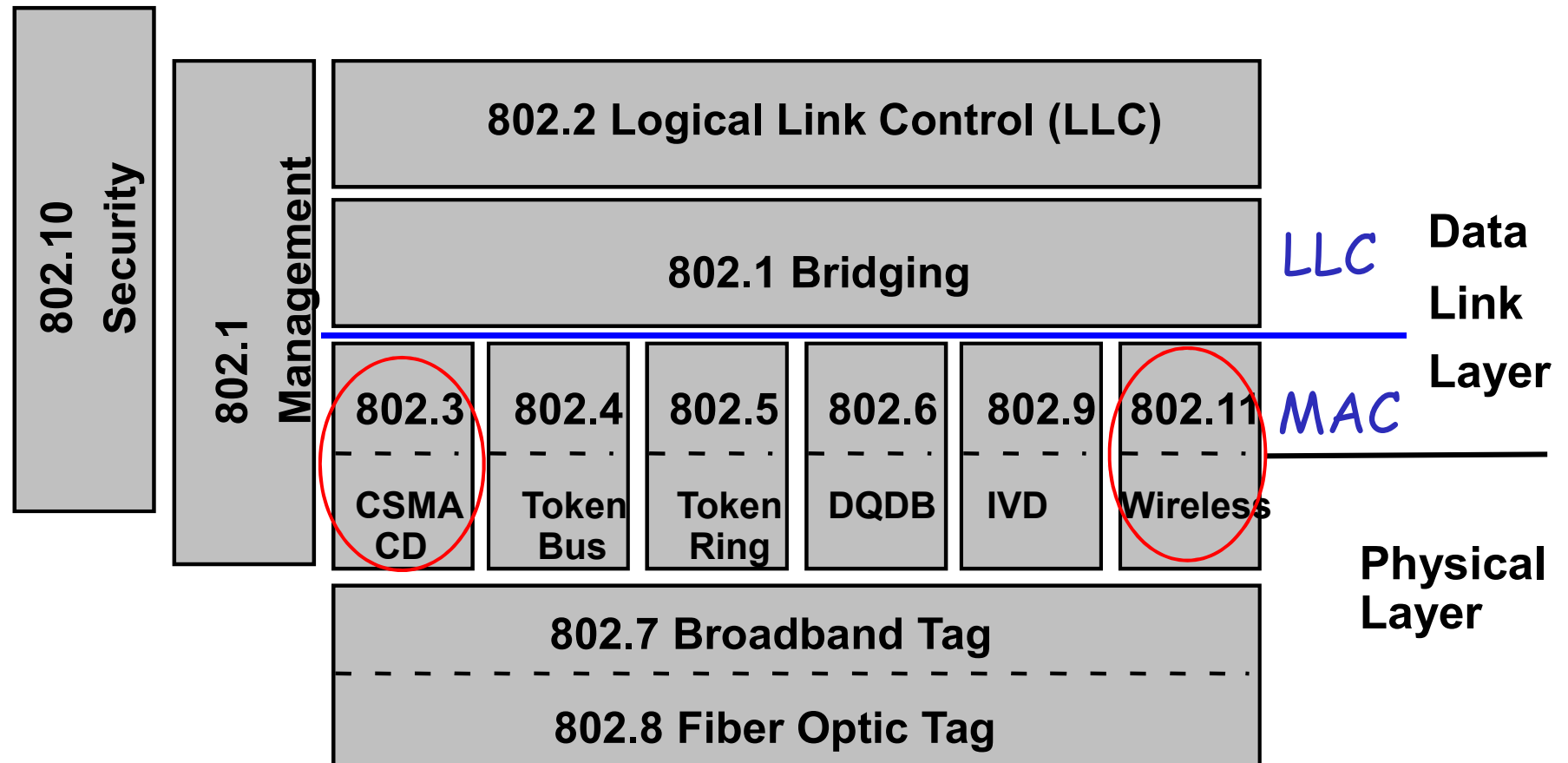


Format of LLC PDU and MAC Frame



[Stallings]

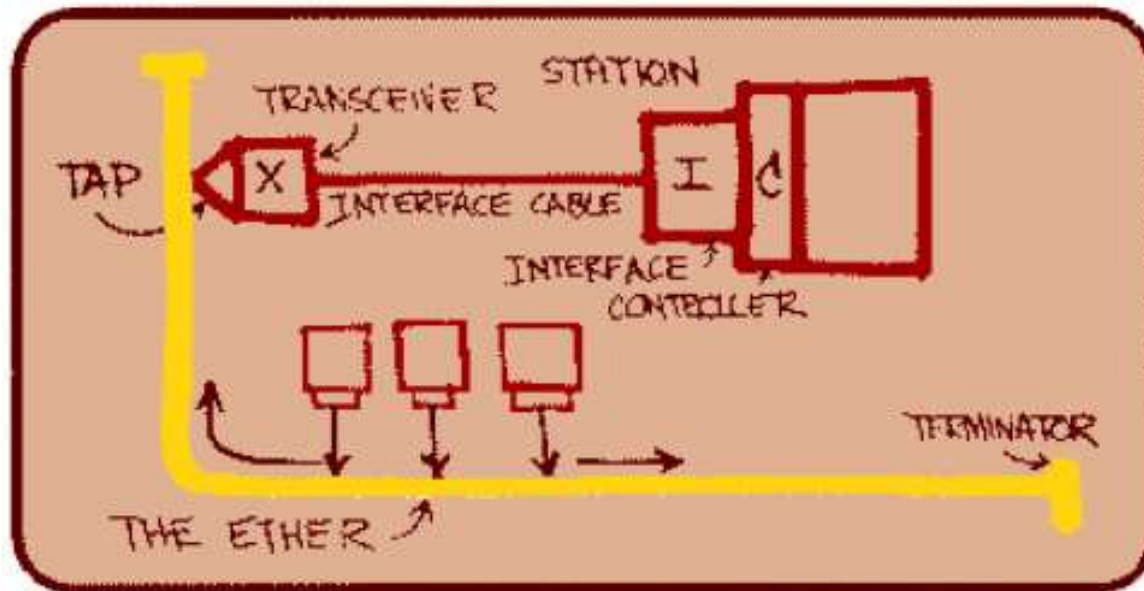
IEEE 802 Standards



Outline

- Channel Allocation Problem
- Multiple Access Protocols
- IEEE 802 Reference Model
- Ethernet
- Wireless LANs
- Data Link Layer Switching

Bob Metcalfe's Vision



June 1976 National Computer Conference
(INTEL had just developed 8080, running at 4.77M)

■ Robert Metcalfe (1946~)

- ◆ 1969 Graduated from MIT
- ◆ 1973 Harvard (Ph.D)
- ◆ 1972~1979 Xerox PARC (Palo Alto Research Center)
 - 1973 Ethernet, 3Mbps
- ◆ 1979~1990 3Com
- ◆ 1990~2000 Column author, 2011~ UT Austin

Early History

1973	Xerox PARC begins development of bus topology LAN, 2.94Mbps
1976	Successful CSMA/CD system to connect 100 workstations on 1 km cable
1980	Digital Equipment Corp, Intel Corp, and Xerox release de-facto Ethernet DIX standard, 10Mbps → Classic Ethernet
1980	IEEE 802 Standardization process begins
1983	IEEE 802.3 Standard Published
1992	Grand Junction Network Company brings up the structure of 100Mbps Ethernet
1998	Standard of Gigabit Ethernet addressed
2002	1 st Standard of 10Gigabit Ethernet published

Ethernet

- Classic Ethernet(传统以太网): Physical Layer
- Classic Ethernet: MAC Sublayer Protocol
- Ethernet Performance
- Switched Ethernet
- Fast Ethernet
- Gigabit Ethernet
- 10G Ethernet

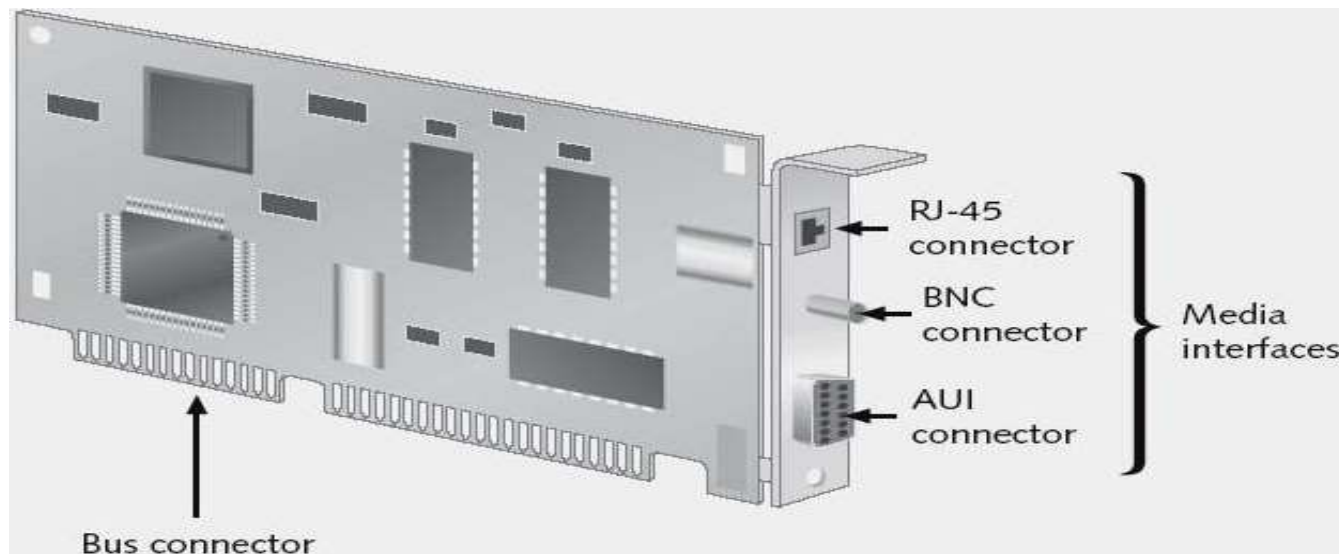
Network Interface Cards (网卡)

■ Tasks of NIC

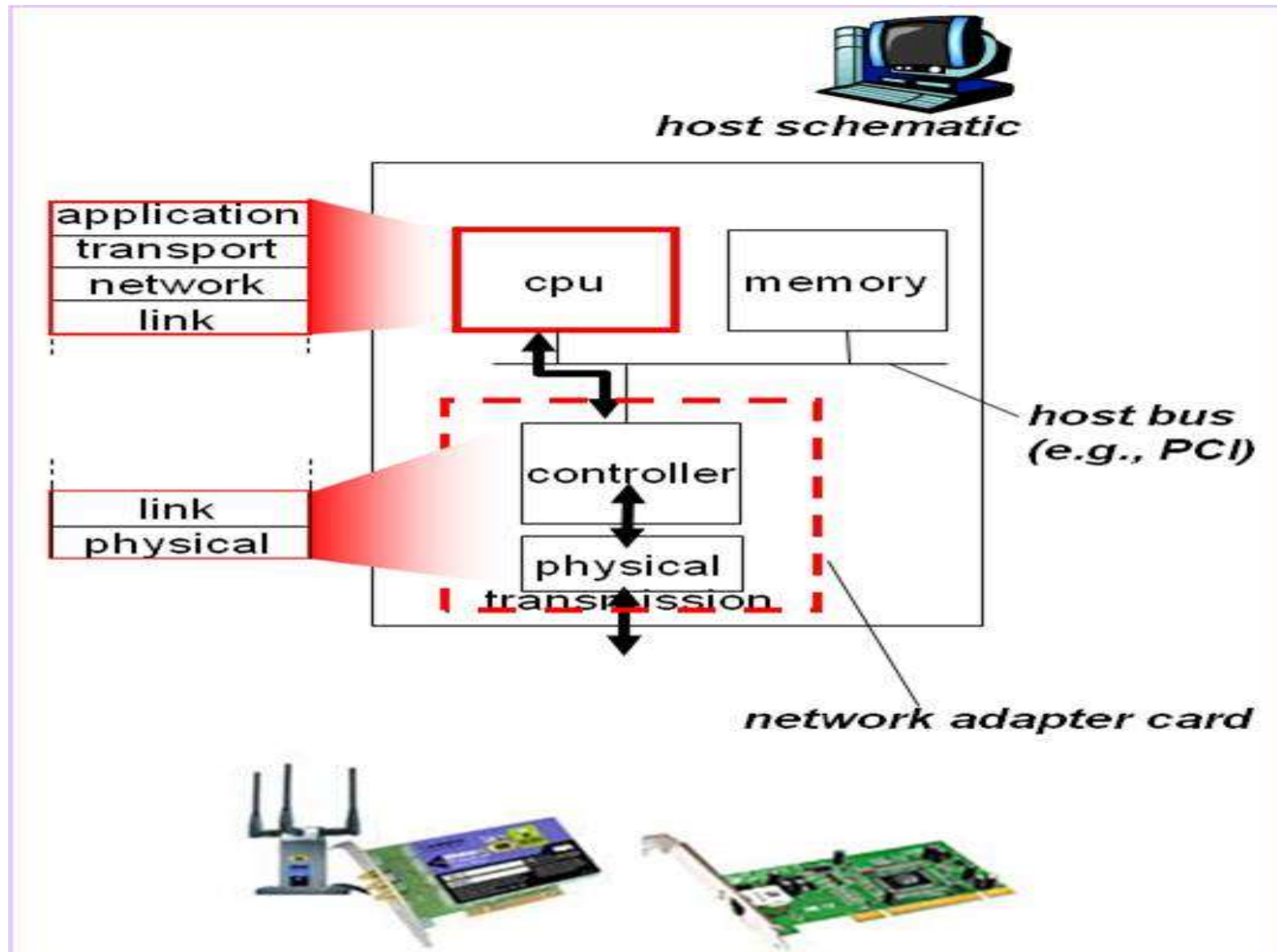
- ◆ Establishes and manages the computer's network connection
- ◆ Encoding: translates digital computer data into signals

■ NIC provides LLC, MAC and physical layer functions

■ Each card has a unique MAC address

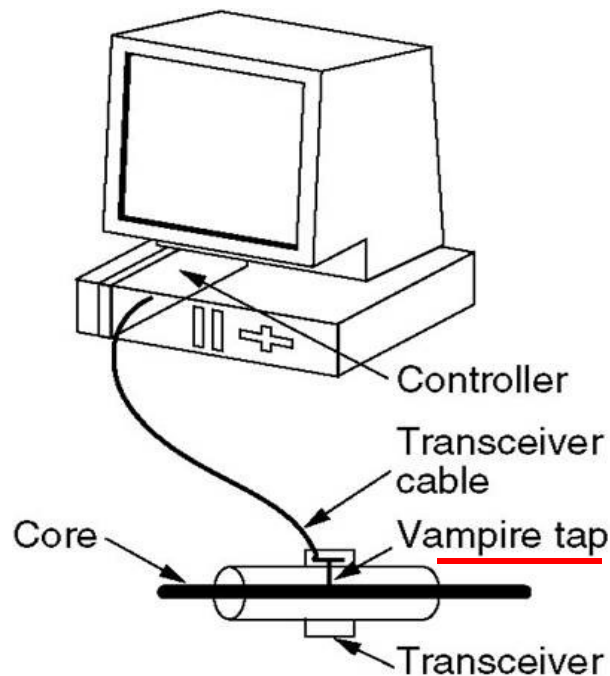


NIC (2)

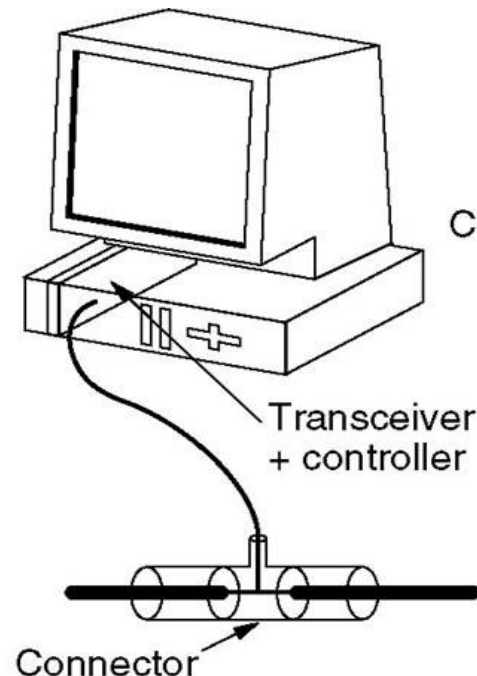


Classic Ethernet (传统以太网) : 10Mbps

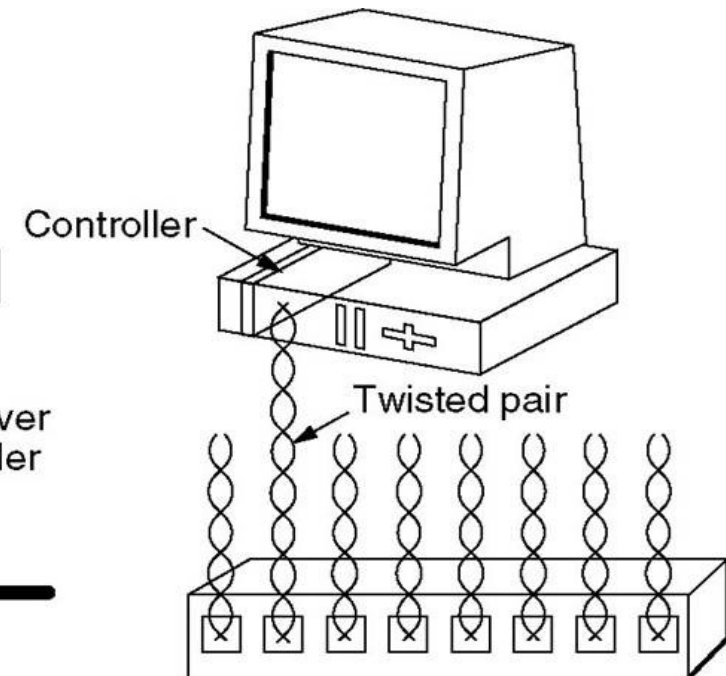
Name	Cable	Max. seg.	Nodes/seg.	Advantages
10Base5	Thick coax	500 m	100	Original cable; now obsolete
10Base2	Thin coax	185 m	30	No hub needed
10Base-T	Twisted pair	100 m	1024	Cheapest system
10Base-F	Fiber optics	2000 m	1024	Best between buildings



10Base5

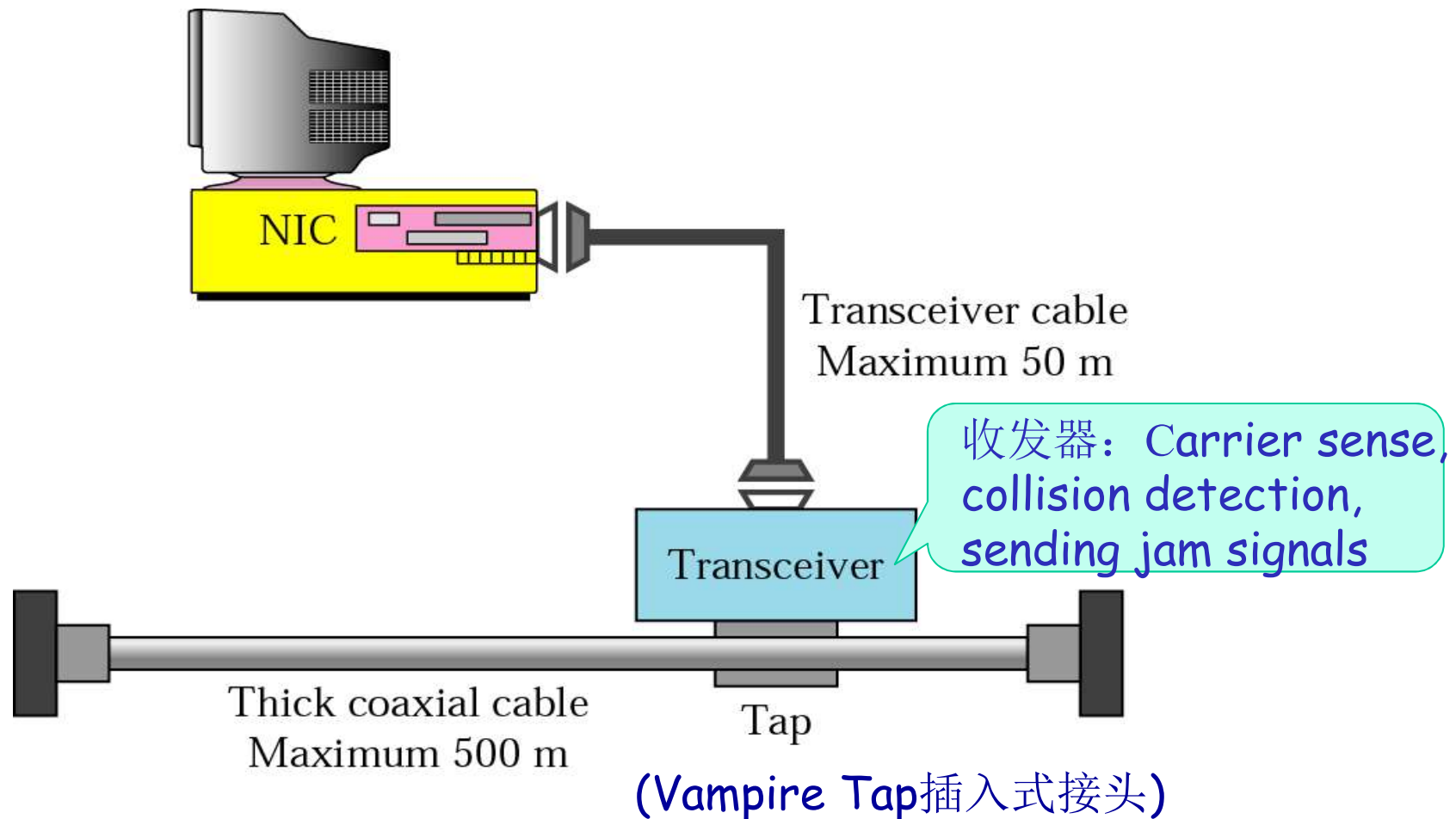


10Base2

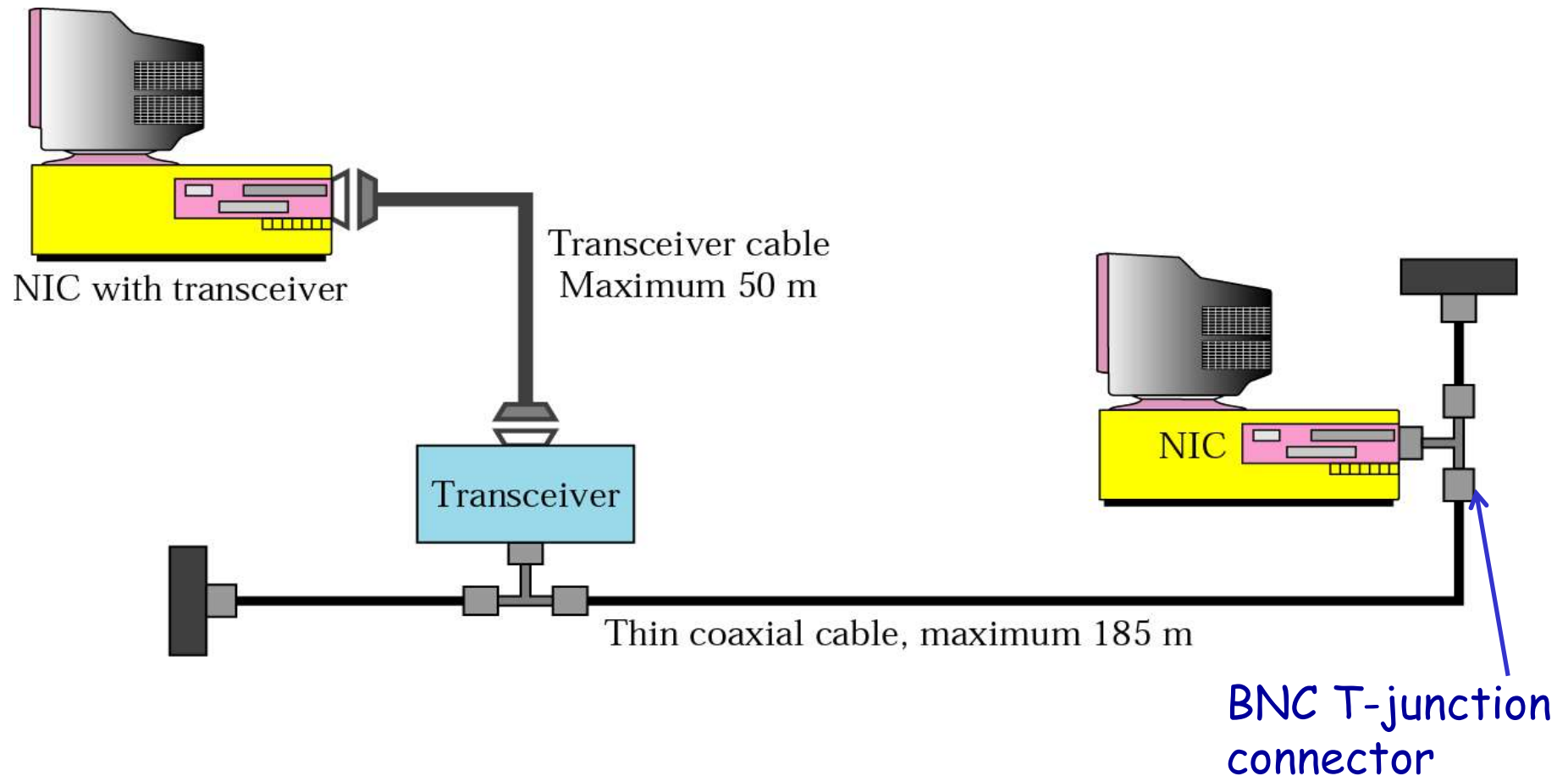


10Base-T

Connection of a Station using 10Base5



Connections of Stations using 10Base 2



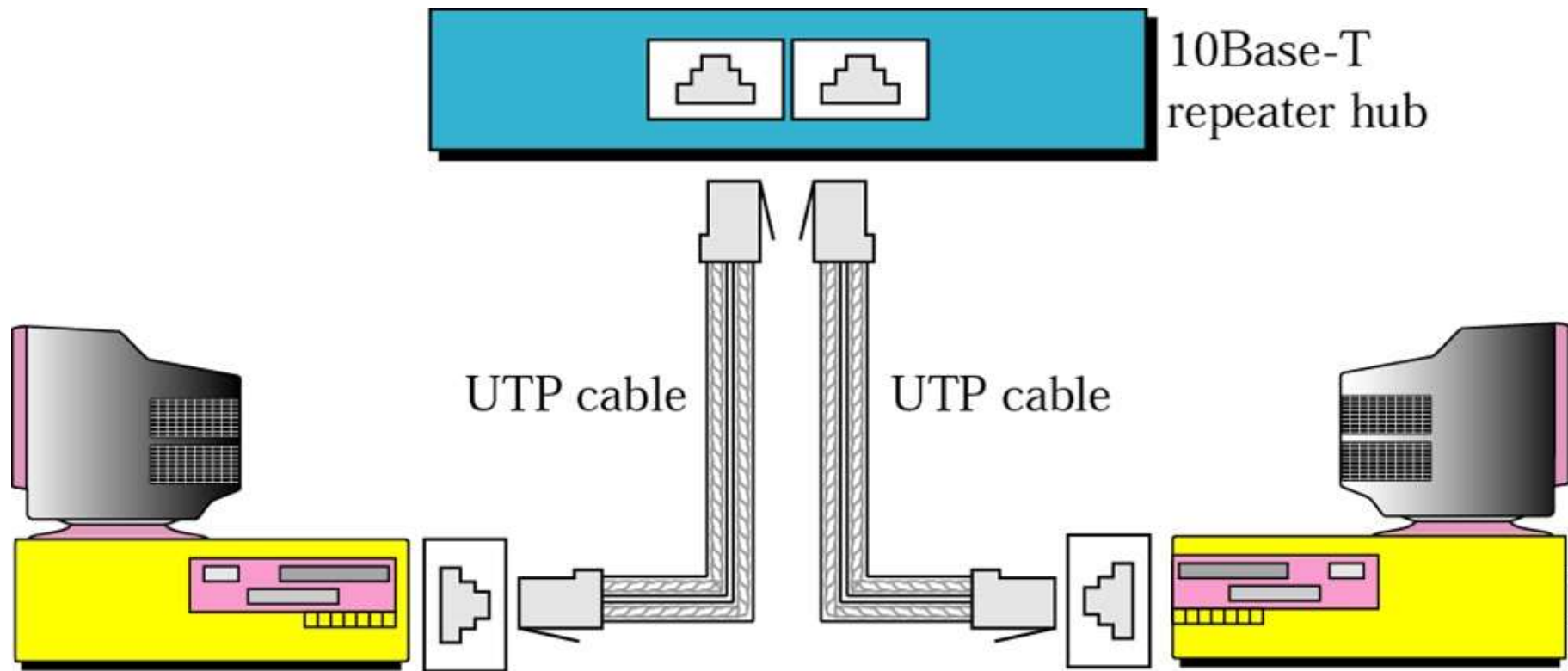
10Base2: Thin Cable Ethernet

- Cable: 5mm, Coax, 185m
- BNC T-junction connector
- Transceiver electronics are put on the NIC



BNC T-junction connector

10BaseT: Connection of Stations

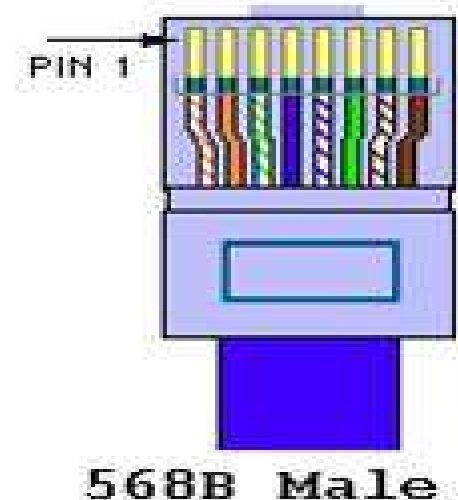
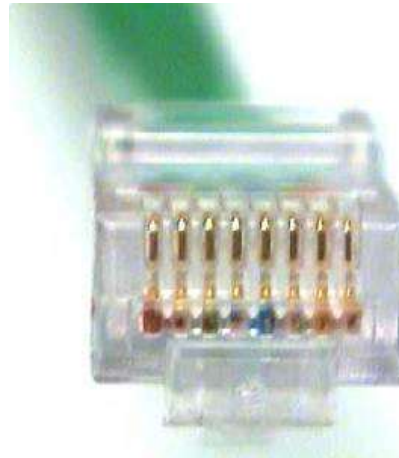


10BaseT

- 10Base-T: No shared cable at all (没有共享链路)
- HUB(集线器)
 - ◆ A box full of electronics
 - ◆ Station is connected to HUB by a **dedicated** cable (Max 100m)
- RJ-45, UTP5 (Unshielded Twisted Pairs)
 - ◆ 两对线，收发各一对
 - ◆ 差分传输：一对线中的一根线以正电平方式传输信号，另外一根线以负电平方式传输同一信号，减少电磁干扰
 - ◆ 同一线对的绞合：减少近端串扰

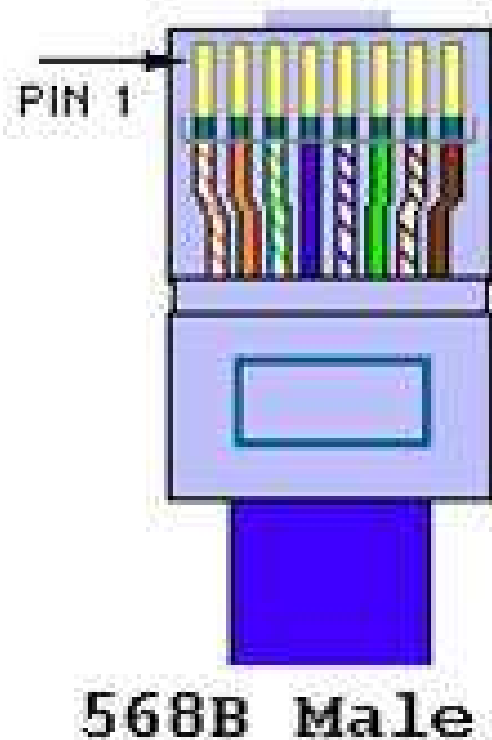
Connection Media: UTP Implementation

- RJ-45 connectors are used for UTP cable
- RJ stand for registered jack, and the number 45 refers to a specific wiring sequence



Wiring standard: TIA/EIA-568B

- T568B is the most commonly used color scheme

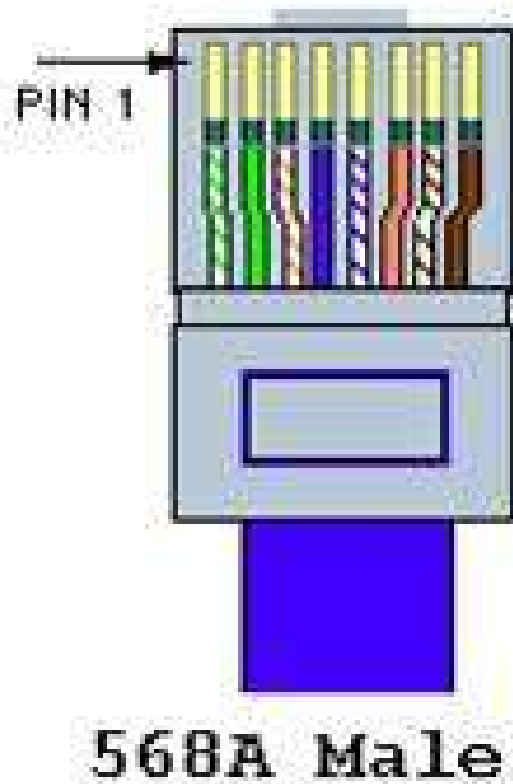


Pin	Color
1	White-Orange
2	Orange
3	White-Green
4	Blue
5	White-Blue
6	Green
7	White-Brown
8	Brown

NIC Transmits on
1 and 2 (Orange)

NIC Receives on
3 and 6 (Green)

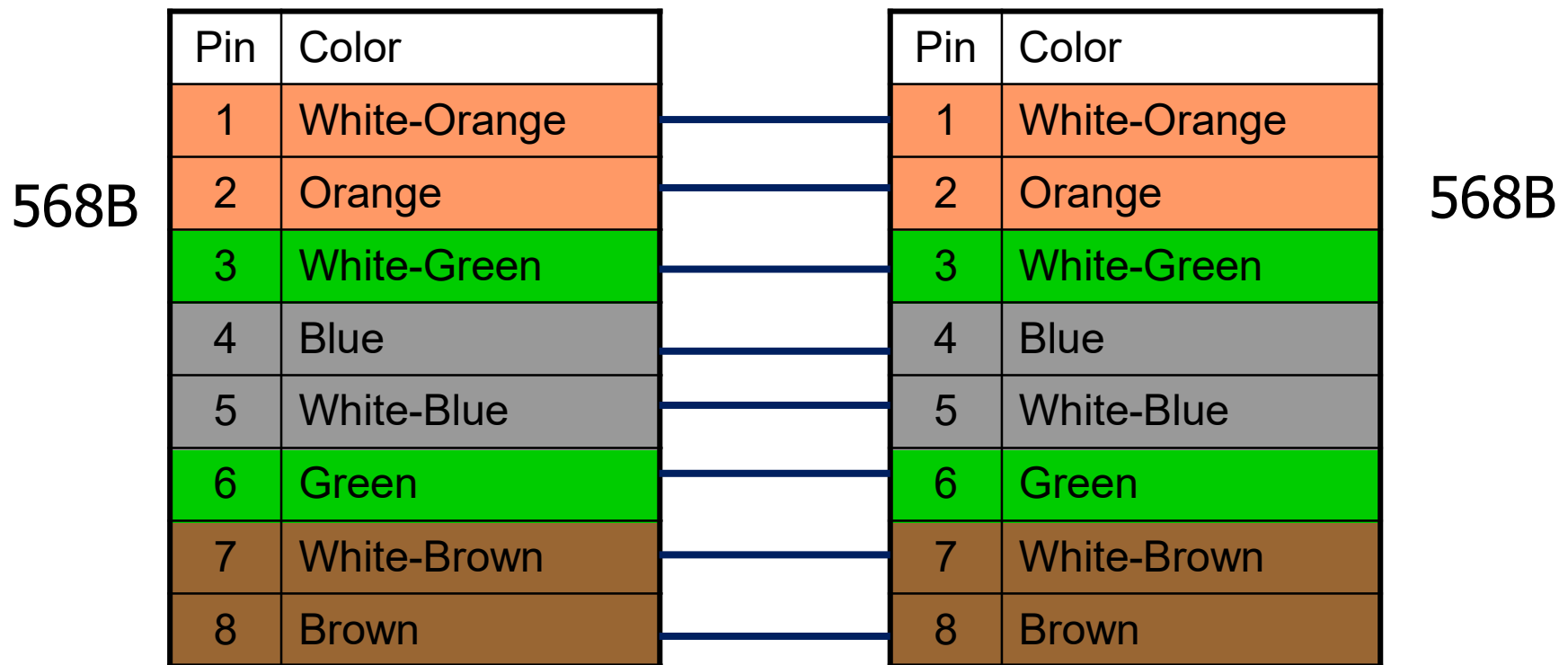
Wiring standard: TIA/EIA-568A



Pin	Color
1	White-Green
2	Green
3	White-Orange
4	Blue
5	White-Blue
6	Orange
7	White-Brown
8	Brown

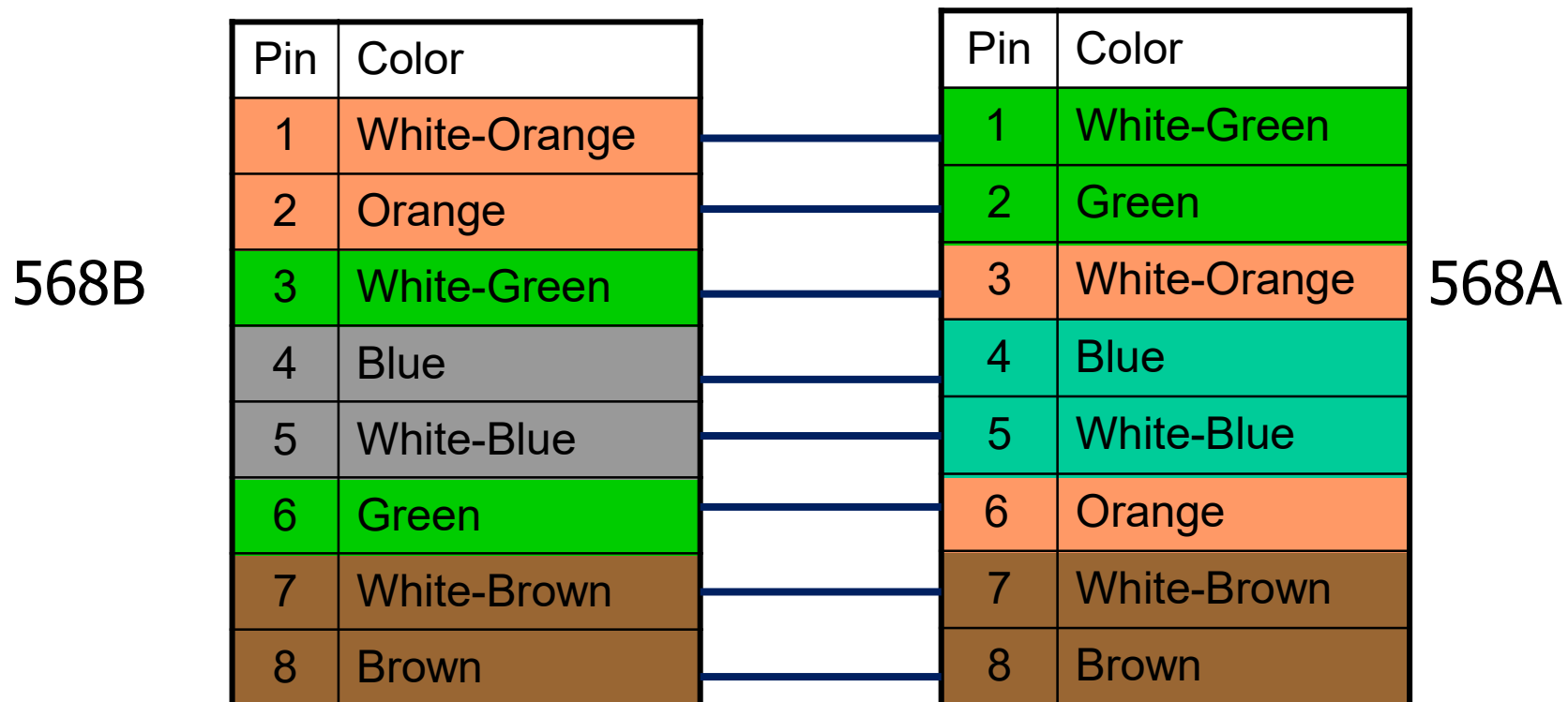
Straight-Through Cables: 直连线

- RJ45 jacks at both ends of a patch cable are terminated with same standard
- Used to connect such a device(PCs or routers) to another different device (hubs or switches)



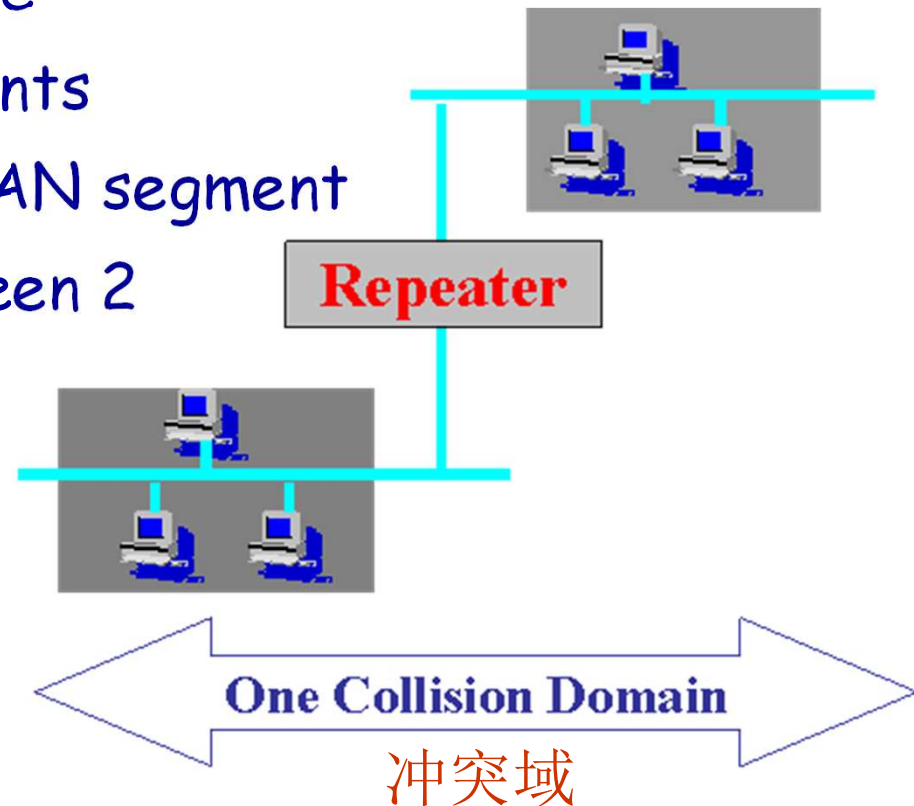
Crossover Cables: 交叉线

- RJ45 jacks at two ends of a patch cable are terminated with different standards
- Pin 1 connected to pin 3, pin 2 connected to pin 6.
- Used to connect **similar** devices: switch to switch, switch to hub, hub to hub, router to router, PC to PC.



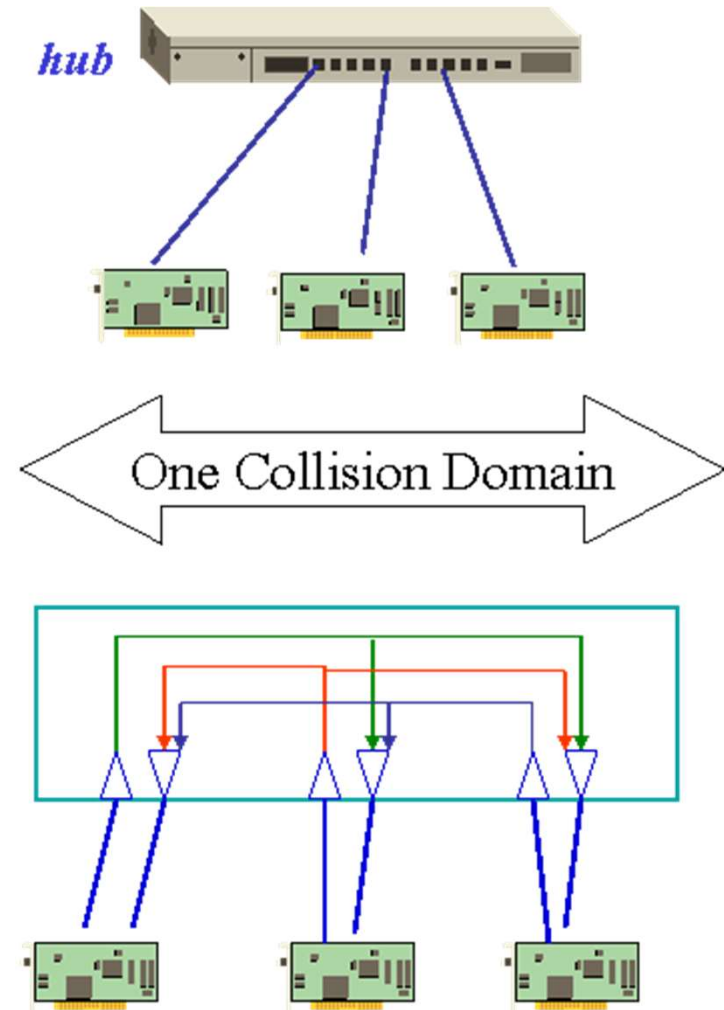
Enlarging Ethernet: Repeaters(中继器)

- **Physics layer** device
- Amplifying and shaping signals, decreasing signal distortion and attenuation, enable signal to propagate longer distance
- Extension of LAN segments
- Up to 4 Repeaters in a LAN segment
- Maximum distance between 2 transceivers: 2.5 km
- Old device, out of date

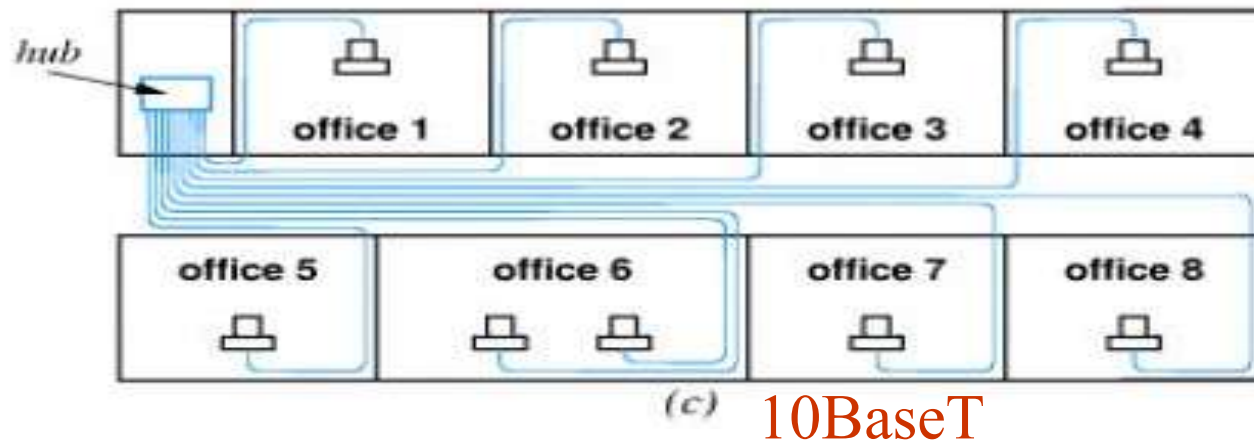
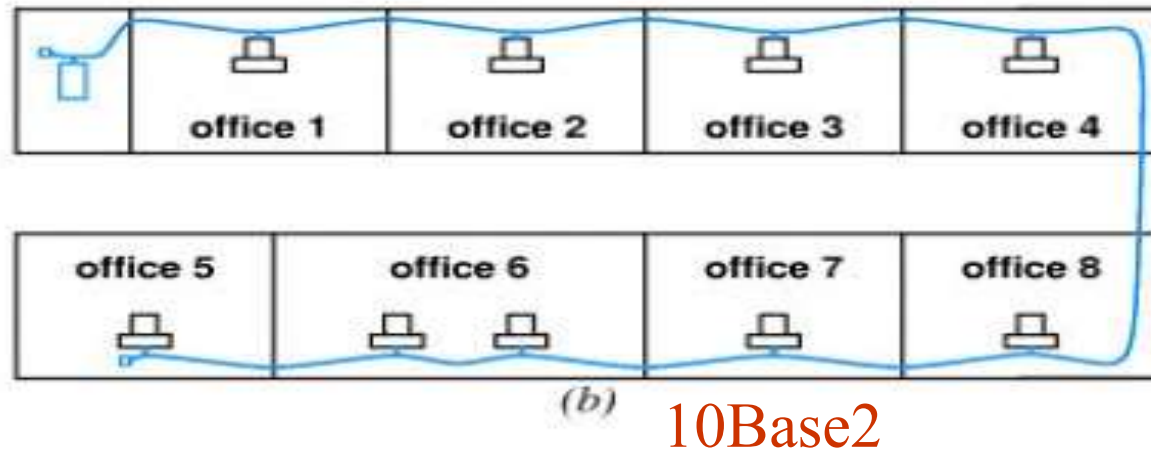
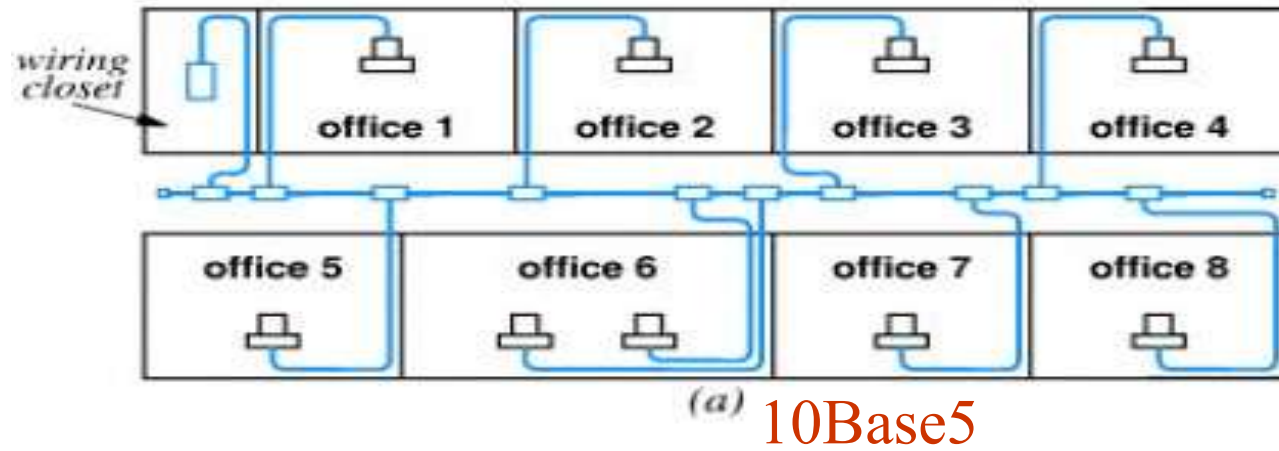


Hubs (集线器)

- A multi-port repeater
- Providing a **star** connection physically, implements a **bus** topology logically
- Devised to overcome problems of Ethernet tap connections, provides more reliable connections
- Supporting 10Mbps or 100Mbps
- Completely replaces repeater
- Configuration limitation
 - ◆ up to 3 hubs in a segment

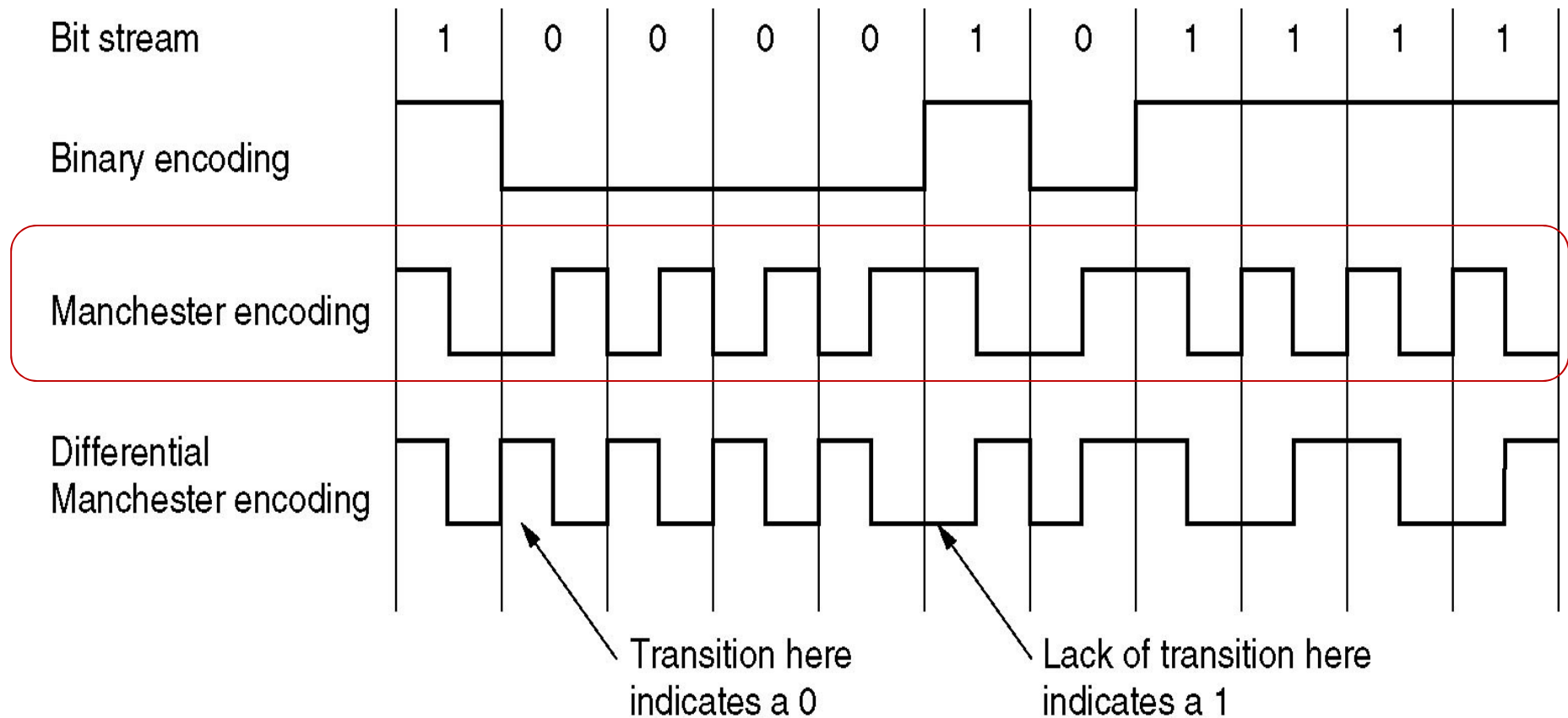


Cabling of 10Mbps Ethernet



Manchester Encoding

- Transition in the middle of each bit

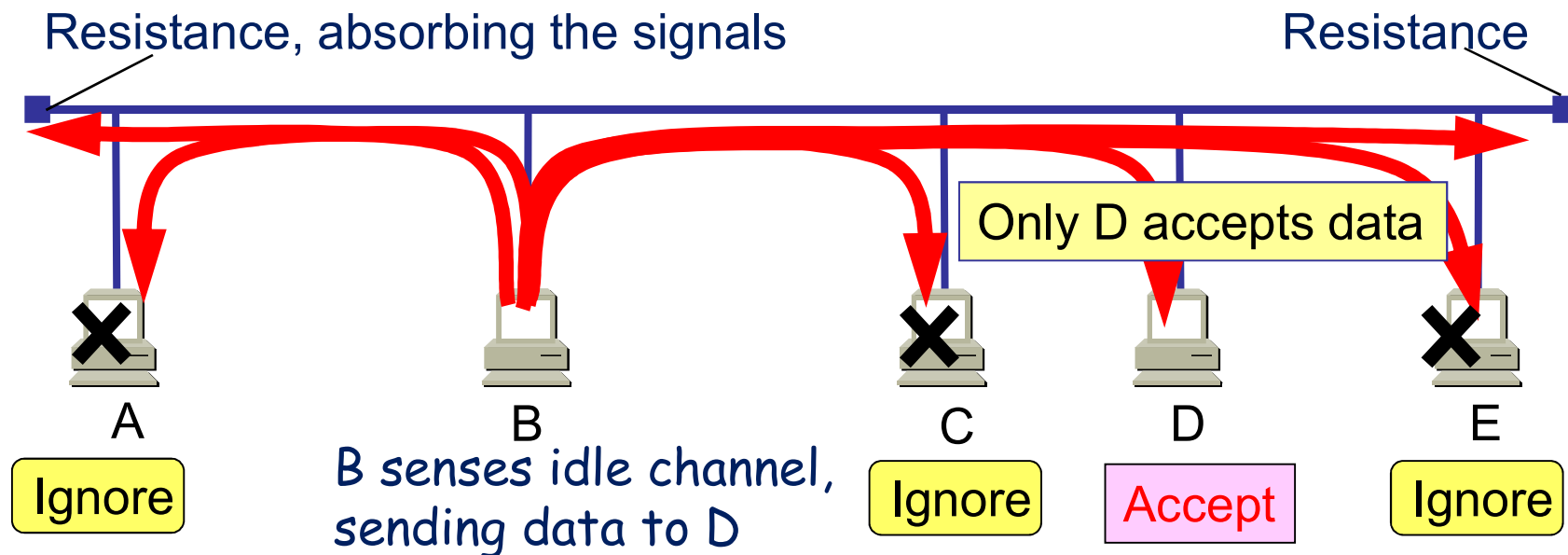


Ethernet

- Classic Ethernet: Physical Layer
- Classic Ethernet: MAC Sublayer Protocol
- Ethernet Performance
- Switched Ethernet
- Fast Ethernet
- Gigabit Ethernet

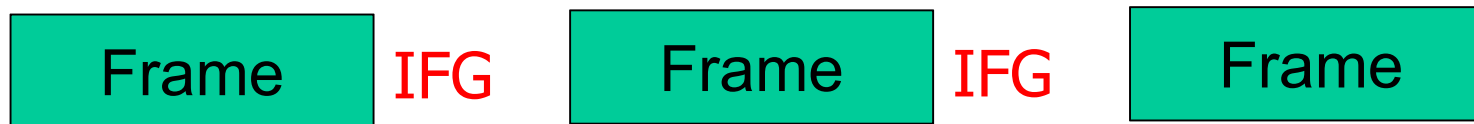
Operations of Ethernet / IEEE 802.3

- Broadcasting link: every station can receive data sent by any other station
 - ◆ Addressing is necessary to identify destination
 - ◆ Only one station copies frame, others ignore the frame



Inter Frame Gap (帧间间隔)

- Also called Interpacket Gap
- 9.6 μ s for 10M bps Ethernet
 - ◆ More precisely, 96 bit time
 - ◆ Used for a station to change from transmit mode to receive mode (*half-duplex*)
 - ◆ In practice, IFG may be much shorter
 - Eg. 4.7 μ s



Operation steps of Ethernet / IEEE 802.3:

1-persistent CSMA/CD (1-坚持的CSMA/CD)

(1)

- Step 1. Sensing the channel first, *"Listen before talking"*. 监听信道是否空闲
- Step 2. If channel is quiet for IFG period, then transmit. *"Talk if quiet"*. 如果空闲时间超过IFG, 则发送
- Step 3. If channel is busy, monitor the channel until it is quiet for IFG period before transmitting. *"Wait for quiet before talking."* 如果信道忙, 持续监听
- Step 4. Monitor the channel continuously during transmission to detect collisions. *"Listen while talking"*. 边发送边检测冲突

Operation steps of Ethernet / IEEE 802.3:

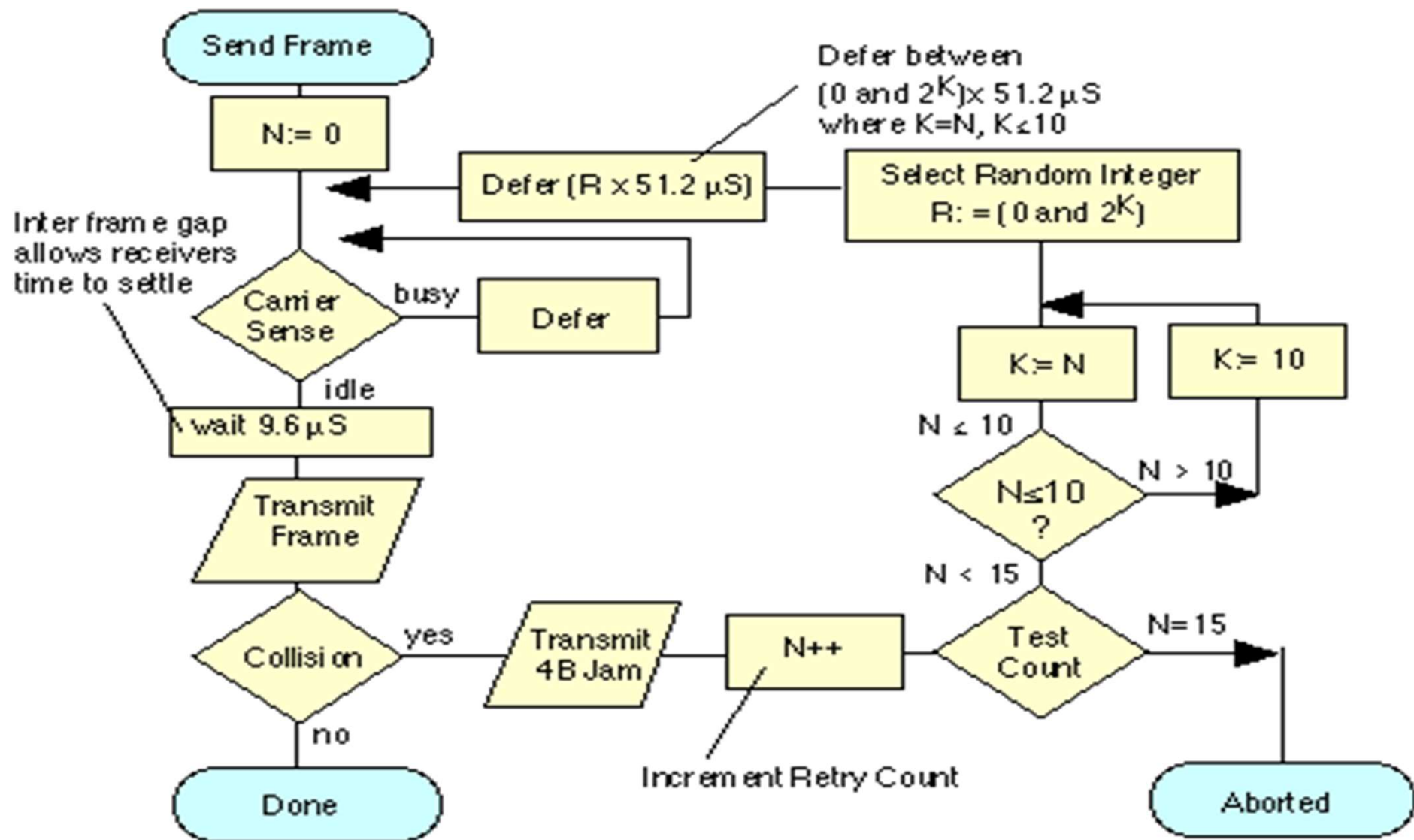
1-persistent CSMA/CD (2)

- Step 5. If collision occurs, the first node recognizes it will send a **jam** signal to ensure that all other stations detect the collision. Transmitters should stop immediately, receivers should reject data, and others wishing to transmit should recalculate their backoff period. *"A buzzer sounds off indicating we have more than one talker at a time."*

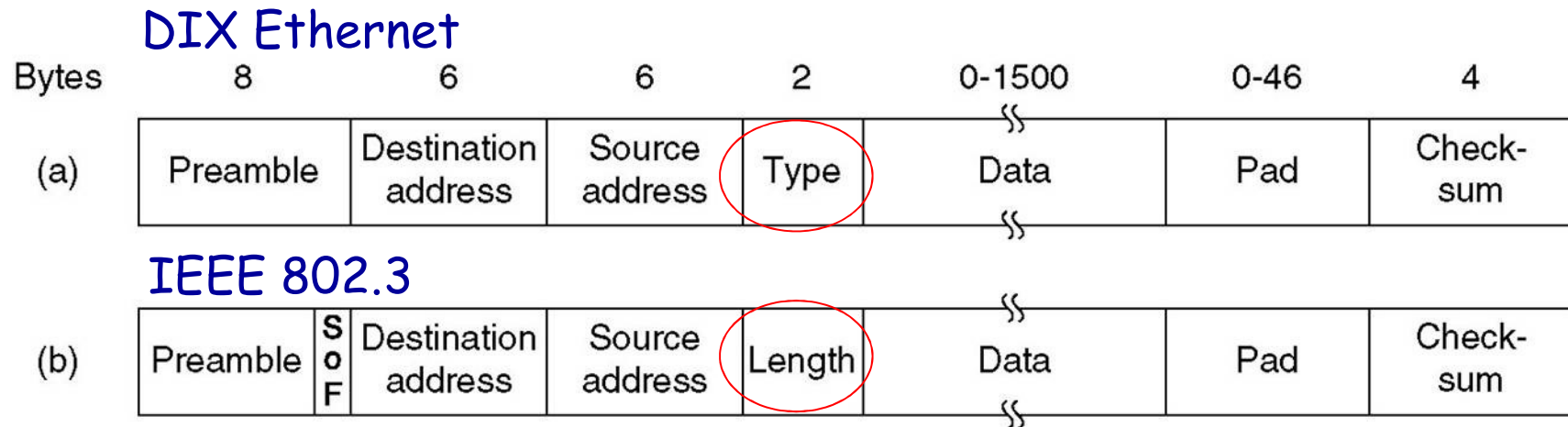
如果发现冲突，发送jam信号

- Step 6. All nodes wish to transmit must now wait a random period called "**backoff**(退避)" and attempt again. After 16 collisions, drop the frame to be transmitted.
收到jam信号，所有站点退避一段时间后重新监听信道

Flow chart of Ethernet / IEEE 802.3



Frame Structure

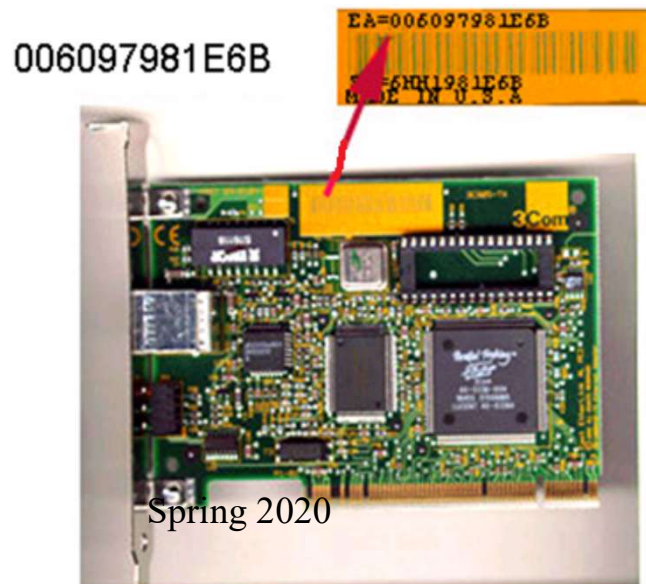


- Preamble(前导码): 8bytes, 101010...101011, used for clock synchronization
- Address: MAC address, 48 bits
- Type: indicates the higher layer protocol, e.g., IP, Novell IPX,... (0x0806:ARP, 0x0800:IP, 0x8137:IPX)
- Length: number of bytes of Data+Pad
- Data: maximum 1500 bytes, minimum 46 bytes (with Pad)
- Checksum: CRC, checked at receiver, if error is detected, the frame is simply dropped

MAC Address (Physical Address)

- 6 bytes, denoted in hex number
 - ◆ eg. 00-60-97-98-1E-6B
- First 24-bits identify the vendor (OUI: Organizationally Unique Identifier), the second 24-bits is a serial number assigned by the vendor
 - ◆ eg. 00-60-97 indentifies 3COM CORPORATION

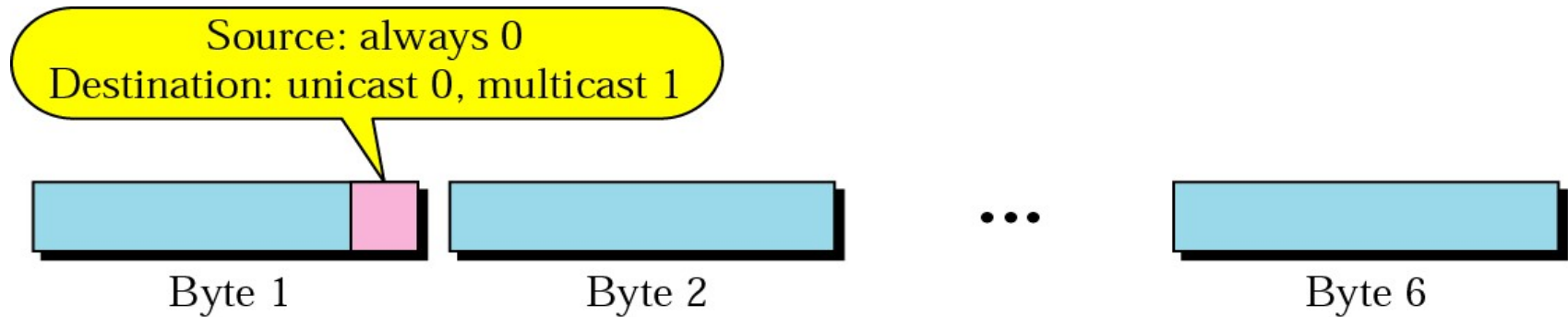
<http://standards.ieee.org/develop/regauth/oui/public.html/>



Computer Networks, MAC Sublayer

MAC Address (Physical Address)

- Multicast address vs. Unicast address

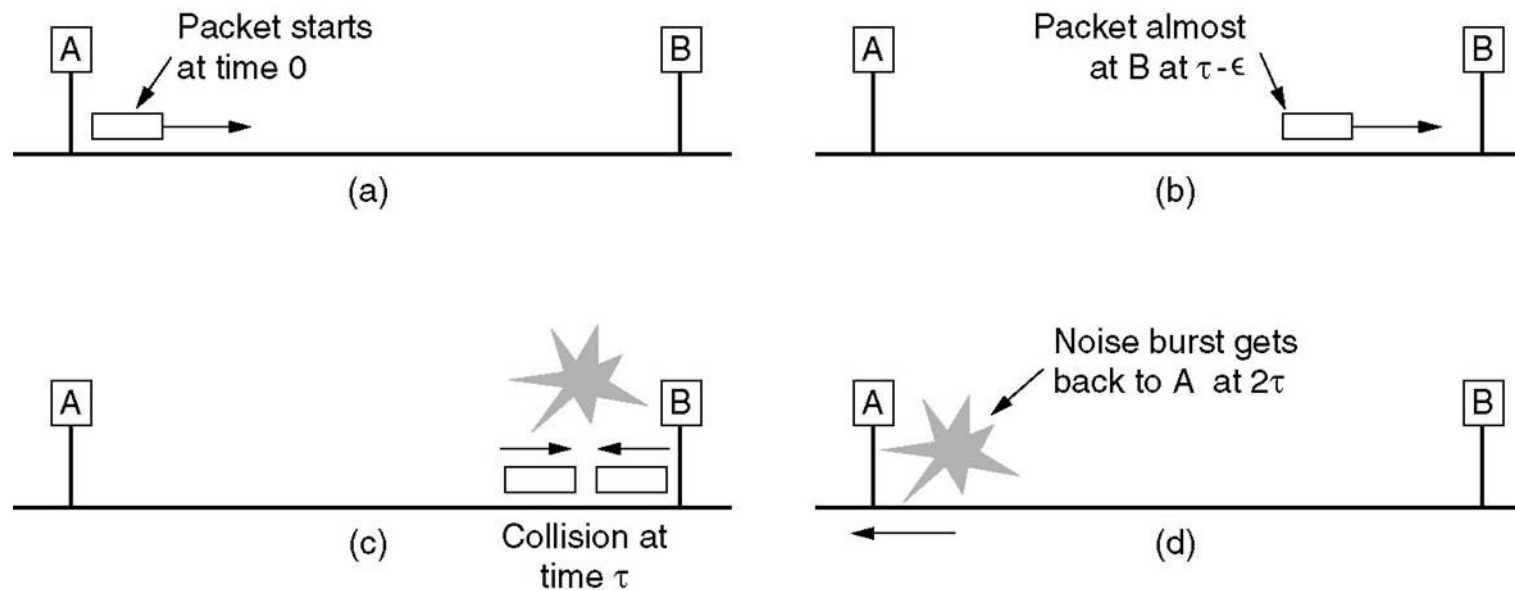


- eg. 09-00-AA-BB-CC-DD is a multicast address
08-00-11-22-33-44 is a unicast address
- Broadcast Address: FF-FF-FF-FF-FF-FF

Minimum Frame Size: 64 Bytes

- Reason: to give a host enough time to detect collisions

- ◆ If host has less than 46 bytes to send, the adaptor pads (adds) bytes to make it 46 bytes



- The transmission time shall be longer than 2τ .

- ◆ For a 10Mbps LAN with max length of 2,500 meters and 4 repeaters, the round-trip time(2τ) is about 51.2 μ s, i.e. 512bit, 64Byte.

Ethernet/IEEE 802.3 Parameters

Slot time(2τ)	512 bit times (51.2 μ s)
Inter Frame Gap	96 bit times (9.6 μ s)
Attempt Limit	16 (tries)
Backoff Limit	10 (exponent)
Jam Size	32 bits
Max Frame Size	1518 bytes
Min Frame Size	64 bytes
Address Size	48 bits (6 bytes)

Ethernet

- Classic Ethernet: Physical Layer
- Classic Ethernet: MAC Sublayer Protocol
- Ethernet Performance
- Switched Ethernet
- Fast Ethernet
- Gigabit Ethernet

Performance of IEEE 802.3

- Premise: heavy and constant load, i.e. k stations always ready to transmit, and retransmission probability is also constant
- Assumed each station emits during a contention slot with probability p

$A = P[\text{Some station emits successfully}]$ and
mean number of slots per contention = $\frac{1}{A}$

- If mean frame time is P sec, Channel efficiency = $P / (P + 2\tau / A)$
- Assumed frame length F , network bandwidth B , cable length L , speed of signal propagation c , for optimal case $A = 1/e$

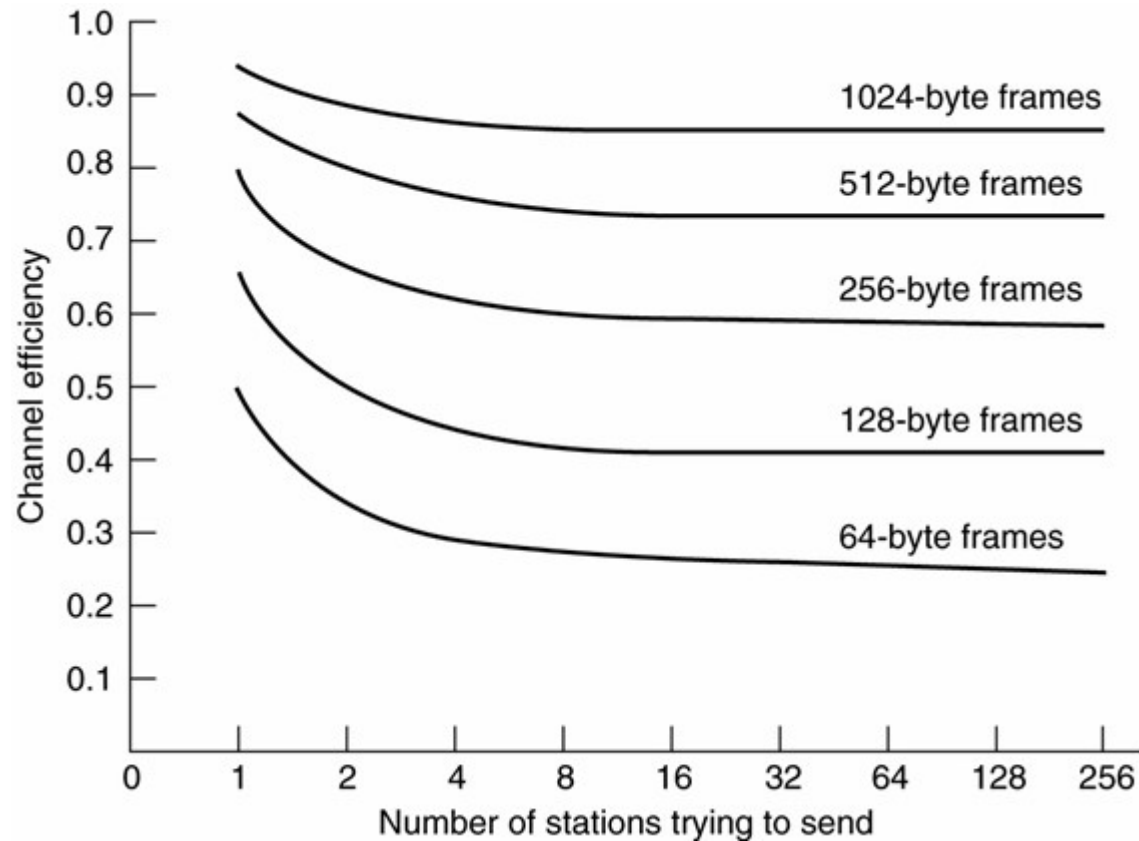
$$\text{Channel efficiency} = \frac{1}{1 + 2BLE/cF}$$

Increasing network bandwidth or distance reduces efficiency for a given frame size

Performance of IEEE802.3(cont.)

Channel efficiency =

$$\frac{1}{1+2BLe/cF}$$



Efficiency of Ethernet at 10 Mbps with 512-bit slot times

How to increase speed of data transmission in Ethernet?

■ Decreasing collision

- ◆ Using switched Ethernet (交换式) instead of shared Ethernet (共享式, classic Ethernet)
- ◆ Decreasing number of computers in a LAN
 - Using bridges(网桥) to connect LANs

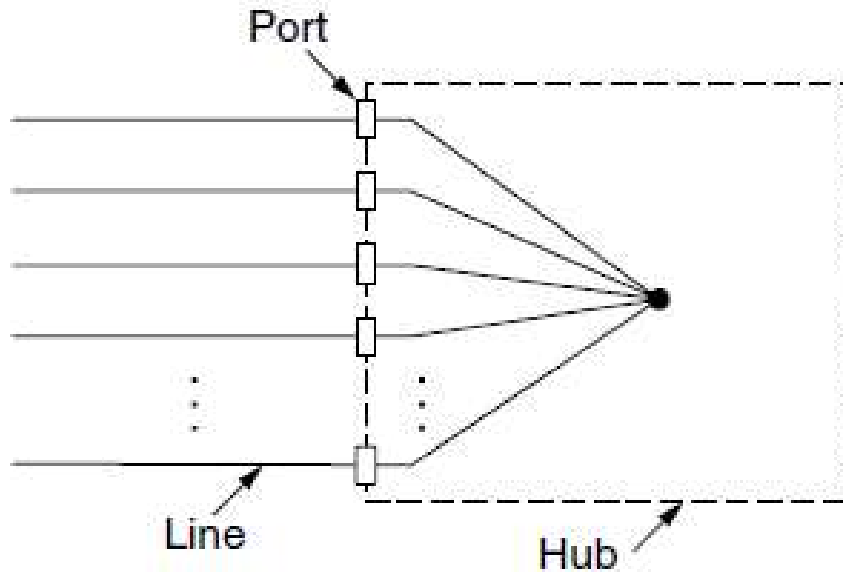
■ Increasing the data rate

- ◆ Fast Ethernet and Gigabit Ethernet

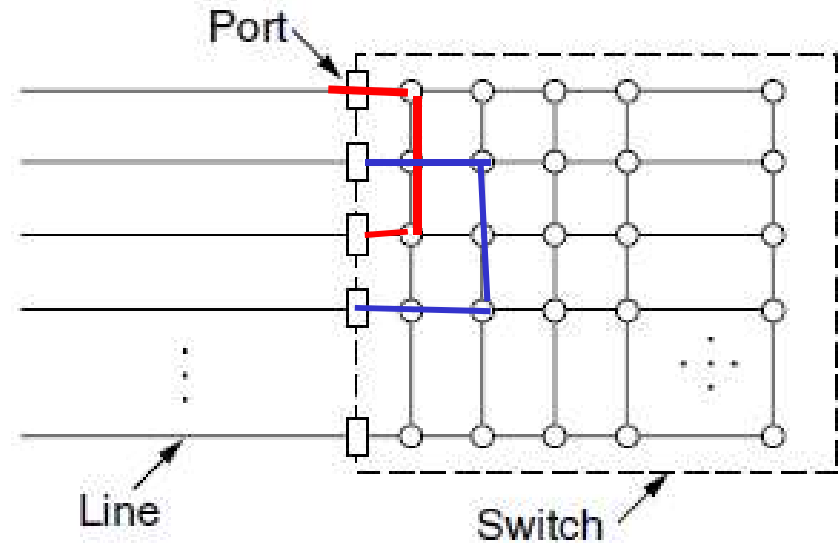
Ethernet

- Classic Ethernet: Physical Layer
- Classic Ethernet: MAC Sublayer Protocol
- Ethernet Performance
- Switched Ethernet
- Fast Ethernet
- Gigabit Ethernet

HUB(集线器) vs. Switch(交换机)



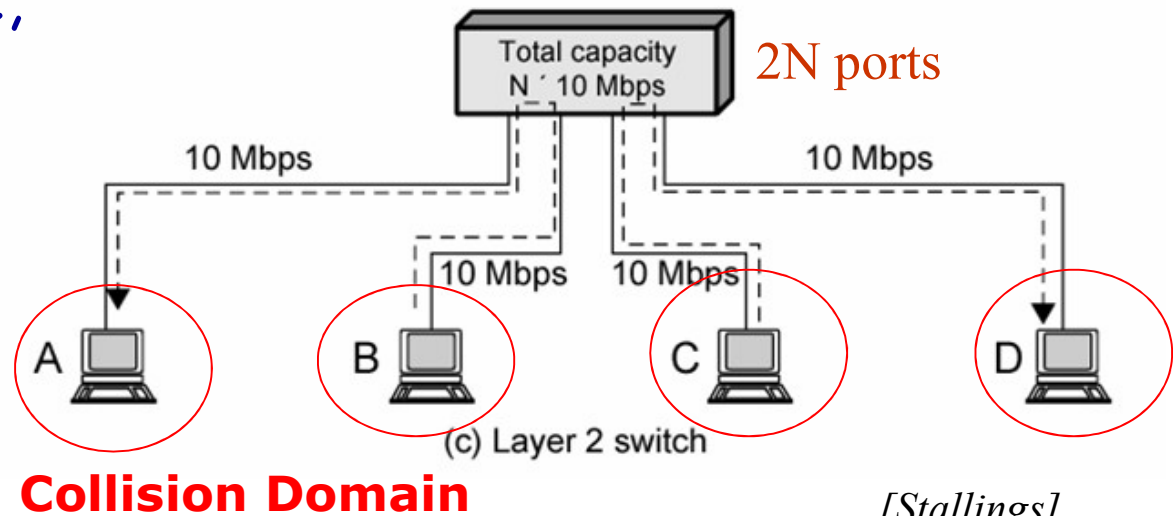
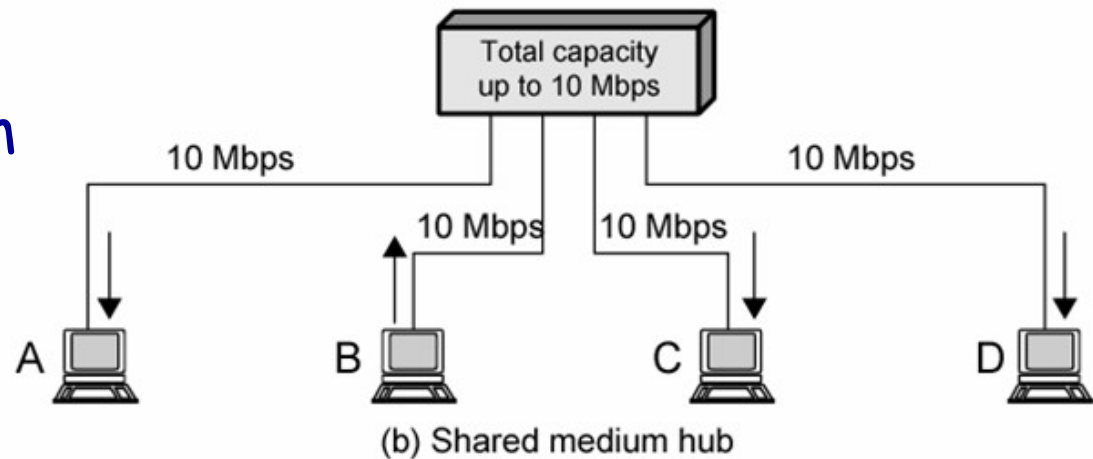
Hubs are logically equivalent
to the single long cable
(逻辑上是总线拓扑)



A switch contains a high-speed backplane that connects all of the ports

Switched Ethernet (交换式以太网)

- Central **switch** instead of hub
- Incoming frame from particular station switched to appropriate output line
- Unused lines can switch other traffic, more than one station transmitting at a time
- Multiplying capacity of LAN
- More secure than classic Ethernet



[Stallings]

Switches(交换机)

How to work out? → See in
Data Link Layer Switching

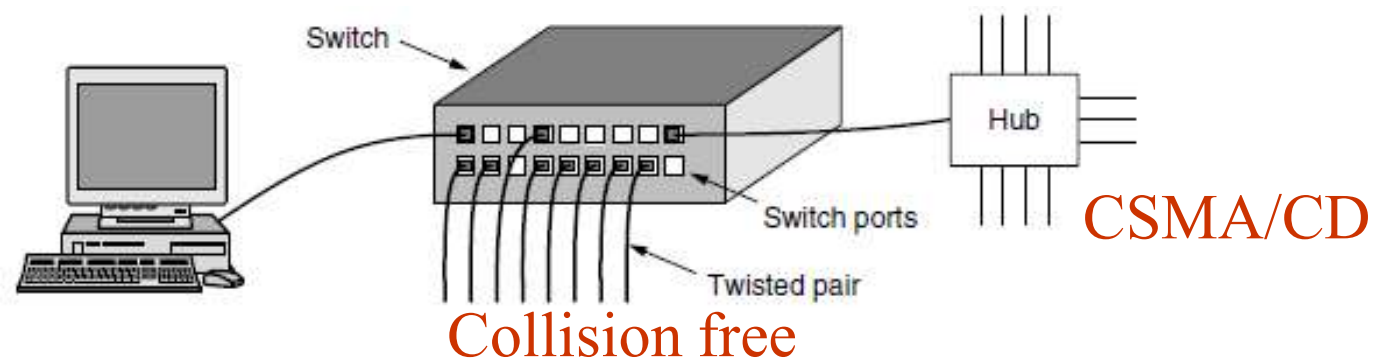
■ Data Link Layer device

- ◆ Switching frames on **destination MAC address**
- ◆ Providing dedicated channels between each pair of users

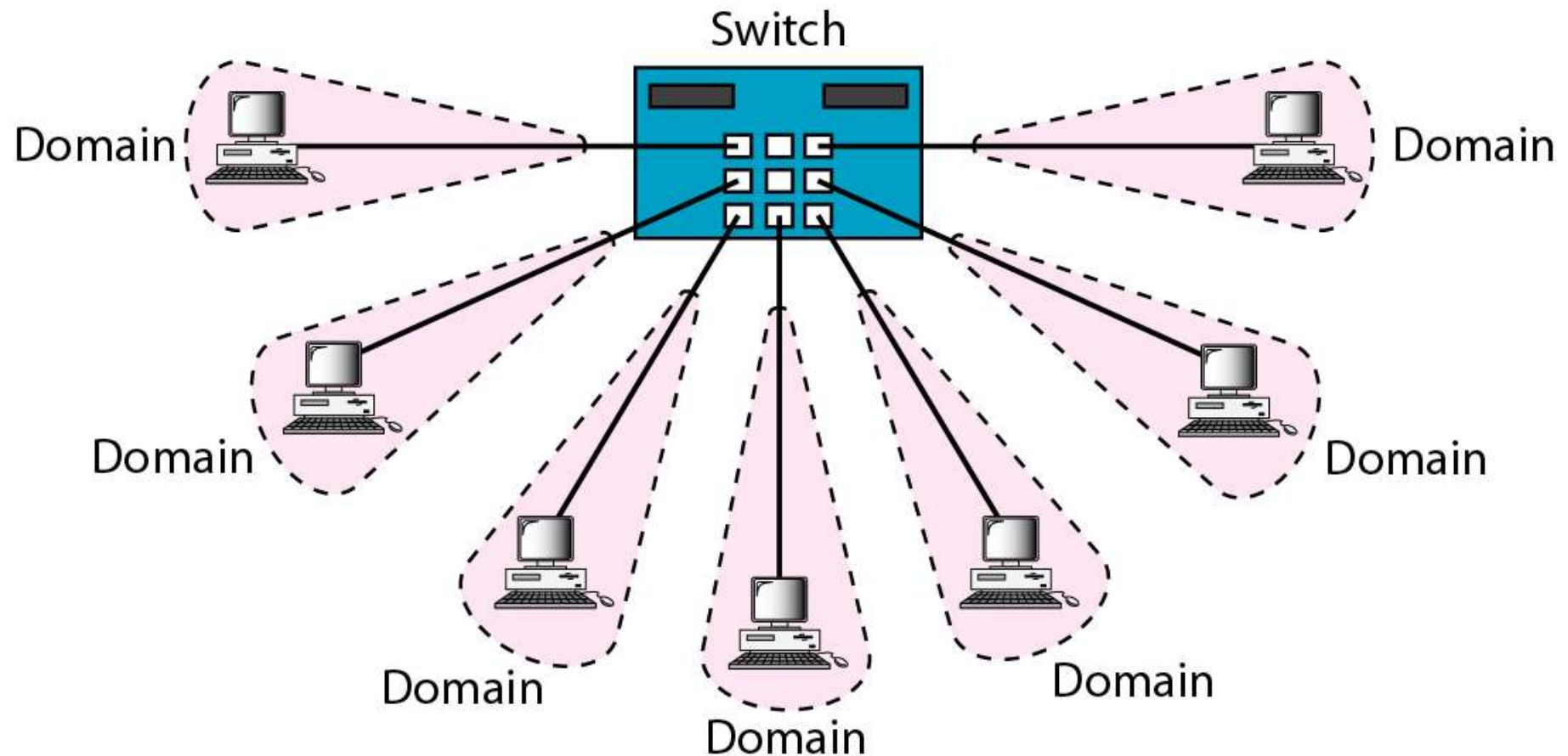
■ Having a high-speed backplane

■ Collision domain

- ◆ With only **one port** per collision domain, collisions are impossible and performance is improved



Collision Domain of Switched Ethernet



Ethernet

- Classic Ethernet: Physical Layer
- Classic Ethernet: MAC Sublayer Protocol
- Ethernet Performance
- Switched Ethernet
- Fast Ethernet Increasing the data rate
- Gigabit Ethernet

Fast Ethernet(快速以太网): IEEE 802.3u

- Keep all the old frame formats, interfaces, and procedural rules of IEEE802.3
- Reduce the **bit time** from 100 ns to **10 ns**
- Minimum frame length remains **64B**
- All fast Ethernet systems use hubs and switches; cables with vampire taps or BNC connectors are not permitted

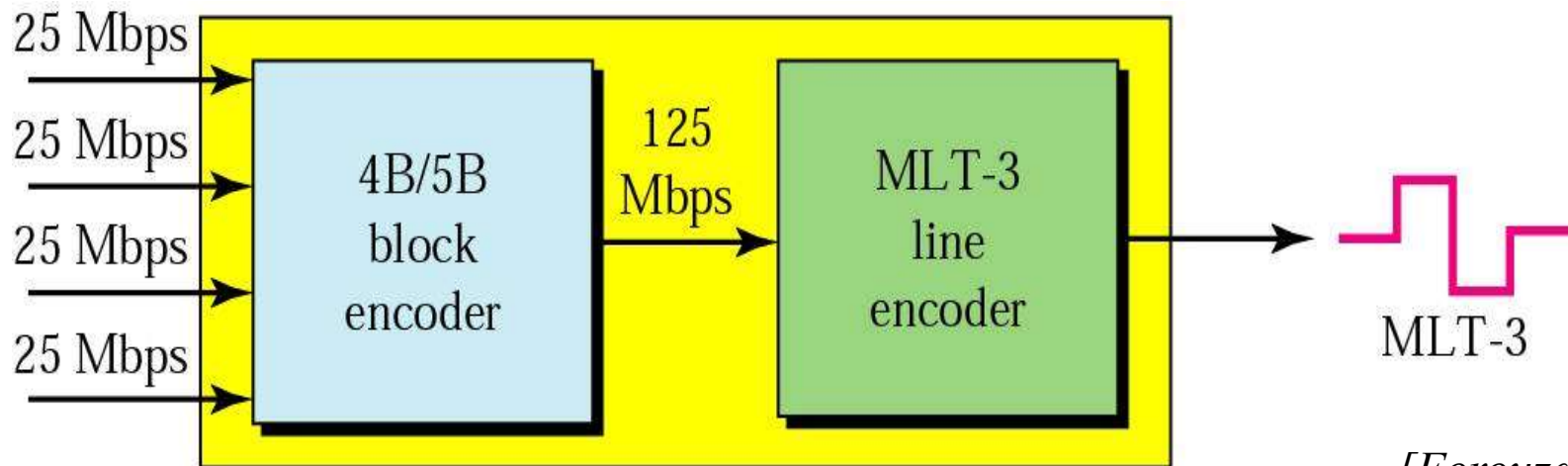
Name	Cable	Max. segment	Advantages
100Base-T4	Twisted pair	100 m	Uses category 3 UTP
100Base-TX	Twisted pair	100 m	Full duplex at 100 Mbps
100Base-FX	Fiber optics	2000 m	Full duplex at 100 Mbps; long runs

100Base-T4

- Uses a signaling speed of 25 MHz
- Requires 4 twisted pairs (UTP3)
 - ◆ One is always to the hub
 - ◆ One is always from the hub
 - ◆ The other two are switchable to the current transmission direction-- Half duplex
- Coding
 - ◆ Manchester encoding is not used
 - ◆ Ternary signals(三元信号) are sent during a clock period
 - ◆ With 25MHz signaling speed, at least 33.3-Mbps per channel(8B/6T)

100Base-TX

- Uses a signaling speed of 125 MHz
- Requires 2 twisted pairs (UTP5)
 - ◆ One is to the hub
 - ◆ One is from the hub (**Full duplex**)
- Coding
 - ◆ 4B/5B: $125 * 4/5 = 100\text{Mbps}$



[Forouzan]

MLT: Multi Level Transmit

Fast Ethernet: Summary

Specification		Frequency (MHz)	Encoding	number of wires	cable for sending	Full duplex	Max seg. length
10BASE-T	UTP3 UTP5	20	Manchester	2 pairs	1 pair	Yes	100m
100BASE-T4	UTP3	25	8B6T-NRZ	4 pairs	3 pairs	No	100m
100BASE-TX	UTP5	125	4B5B-MLT3	2 pairs	1 pair	Yes	100m
100BASE-F	Fiber	125	4B5B-NRZ	2 pairs	1 pair	Yes	2000m

Interconnection Devices for Fast Ethernet

■ Hub (Half-duplex communication)

- ◆ Incoming lines are logically connected, forming a single collision domain
- ◆ The system works just like **classic Ethernet**

■ Switch

- ◆ Incoming frame is buffered and passed to the destination
- ◆ 100Base-FX cables are too long for collision algorithm, must be connected to switches

10-Mbps & 100-Mbps Autoconfiguration

- 803.3u provides a way for two stations to negotiate the speed (10 or 100 Mbps) and duplexity (half or full)
- Most fast Ethernet products use this feature to autoconfigure themselves

Ethernet

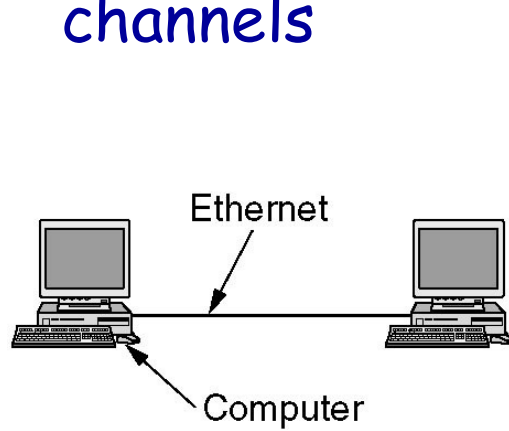
- Classic Ethernet: Physical Layer
- Classic Ethernet: MAC Sublayer Protocol
- Ethernet Performance
- Switched Ethernet
- Fast Ethernet
- Gigabit Ethernet

IEEE 802.3z Goals

- Make Ethernet go 10 times faster
- Remain **backward compatible** with all existing Ethernet standards
 - ◆ **Unacknowledged datagram service** with both unicast and multicast
 - ◆ Use 48-bit addressing scheme already in use
 - ◆ Maintain the same frame format, including the minimum and maximum frame sizes

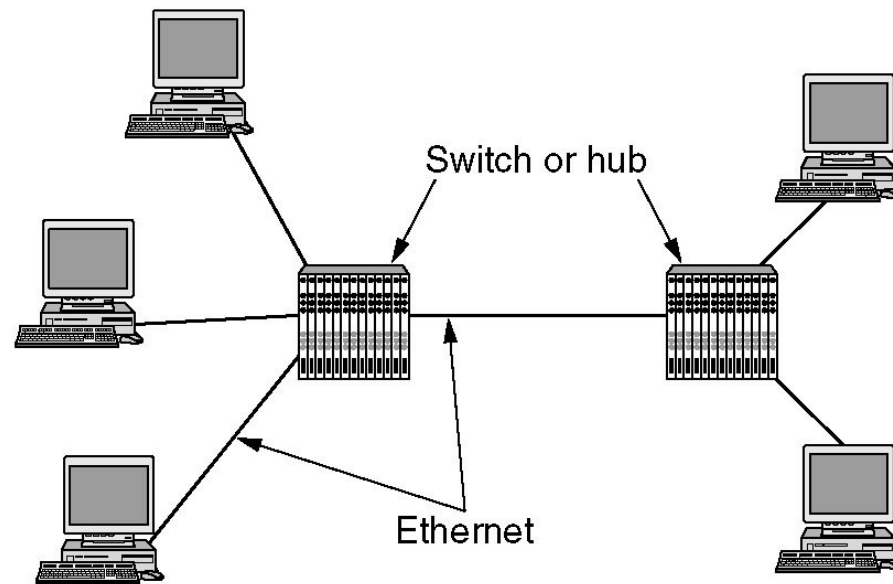
Gigabit Ethernet(千兆以太网)

- Allows for point-to-point links and shared broadcast channels



(a)

A two-station Ethernet



(b)

A multistation Ethernet

Full-duplex: computer connecting to switch
Half-duplex: computer connecting to hub

Switch : Full-duplex Mode

- Switch (Full-duplex Mode)
 - ◆ Since no contention is possible, the **CSMA/CD protocol is not used**
 - ◆ Maximum length of the cable is determined by signal strength issues
 - ◆ Switches are free to mix and match speeds.
 - ◆ Autoconfiguration is supported

Hub: Half-duplex Mode

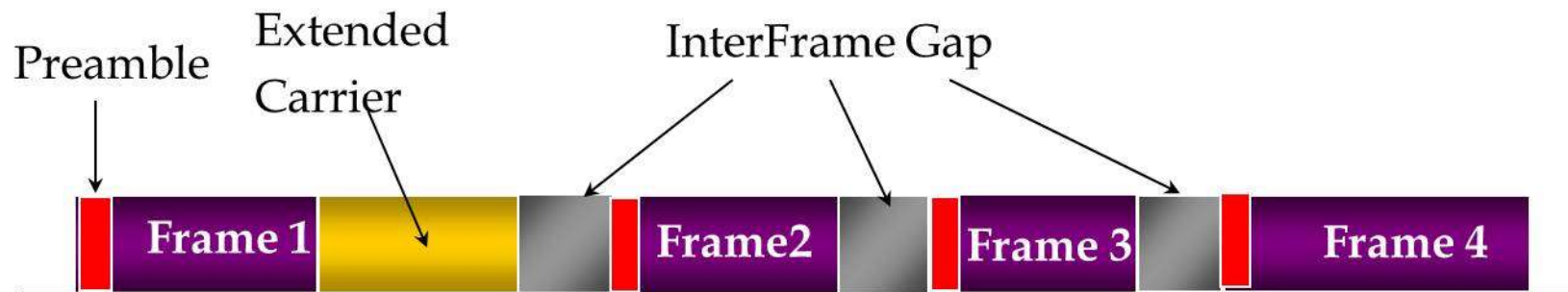
- Simulating classic Ethernet
 - ◆ Collisions are possible, so **CSMA/CD** protocol is required
- Problem: A radius of 25 meters
- Solutions
 - ◆ **Carrier extension**(载波扩展)
 - Add padding **after** the normal frame to extend the frame to **512 bytes** (Padding is added by the sending hardware and removed by the receiving hardware)
 - ◆ **Frame bursting**(帧突发)
 - Allows a sender to transmit a concatenated sequence of multiple frames in a single transmission

Remain the radius of the network to 200 meters

Carrier Extension vs. Frame Bursting



Carrier Extension 载波扩展



Frame Bursting 帧突发

Gigabit Ethernet cabling

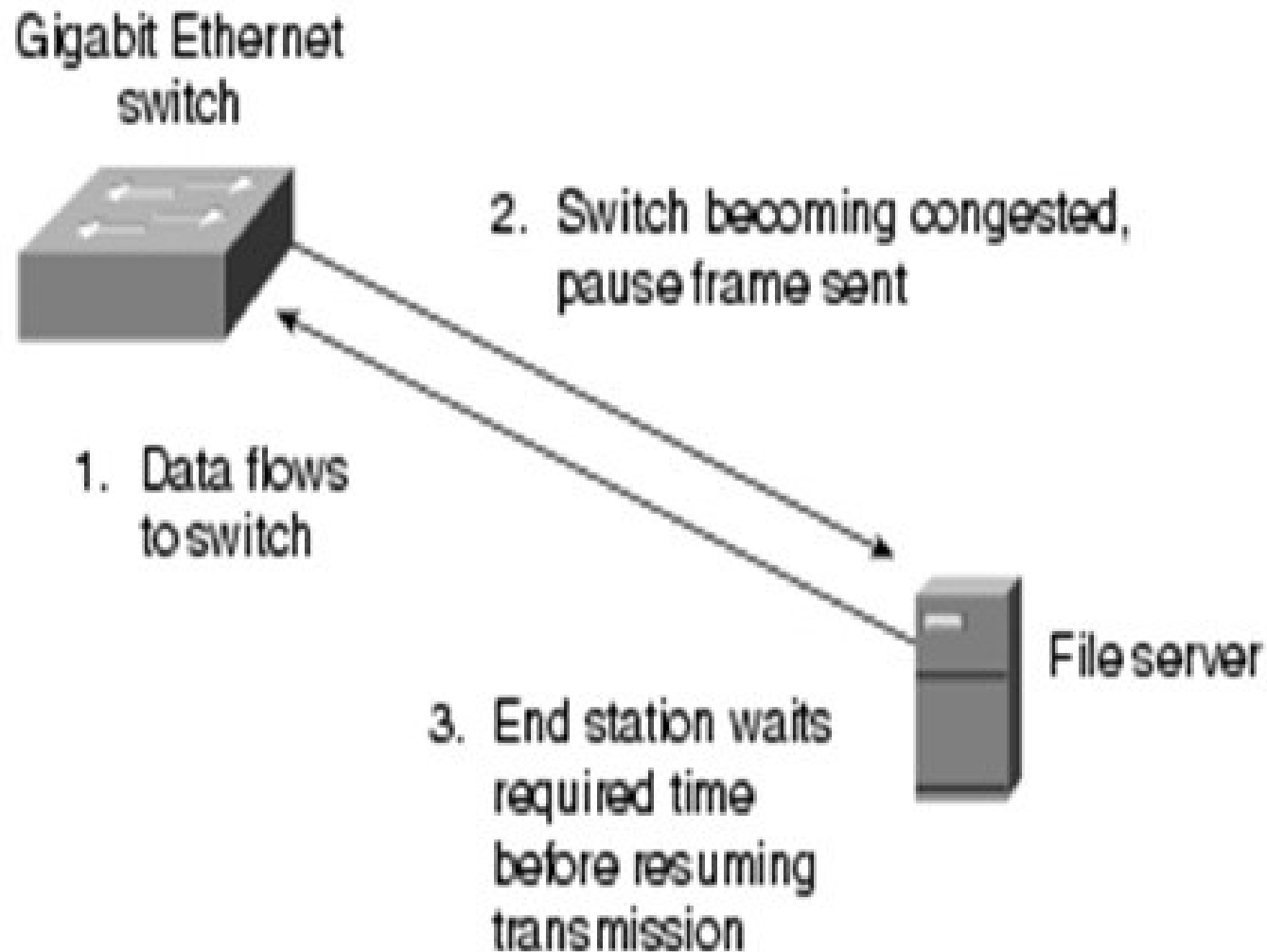
Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

- SX: Short Wavelength
- LX: Long Wavelength
- CX: Copper Cable

Flow Control of IEEE802.3x (p314)

- Introduced for Fast and Gigabit Ethernet
- Problem: 1 Gbps is quite fast
 - ◆ If a receiver is busy with some other task for 1 ms, 1953 frames may have accumulated, and buffer overruns are very likely
- Control frames
 - ◆ One end sending a special **control frame** to the other end telling it to **pause** for some period of time
 - ◆ Normal Ethernet frames, containing a **type of 0x8808**
- Flow control
 - ◆ **PAUSE frames** are used to enable or disable frame transmission, 512ns~33.6ms

Example of Flow Control in Gigabit



Courtesy of Cisco Systems Documentation online

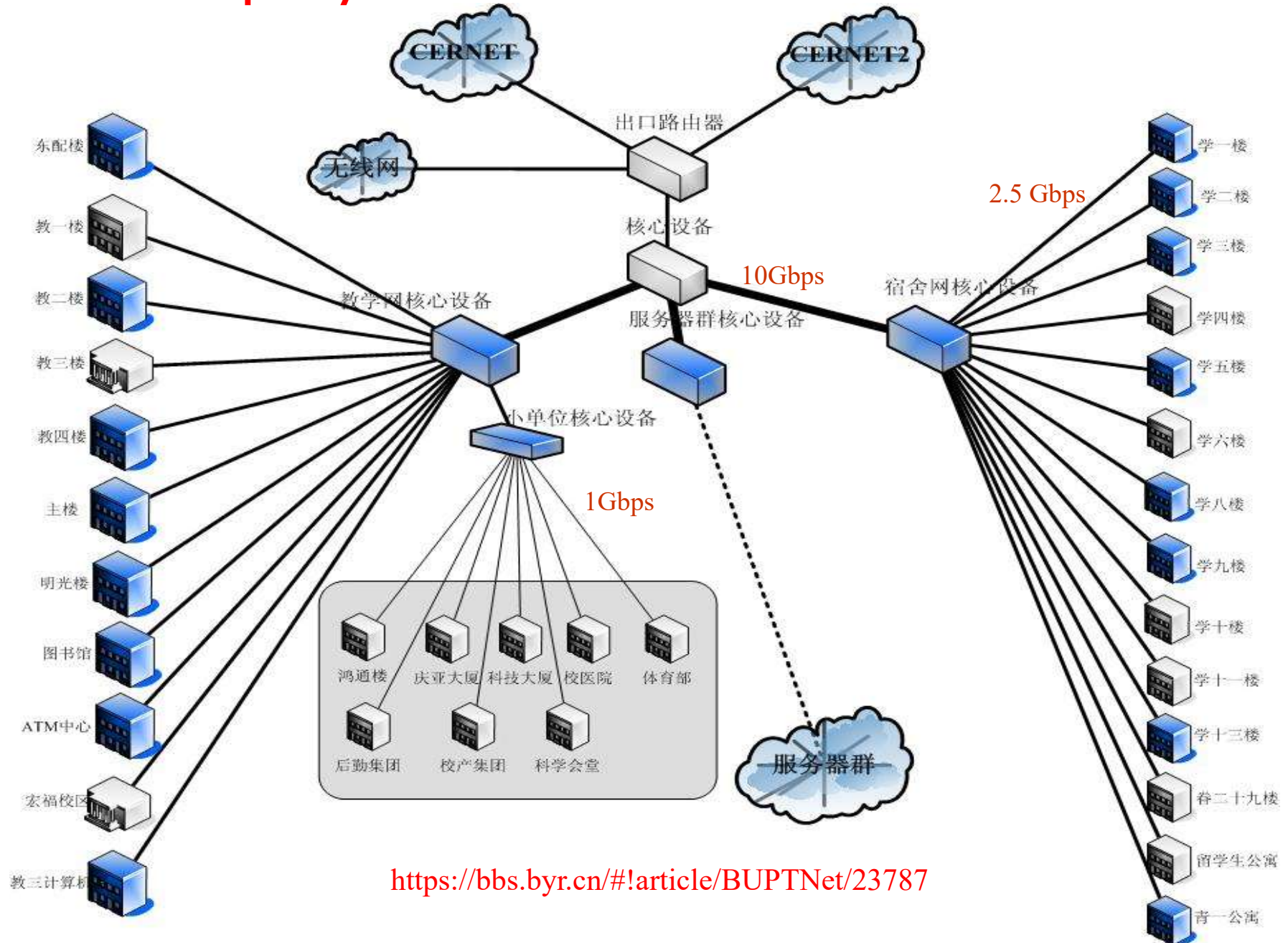
10Gbps Ethernet: IEEE 802.3a

- High-speed, local backbone interconnection between large-capacity switches
- Server farm
- Campus wide connectivity
- Enables Internet service providers (ISPs) and network service providers (NSPs) to create very high-speed links at very low cost
- Allows construction of MANs and WANs
 - ◆ Connect geographically dispersed LANs between campuses or points of presence (PoPs)
- 10-Gbps Ethernet competes with ATM and other WAN technologies

10G Ethernet: Design Objectives

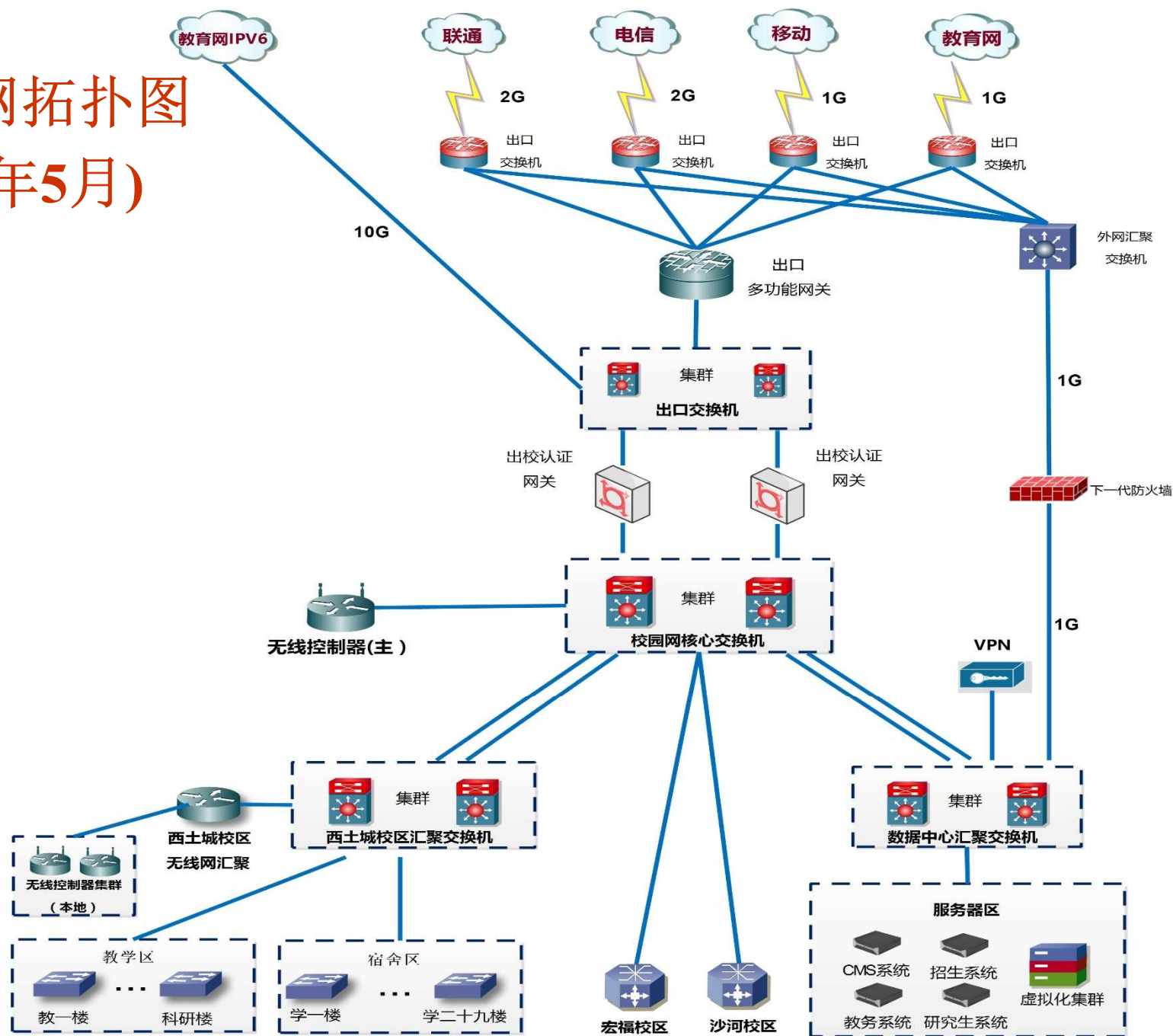
- Preserve the 802.3 Ethernet frame format at MAC interface
- Preserve the minimum and maximum frame size of the current 802.3 standard
- Support full duplex only(Not using CDMA/CD)
- Support star-wired LANs using point-to-point links and structured cabling topologies
- Support a MAC data rate of 10 Gbps
- Define two familiar physical interfaces: LAN & WAN
- ...

Deployment of 10G Ethernet



<https://bbs.byrcn/#!/article/BUPTNet/23787>

校园网拓扑图 (2018年5月)



Summary of Ethernet

- 1-persistent CSMA/CD
- Classic Ethernet: 10Base5, 10Base 2 and 10BaseT
- Operation steps of Ethernet
- MAC address: 48bits
- Frame size: 64-1518B
- Repeater, Hub and Switch: layers, collision domain, forwarding table of switch
- Fast Ethernet and Gigabit Ethernet: similarity and difference with 10Mbps Ethernet
- 10G Ethernet: usage, objectives

	经典以太网	快速以太网	千兆以太网	万兆以太网	交换式以太网
传输速率(Mbps)	10	100	1,000	10,000	
相关标准	IEEE802.3	IEEE802.3u	IEEE802.3z	IEEE802.3a	
典型网络种类、介质及拓扑结构	总线:10base5, 10base2; 星型:10baseT	星型,点到点	星型,点到点	星型, 点到点, 光纤	星型
介质访问控制技术	CSMA/CD	CSMA/CD	CSMA/CD	不使用	无冲突
工作模式	半双工: 中继器、集线器	半双工:集线器 全双工:交换机	半双工:集线器 全双工:交换机	全双工:交换机	全双工:交换机
编码方案	曼彻斯特码	MLT-3, 4B/5B,8B/6T	8B/10B等		
典型帧长度	最小64B, 最大1518B	最小64B, 最大1518B	帧突发最小64B, 载波扩展512B 最大1518B		
其他	1、最大网络长度2500m 2、帧间隔:9.6微秒 3、48bit地址	1、地址、帧格式和长度等不变; 2、自动协商	1、地址、格式和帧长度不变; 2、自动协商 3、载波扩展200m	应用: 校园网的骨干网、MAN、与WAN互联	1、无冲突: 引入交换机, 分割冲突域 2、更安全: 过滤、隔离流量

Reasons for Longevity of Ethernet

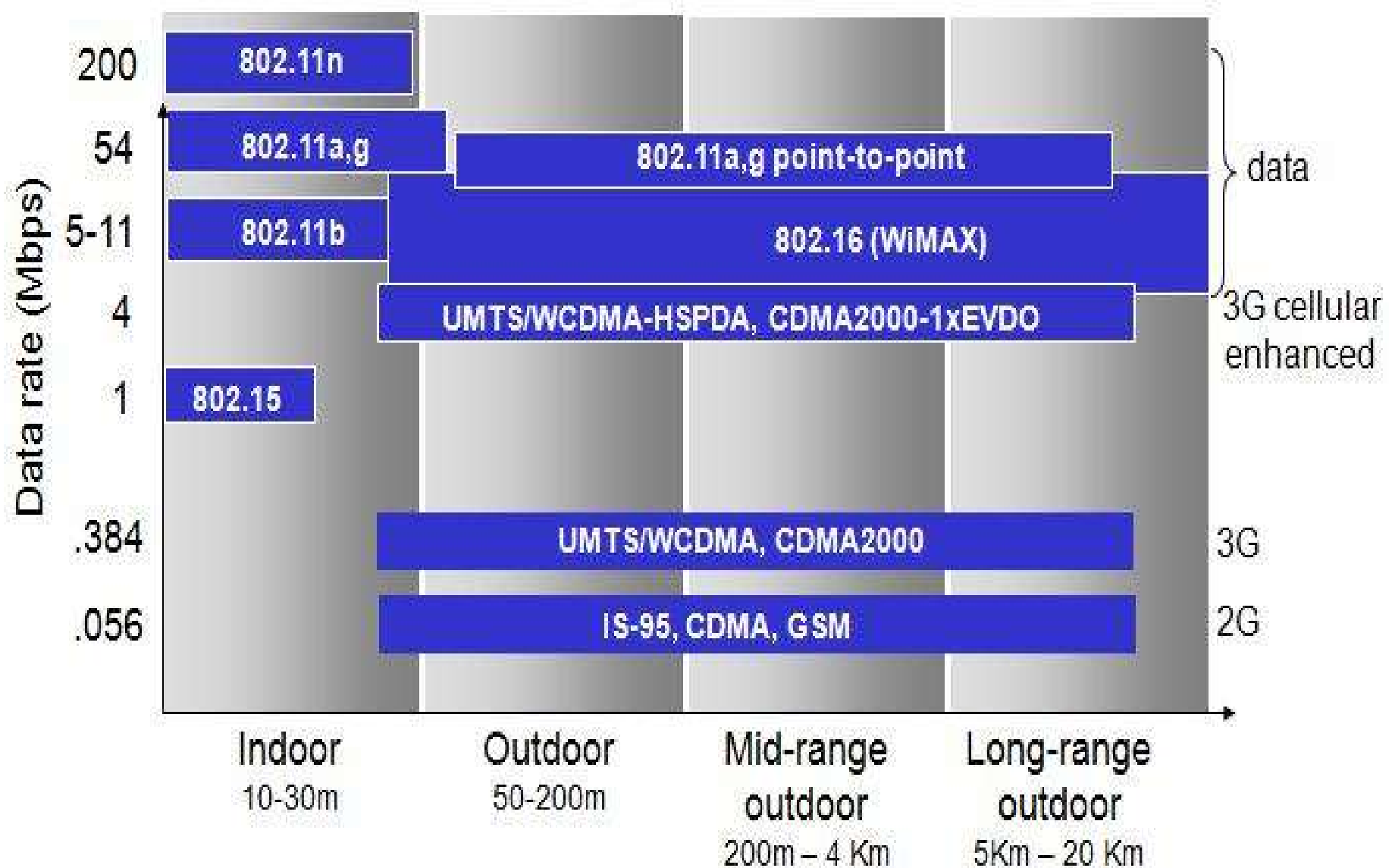
KISS(Keep It Simple, Stupid)

- Reliable: star topology with hub or switch
- Cheap
- Easy to maintain
 - ◆ No software to install
 - ◆ No configuration tables to manage
 - ◆ Adding new hosts is as simple as just plugging them in
- Interworking easily with TCP/IP
- Able to evolve in certain crucial ways
- Having most markets of LAN or MAN

Outline

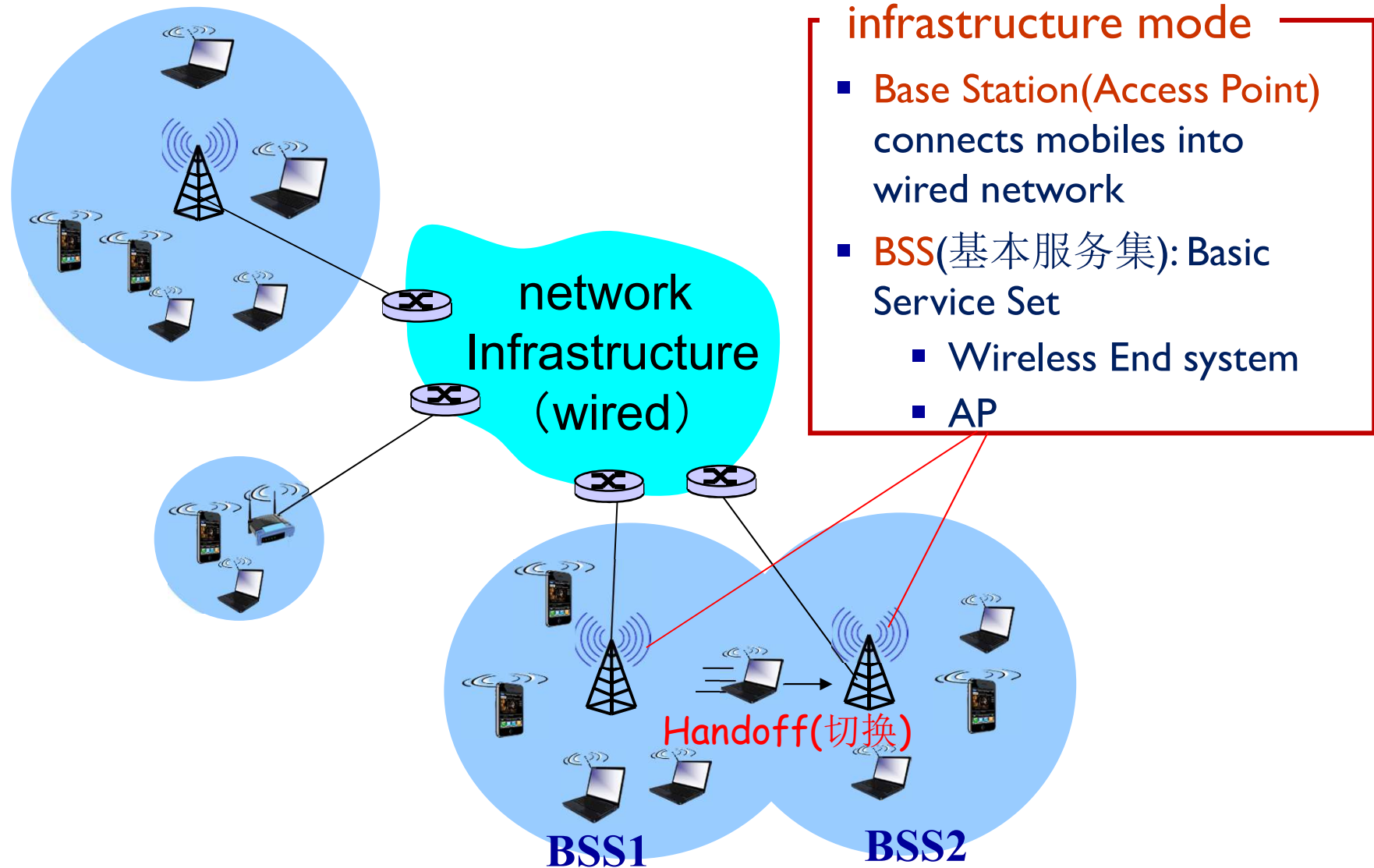
- Channel Allocation Problem
- Multiple Access Protocols
- IEEE 802 Reference Model
- Ethernet
- Wireless LANs
 - ◆ IEEE 802.11 (WiFi)
 - ◆ Broadband wireless: IEEE 802.16 (WiMax)
 - ◆ Bluetooth: IEEE 802.15
- Data Link Layer Switching

Wireless link standards



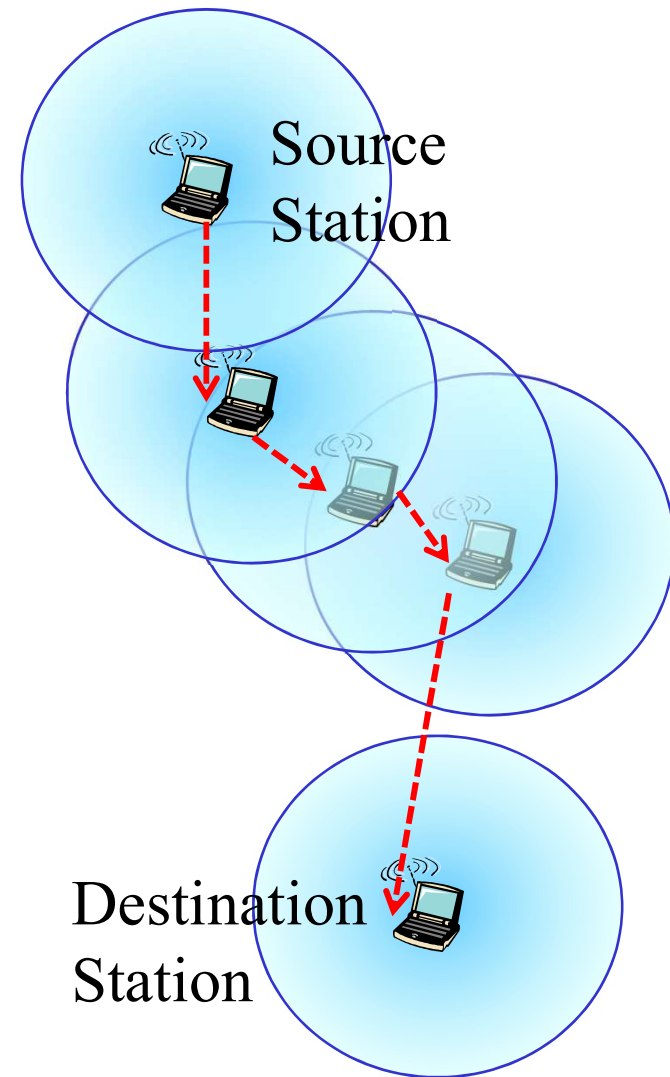
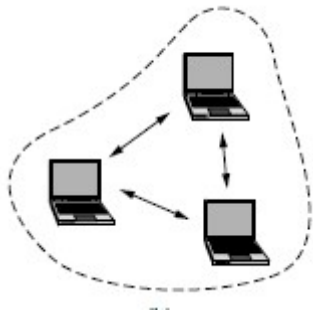
[Kurose]

Wireless LAN: Infrastructure Mode

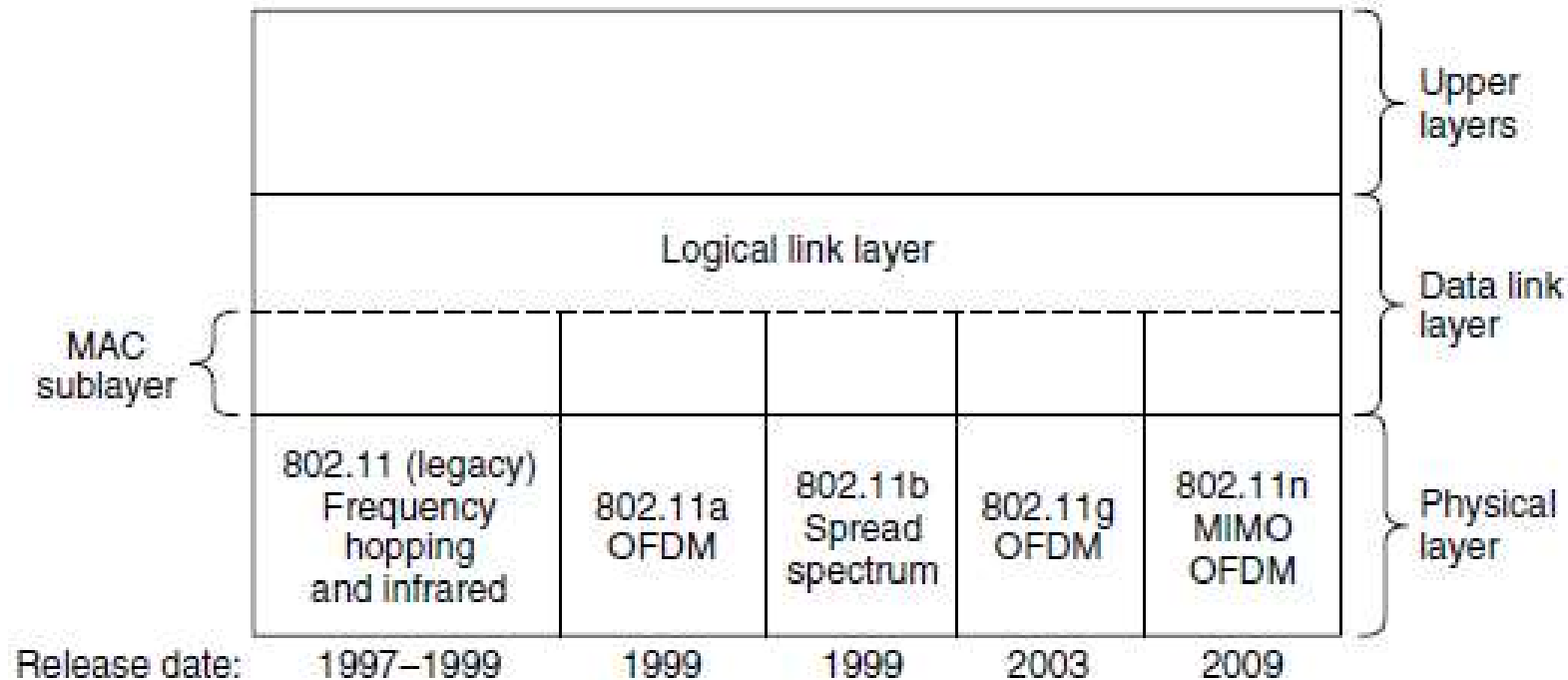


Wireless LAN: Ad hoc Mode

- No base stations
- Nodes can only transmit to other nodes within link coverage
- Nodes organize themselves into a network:
route among themselves



IEEE802.11 Protocol Layer



- FHSS: Frequency Hopping Spread Spectrum 跳频扩频
- DSSS: Direct Sequence Spread Spectrum 直接序列扩频
- OFDM: Orthogonal Frequency Division Multiplexing 正交频分复用
- MIMO: Multiple Input Multiple Output 多输入多输出

物理层编码技术不属于课程范畴！

Widely Used 802.11 WLANs

■ 802.11b

- ◆ 2.4-5 GHz range
- ◆ up to 11 Mbps

■ 802.11a

- ◆ 5-6 GHz range
- ◆ up to 54 Mbps

■ 802.11g

- ◆ 2.4-5 GHz range
- ◆ up to 54 Mbps

■ 802.11n: multiple antennae

- ◆ 2.4-5 GHz range
- ◆ up to 600 Mbps

-
- All use CSMA/CA for multiple access
 - All have base-station and ad-hoc network versions

802.11 MAC Sublayer: Multiple access

■ How to avoid collision?

◆ Collision Detection does not work!

- difficult to receive (sense collisions) when transmitting due to weak received signals
- Hidden station problem

■ 2 Modes of operation

◆ Point coordination function (PCF, 点协调功能)

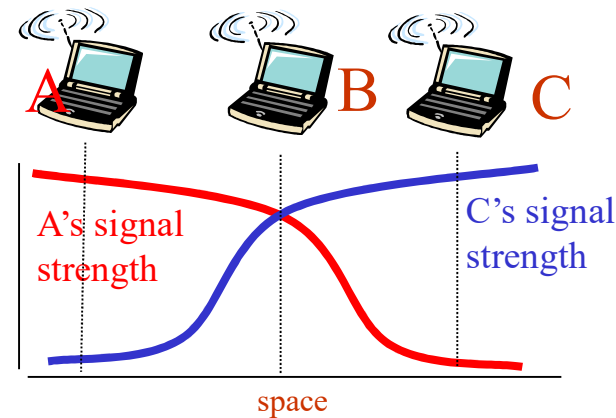
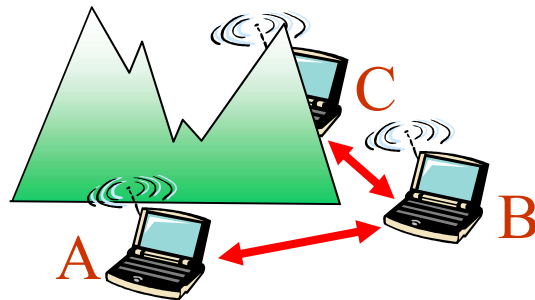
- Controlled by base station
- Polling-based priority

◆ Distributed coordination function (DCF, 分布协调功能)

- No central control
- CSMA/CA(Collision Avoidance)

802.11 MAC: CSMA/CA

- CSMA - sense before transmitting
 - ◆ don't collide with ongoing transmission by other node
- Explicit ACK
- Exponential backoff
- Optional CA: Collision Avoidance
 - ◆ Using RTS and CTS



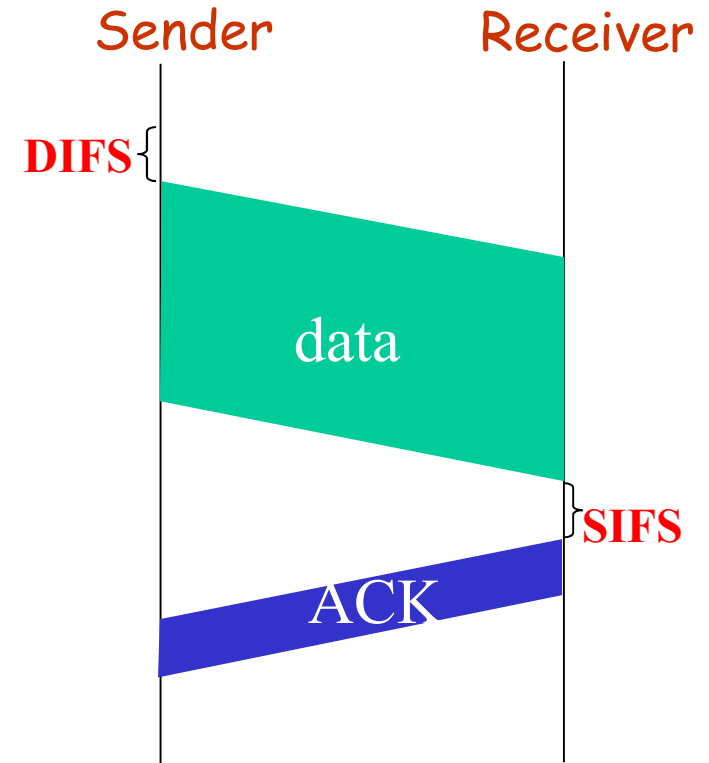
802.11 MAC : A Brief View

802.11 sender

1. If sense channel idle for **DIFS** then transmit entire frame
2. If sense channel busy then start random **backoff** time, timer counts down while channel idle; transmit when timer expires.
If no **ACK**, **increase random backoff interval(double)**, repeat 2

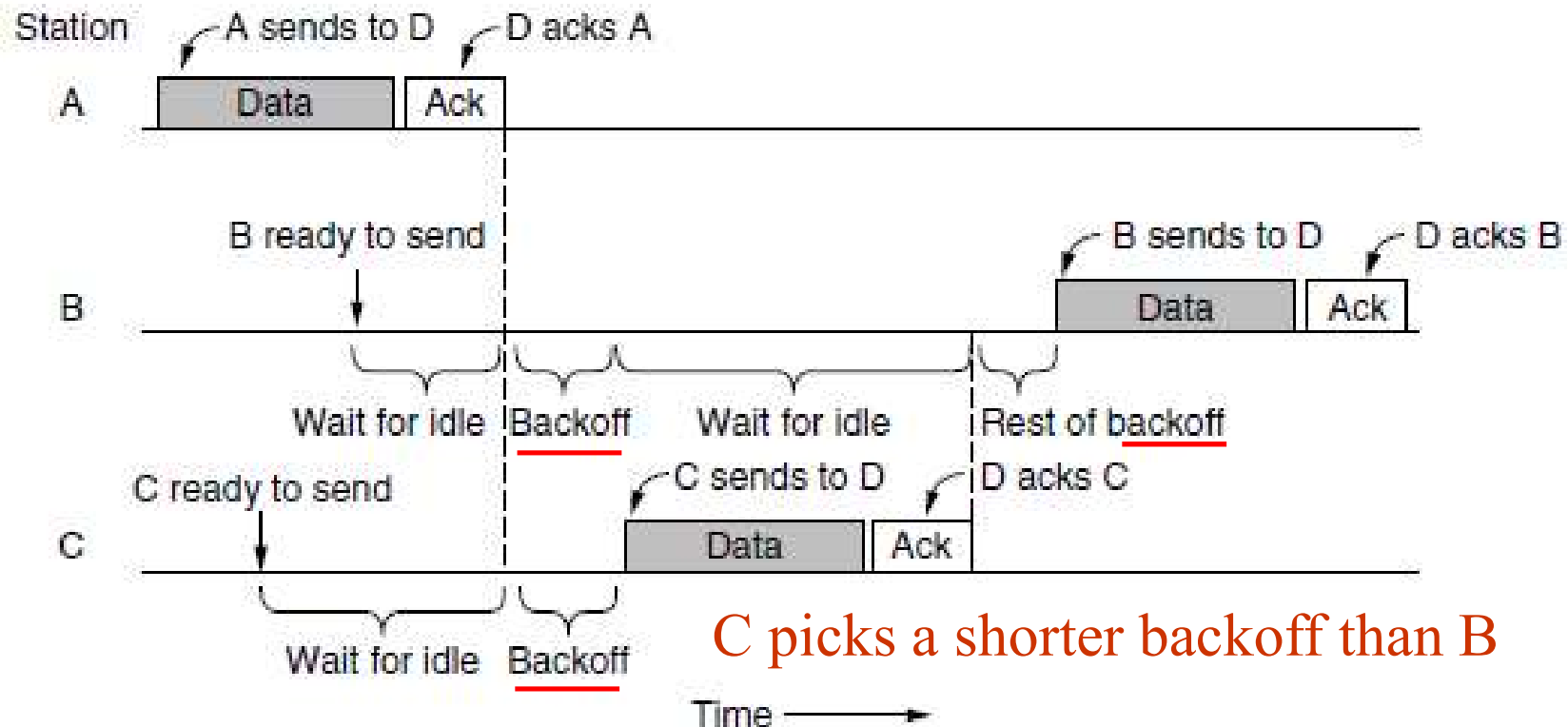
802.11 receiver

- if frame received OK, return **ACK** after **SIFS**



DIFS: DCF Inter-Frame Space
SIFS: Short Inter-Frame Space

CSMA/CA: Example



■ Compared with Ethernet

- ◆ Starting backoffs(退避) early helps to avoid collisions.
- ◆ Acknowledgements are used to infer collisions.

Hidden Terminal (隐蔽站) Problem and Exposed Terminal (暴露站) Problem

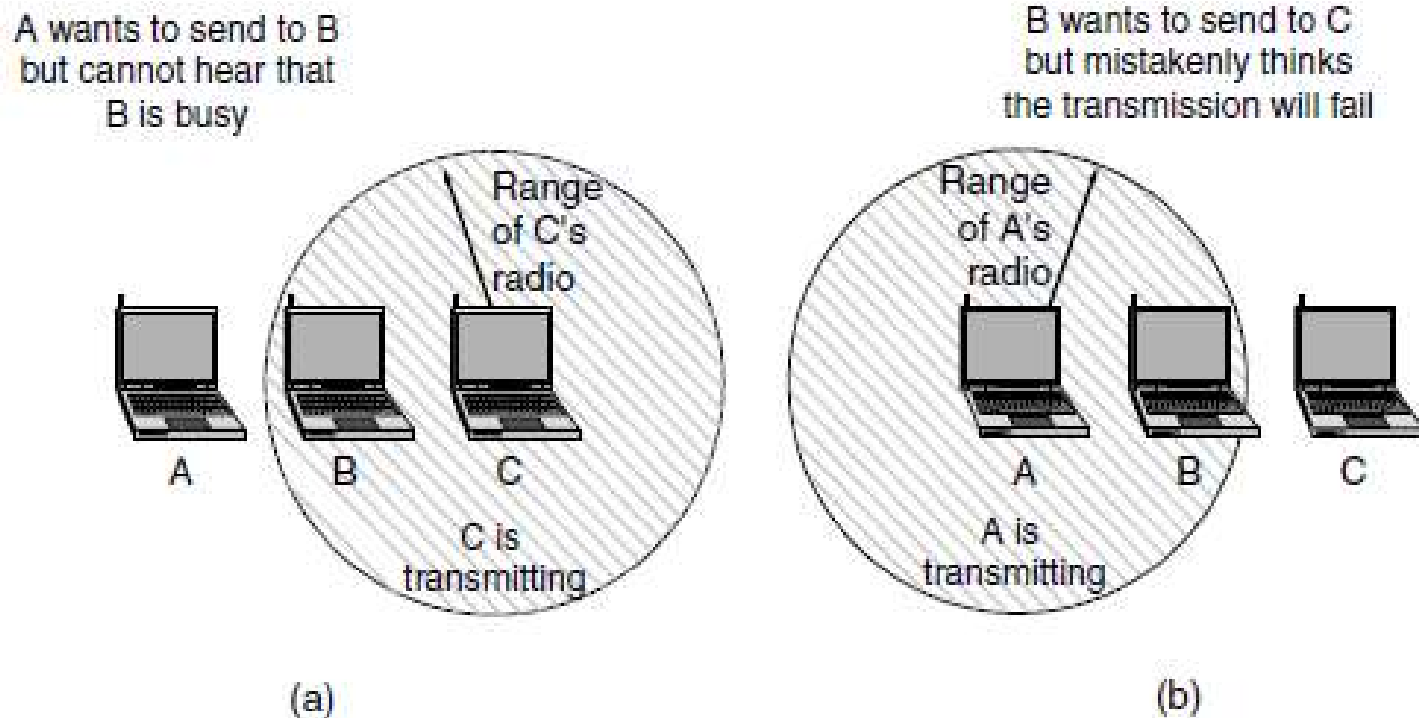


Figure 4-26. (a) The hidden terminal problem. (b) The exposed terminal problem.

802.11 solutions to Hidden Terminal: **RTS/CTS + NAV**

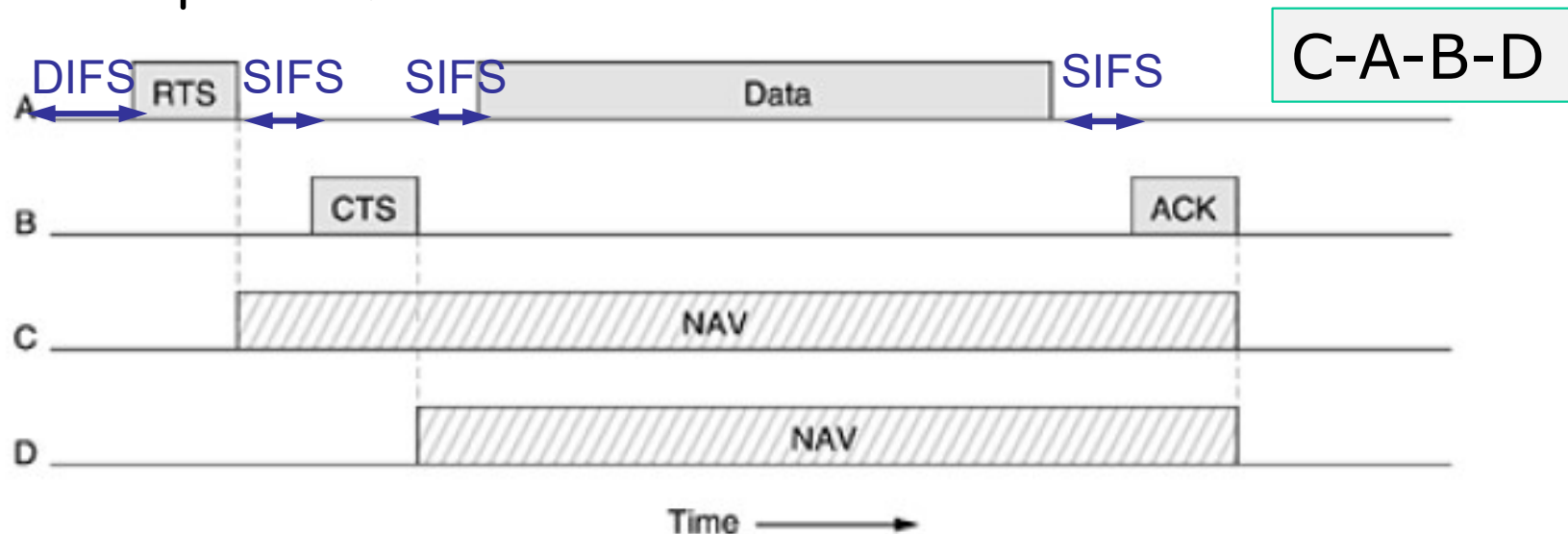
802.11 MAC: Virtual sensing

Physical sensing

- Checks the medium to see if there is a valid signal

Virtual channel sensing

- Each station keeps a logical record of when the channel is busy by tracking **NAV (Network Allocation Vector)**
- Each frame carries a **NAV field** that says how long the sequence of which this frame is part will take to complete. 本帧所属的通信将占用信道的时间



通过NAV(网络分配向量)确定退避时间，以避免冲突

802.11 MAC: Fragment burst(分段突发)

■ Frame success probability

- ◆ p is the error rate of 1 bit

短帧出错概率低！

$$S = (1 - p)^n$$

$$n = 1518B = 12144 \text{ bits}$$

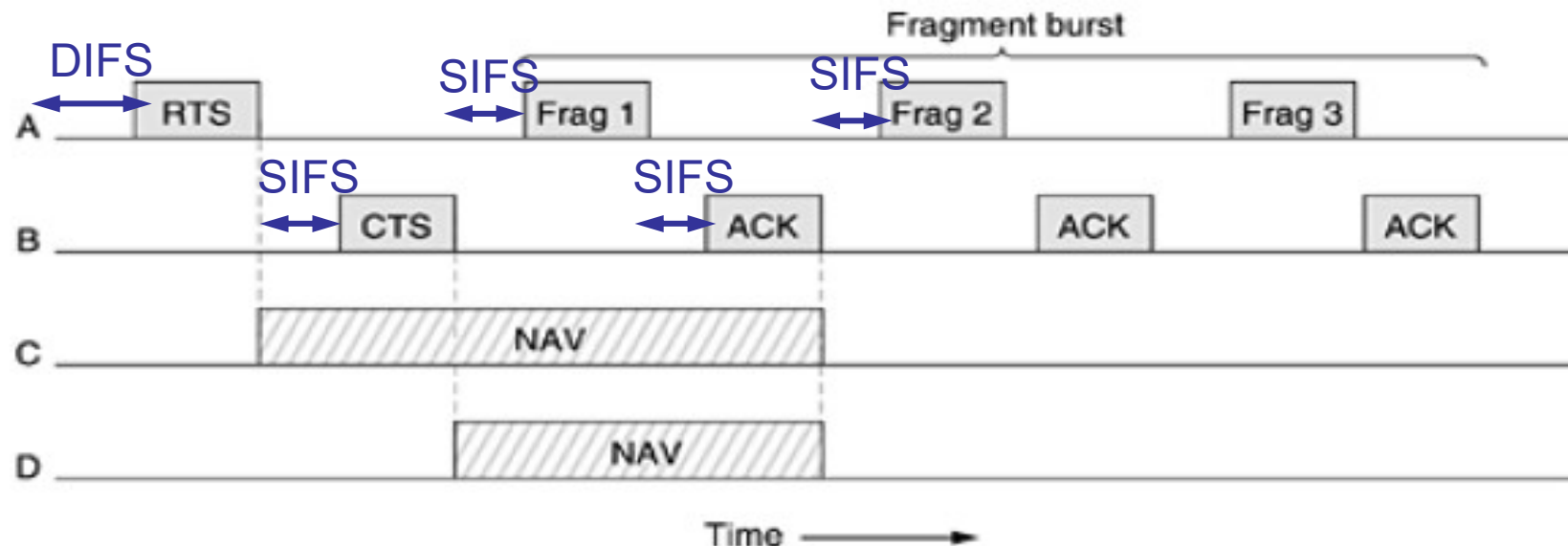
$$p = 10^{-4} \quad S = 0.296$$

$$p = 10^{-5} \quad S = 0.885$$

$$p = 10^{-6} \quad S = 0.988$$

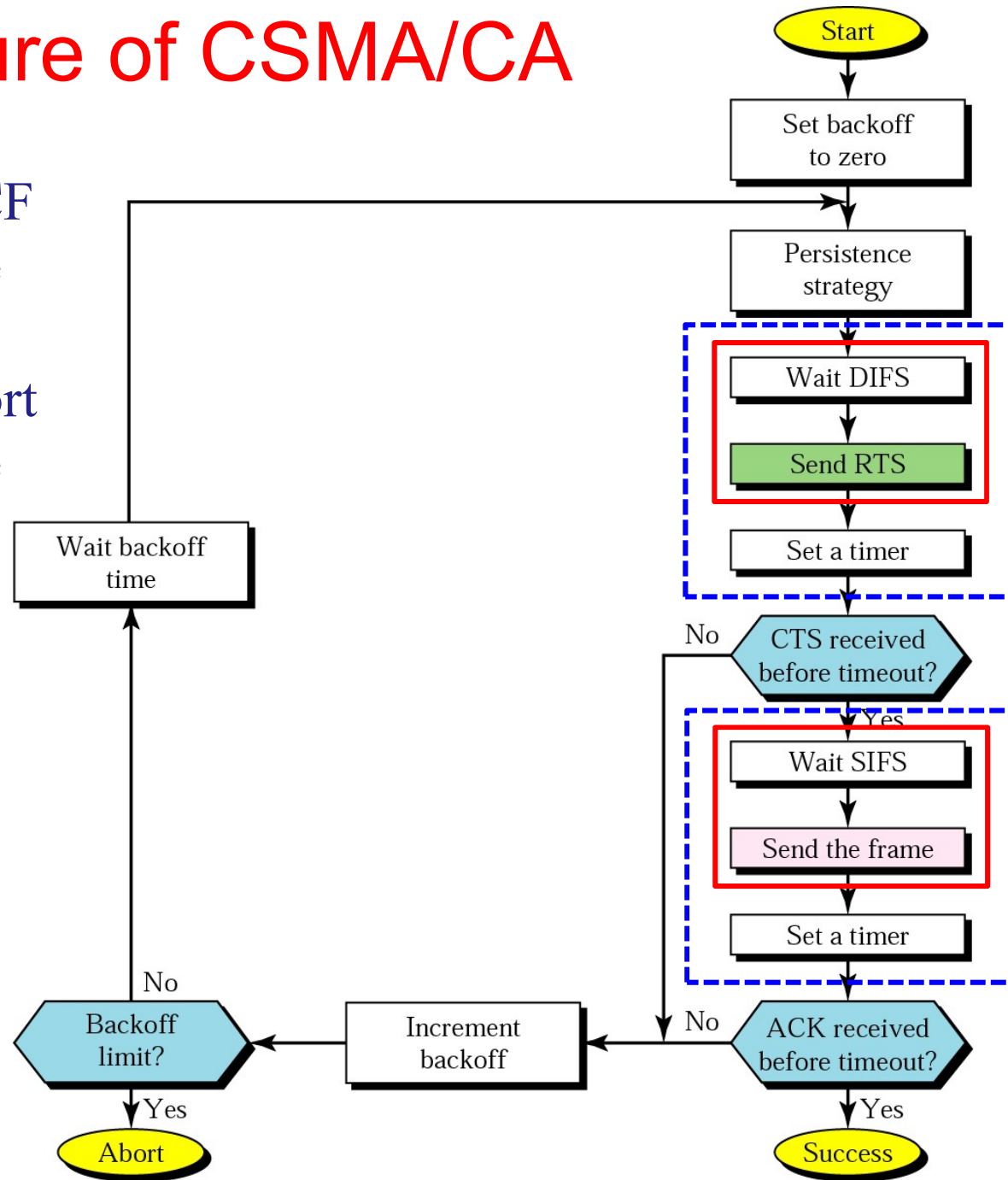
■ Fragment burst

- ◆ The solution is to fragment data to small pieces
- ◆ Each fragment is sent with individual sequence number and checksum by **stop-and-wait** protocol



Procedure of CSMA/CA

- DIFS: DCF Inter-Frame Space
- SIFS: Short Inter-Frame Space



[Forouzan]

802.11 MAC: PCF

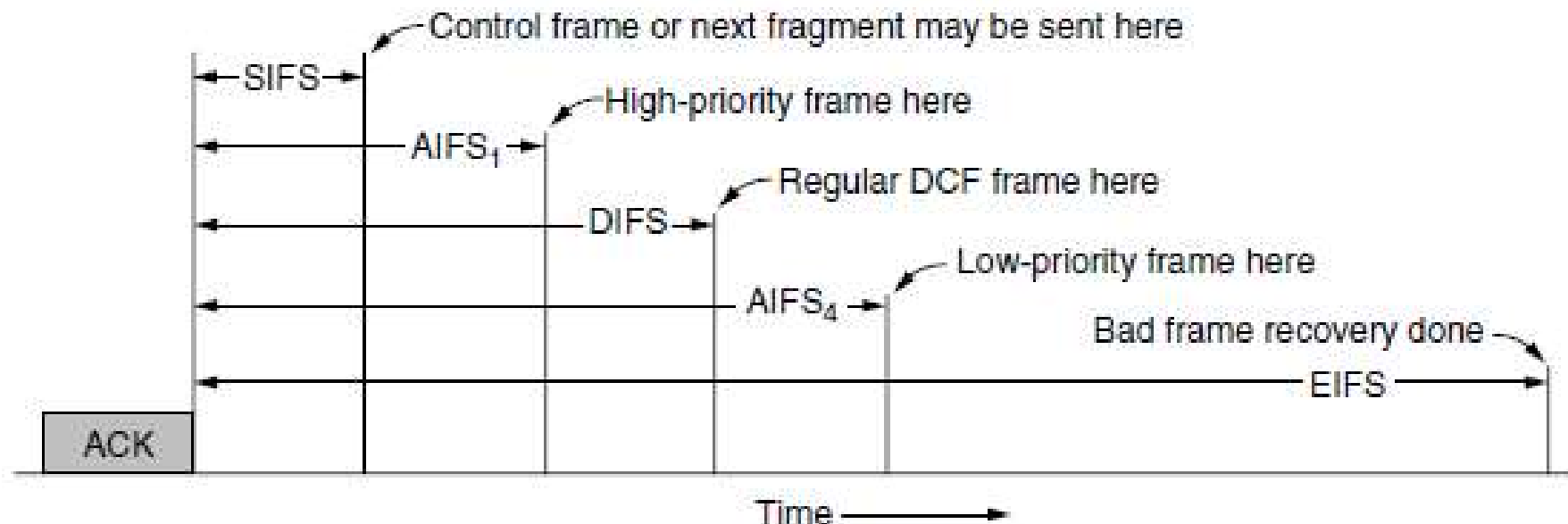
- Point Coordination Function: Centralized control
 - ◆ A base station controls data transfer by polling
 - ◆ A beacon frame is broadcast by the base station in a cell periodically (10-100 times per second)
 - ◆ Beacon invites new stations to sign up for polling service
 - ◆ After sign up, a certain bandwidth will be guaranteed for station
 - ◆ Power management
 - Base station can direct the station to go to sleep mode and also awakened it when necessary

PCF: 由基站集中轮询

Neither PCF nor CA used in practice

IEEE802.11e: 5 Interframe Spaces (IFSs)

- Used to provide **priority services**
 - ◆ DIFS: time waiting before frame transmission
 - ◆ SIFS: time waiting before ACK transmission
 - ◆ AIFS₁ (Arbitration InterFrame Space): time waiting before sending high-priority traffic
 - ◆ AIFS₄: time waiting before sending low-priority traffic
 - ◆ EIFS: time waiting before retransmission

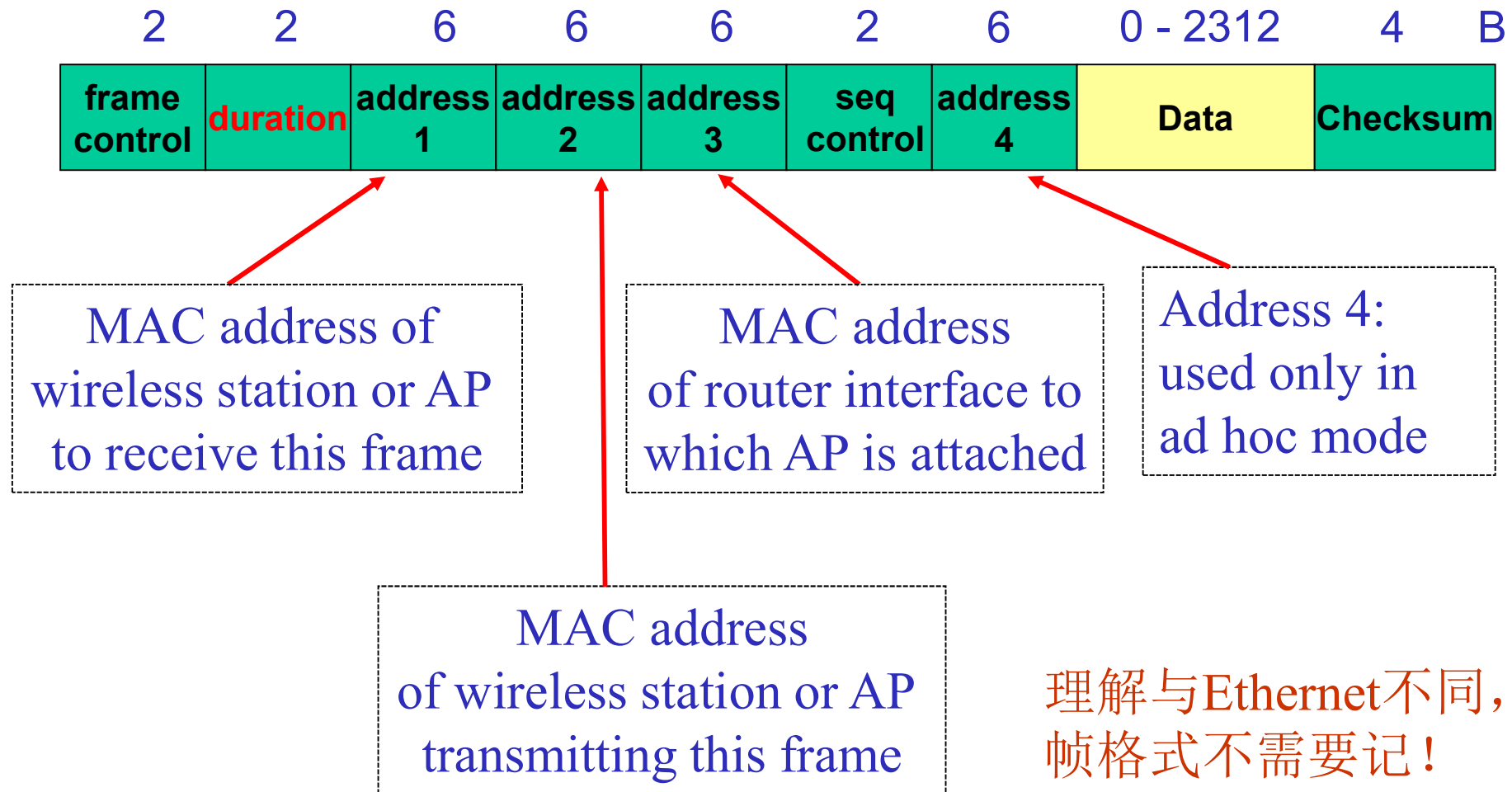


通过为不同类型的帧设置不同长度的帧间间隔IFS，实现优先级服务

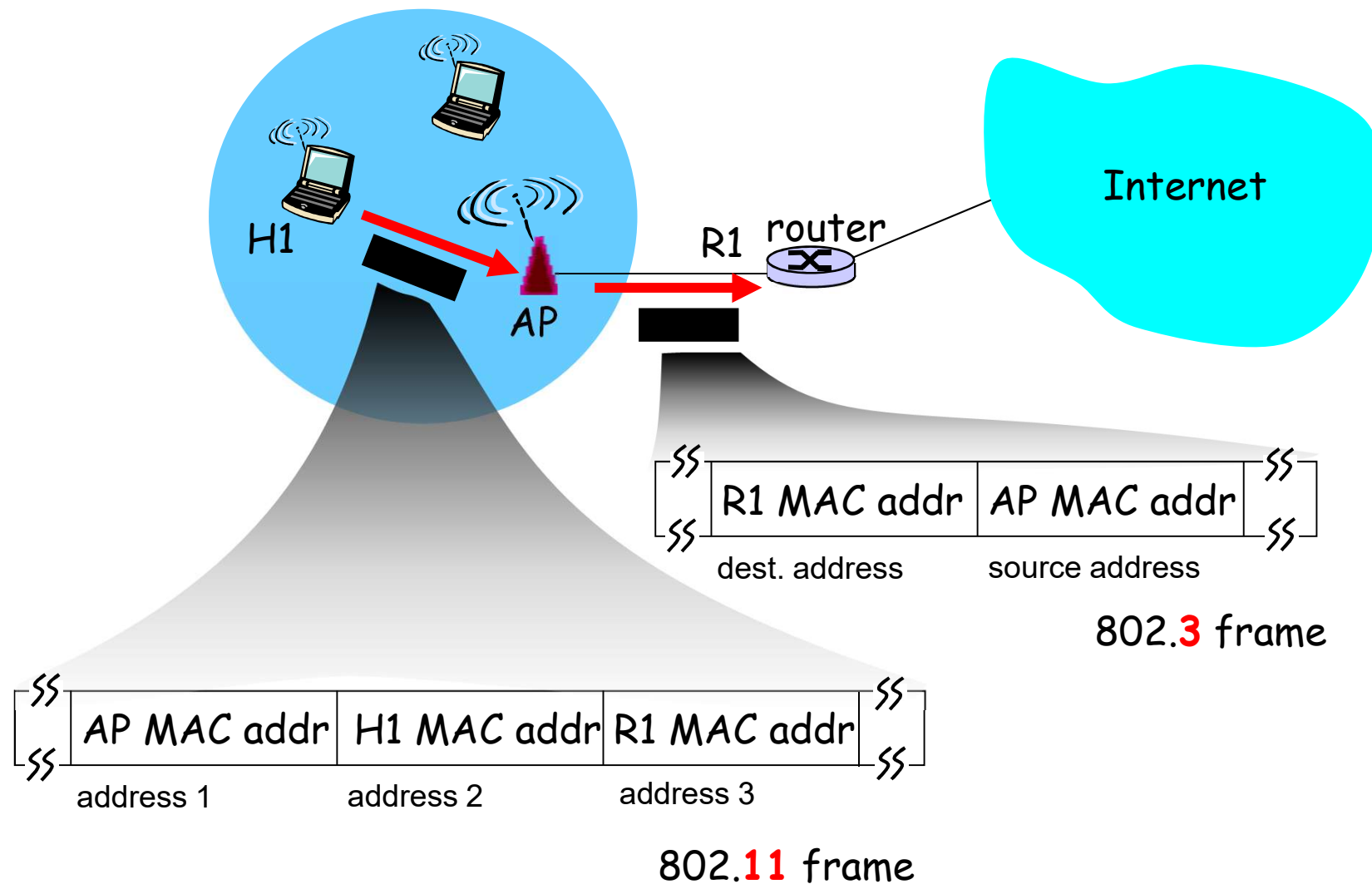
QoS Extension: TXOP

- Transmission Opportunity(发送机会)
- Original problem: Rate anomaly(速率异常)
 - ◆ A station sends one frame at a time
 - ◆ If a fast station and a slow station send at same time, the fast one has to slow down.
 - ◆ 示例：两个站点，发送速率分别为6Mbps和54Mbps
每个站点每次只发一帧，
平均速率= $1 / (1/6 + 1/54) = 5.4\text{Mbps}$
- With transmission opportunities, each station gets an **equal amount of airtime**, not an equal number of frames
 - ◆ 示例：上述两站点各占用一半信道（TDM），速率分别为3Mbps和27Mbps

802.11 Frame: Addressing



Example of 802.11 addressing



Review of IEEE802.11

- Wifi
- 物理层有多种编码技术(超出本课程范畴)
- MAC子层: ~~≠~~MACA
 - ◆ PCF: 集中式, 基站轮询
 - ◆ DCF: 分布式, CSMA/CA
 - RTS/CTS 可选
 - 依靠NAV确定退避时间, 避免冲突
 - 分段突发: 把长帧分成多个短片段发送, 降低出错概率
 - TXOP
 - 不同帧监听时的退避时间不同
(DIFS vs. SIFS, 为什么?)

Outline

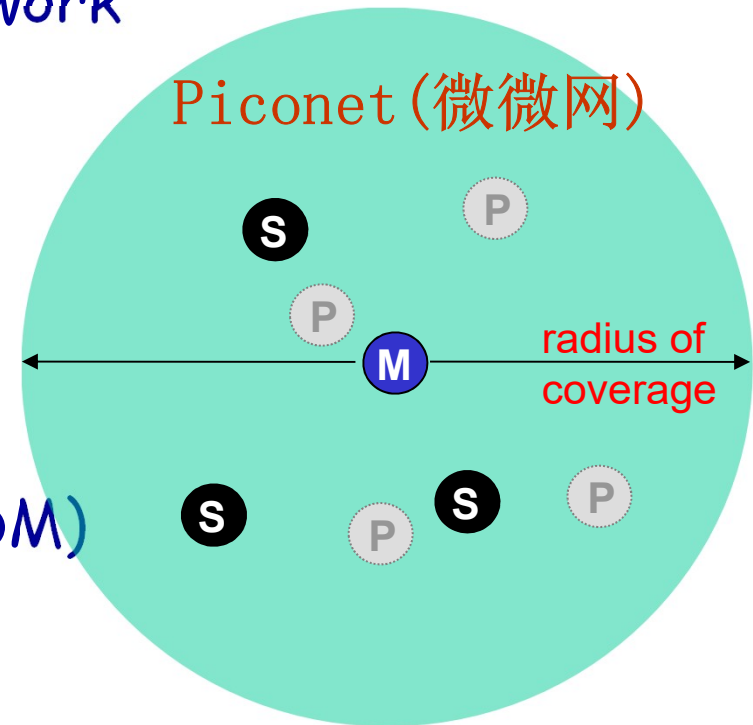
- Channel Allocation Problem
- Multiple Access Protocols
- IEEE 802 Reference Model
- Ethernet
- Wireless LANs
 - ◆ IEEE 802.11 (WiFi)
 - ◆ Broadband wireless: IEEE 802.16 (WiMax)
(self-study)
 - ◆ Bluetooth: IEEE 802.15
- Data Link Layer Switching

Outline

- Channel Allocation Problem
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IEEE802.15: Bluetooth (蓝牙)

- Used in PAN: Personal area network (个域网)
- Less than 10 m diameter
- Replacement for cables (mouse, keyboard, headphones)
- Ad hoc: no infrastructure
- Master/Slaves: (Centralized TDM)
 - ◆ Slaves request permission to send
 - ◆ Master grants requests
- 802.15: evolved from Bluetooth specification
 - ◆ 2.4-2.5 GHz radio band
 - ◆ 1Mbps for v1.2, 3Mbps for v2.0



- Ⓜ Master device
- Ⓢ Active slave
- Ⓟ Parked slave(inactive)

Outline

- Channel Allocation Problem
- Multiple Access Protocols
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- Ethernet
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Data Link Layer Switching

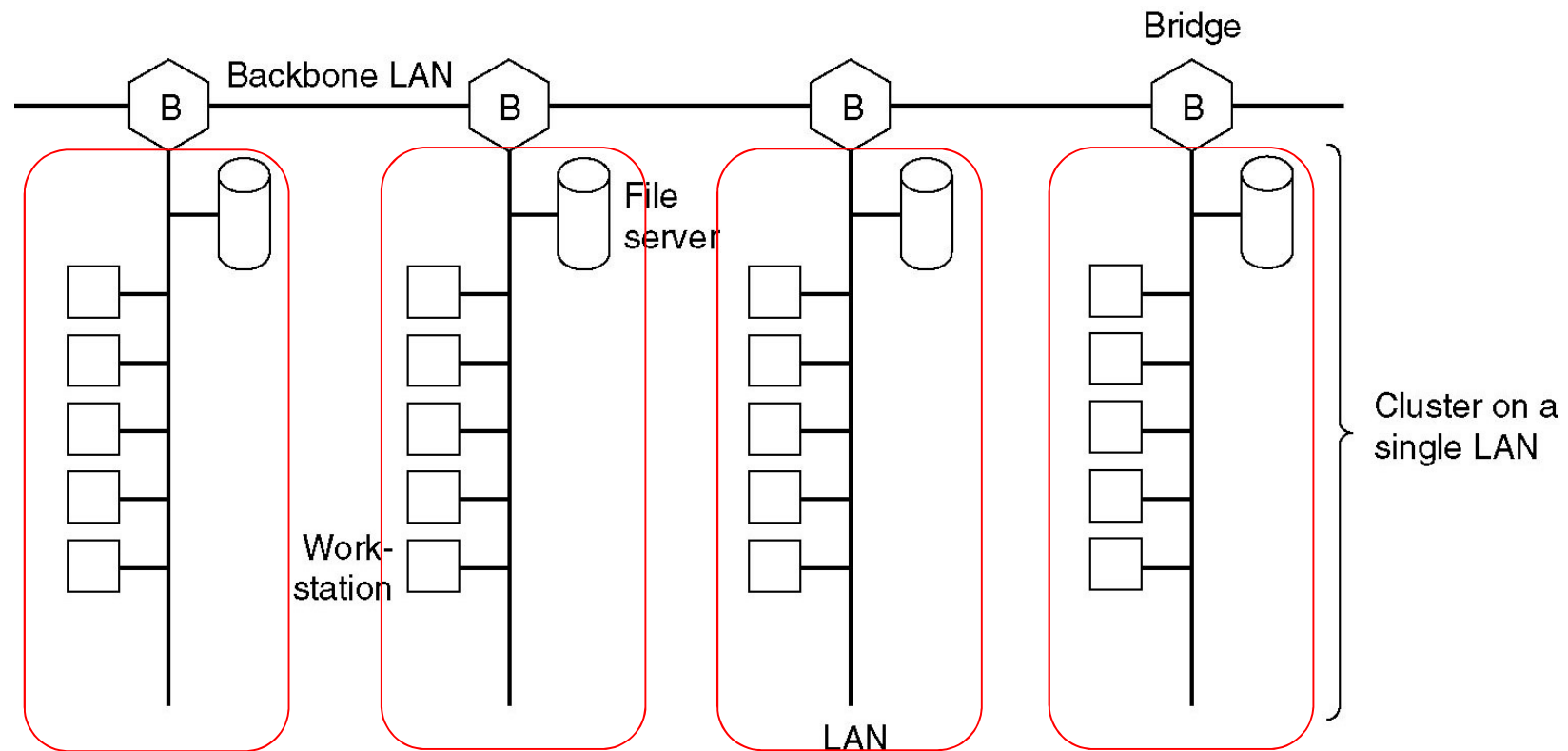
- Bridges from 802.x to 802.y
- Learning Bridges and LAN Switches
- Spanning Tree Bridges (自阅)
- Summary of connection devices
- Virtual LANs

Why Interconnecting Multiple LANs?

- The need for interaction
- The organization may be geographically spread over several buildings
- Split a single LAN into separate LANs to accommodate the load (multiple collision domains)
- Physical distance($\leq 2500\text{m}$)
- Reliability(可隔离故障)
- Security (promiscuous mode, 混杂模式)

Multiple LANs connected by a backbone

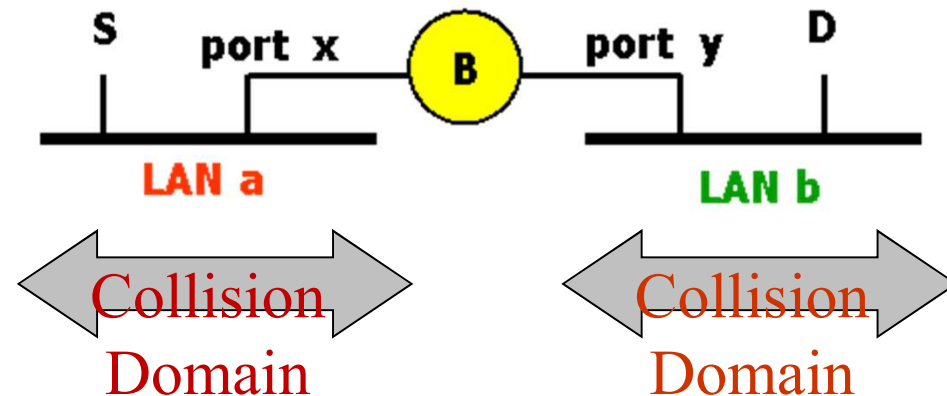
Able to handle a total load higher than the capacity of a single LAN



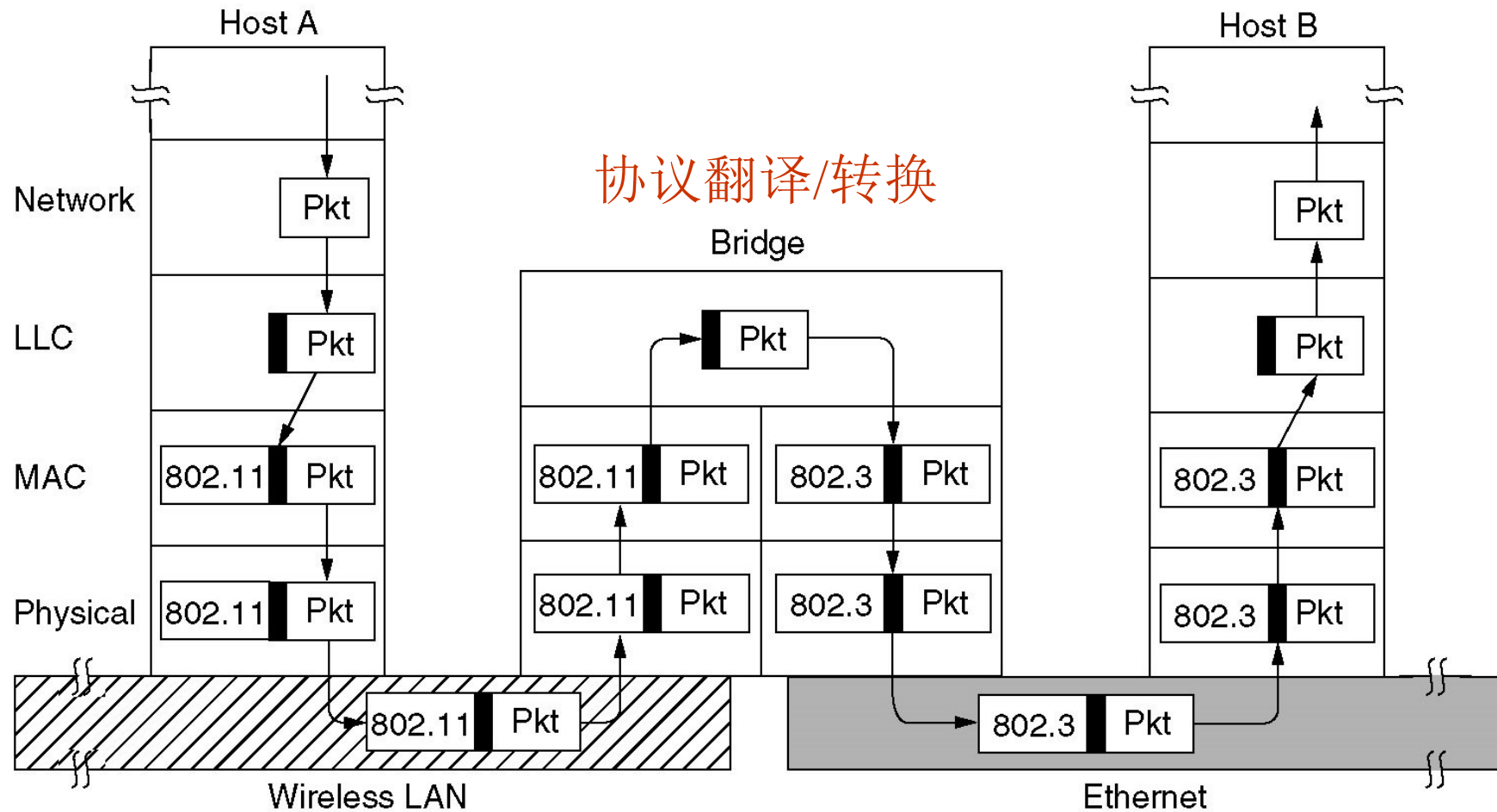
Collision domain

What is Bridge(网桥)?

- Bridges are used to interconnect LANs at **data link layer**.
- Frame forwarding from one LAN to another is based on the **destination MAC address** without making any changes to the frame.
- A bridge uses the address of the destination to decide the adjacent LAN over which the frames to the destination should be forwarded.



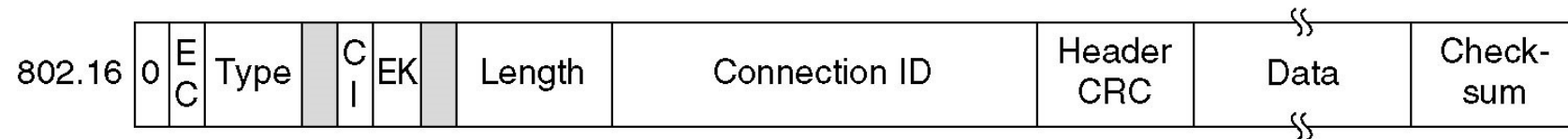
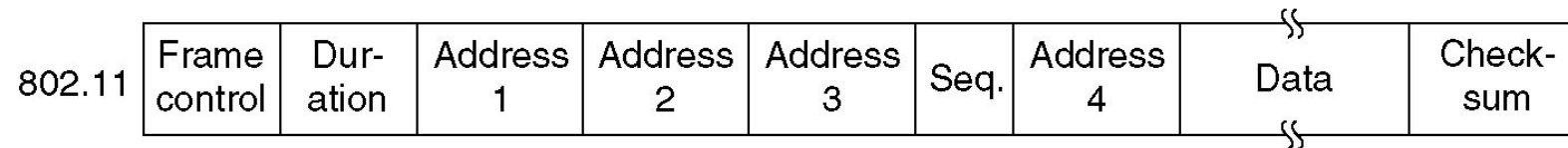
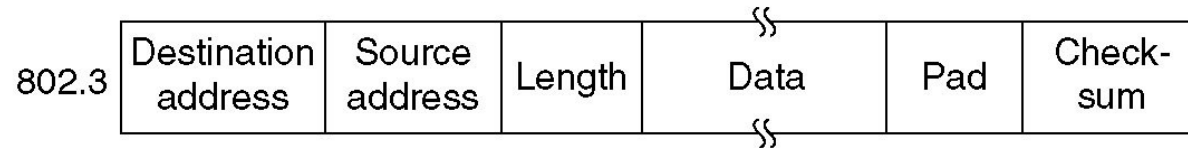
A LAN Bridge from 802.11 to 802.3



Bridge: Problems

- Different LANs uses different frame format
- LANs do not run at the same data rate
- Different LANs have different maximum frame lengths.
- Security: Wireless networks support encryption in the data link layer, Ethernet does not
- Delay increased
- No flow control function. Frames might be lost in case of heavy load
- **Broadcast storm (广播风暴)** can not be avoided
- QoS (Quality of service) can not be guaranteed

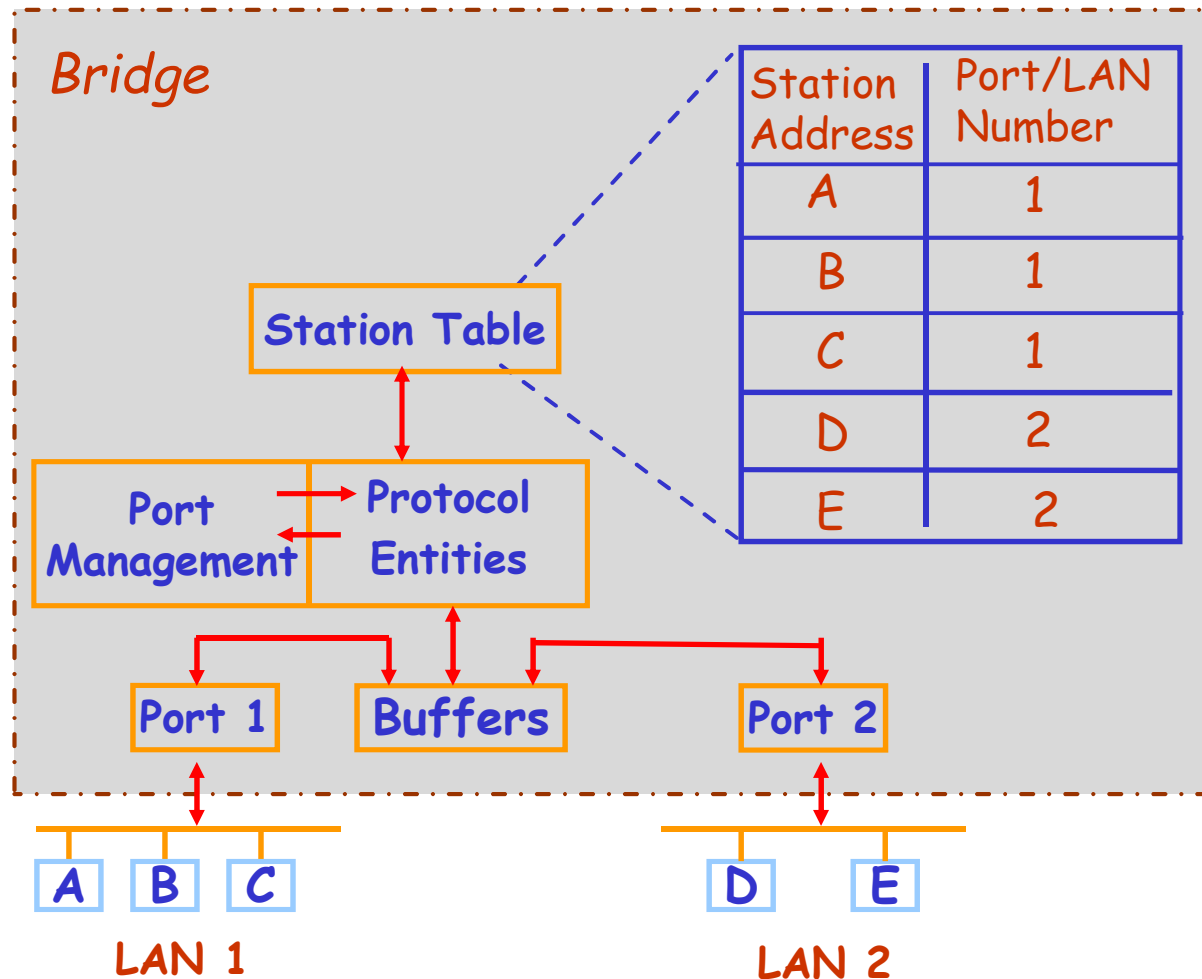
Different frame formats



Learning Bridges(学习网桥)

- Frame routing and forwarding is determined by bridges, which is transparent to hosts, also called transparent bridges(透明网桥)
- Plug and play 即插即用
- Operates in promiscuous mode(混杂模式)
 - ◆ Accepting every frame transmitted on all the LANs it attached
- For an incoming frame from port-x, checking **Station Table**(Forwarding Table/Hash Table, 站表/转发表)
 - ◆ If destination port = port-x, **discard** the frame
 - ◆ If destination port $\neq$ port-x, **forward** the frame
 - ◆ If destination port is unknown or for broadcast address, **flooding**
 - Outputting on all the ports connected except the one frame arrived on

Transparent Bridge: Operations

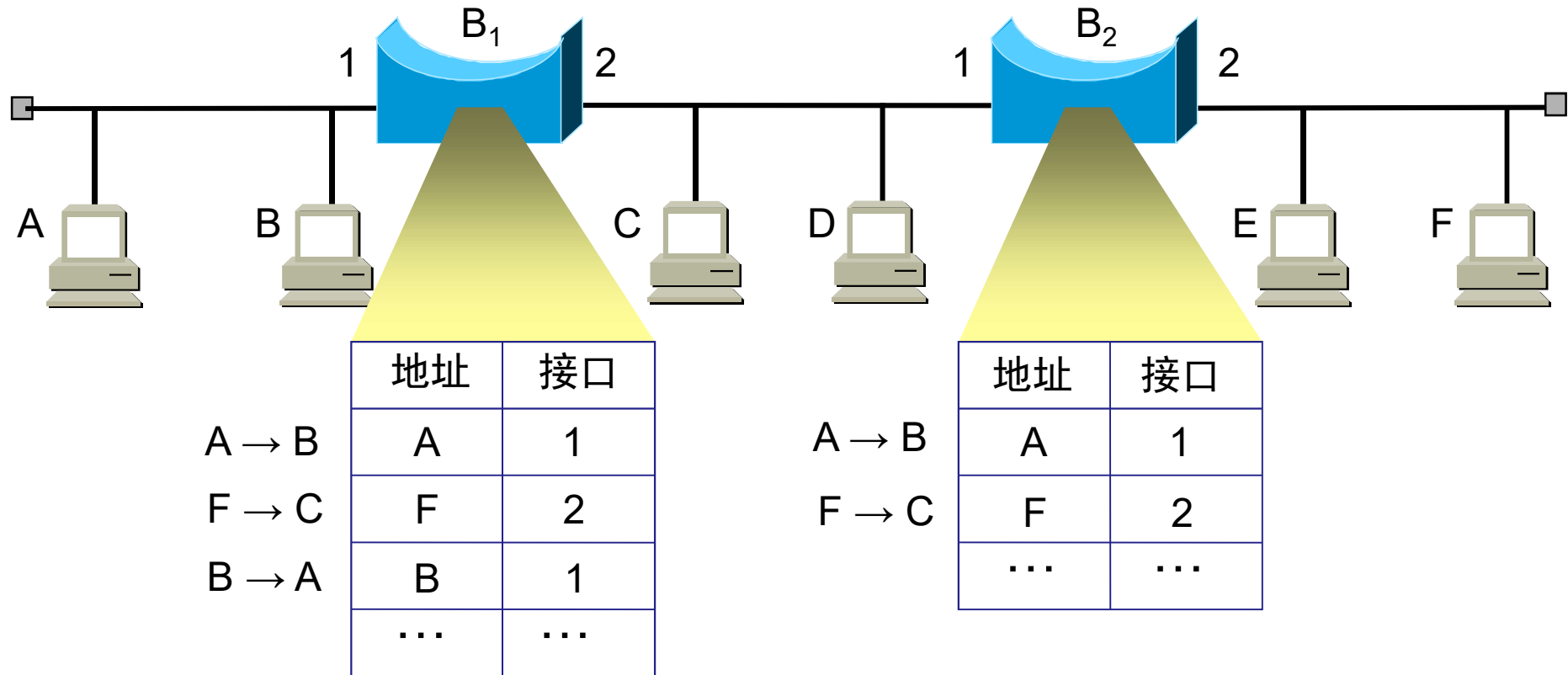


- (1) Receiving frames
- (2) Buffering
- (3) Looking up cache
- (4) Discarding frames sending to same LAN, else forwarding to relative port

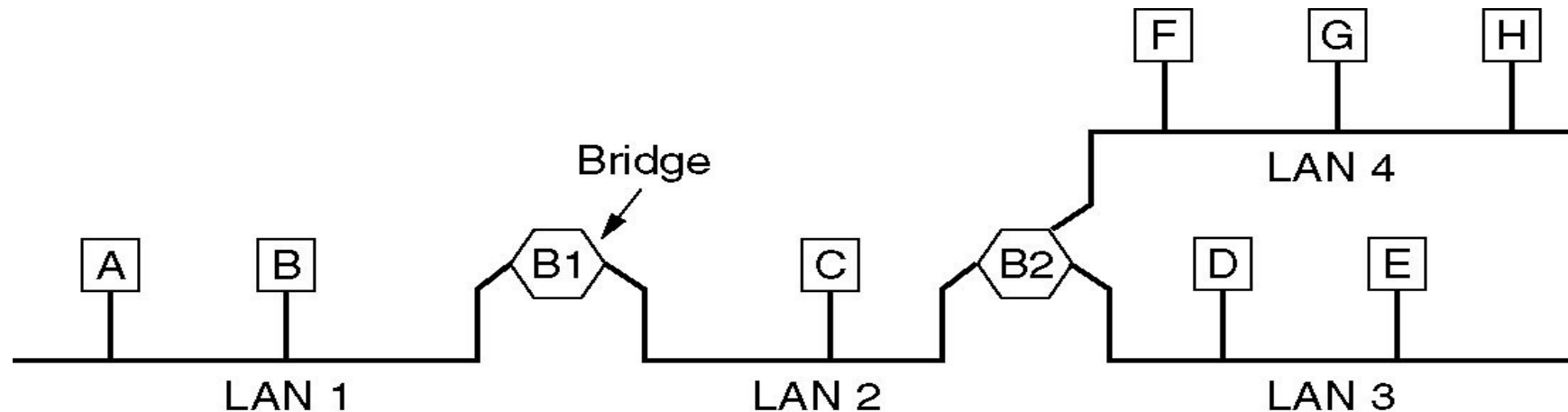
How to build Station table

- Backward Learning(逆向学习)
- Init
 - ◆ When the bridges are first plugged in, all the hash tables are empty
- Adding
 - ◆ Bridges operate in promiscuous mode, so they see every frame sent on any of their LANs
 - ◆ By looking at **the source address**, they can tell which machine is accessible on which LAN

Building a Station Table: Example



Building a Station Table: Example 2

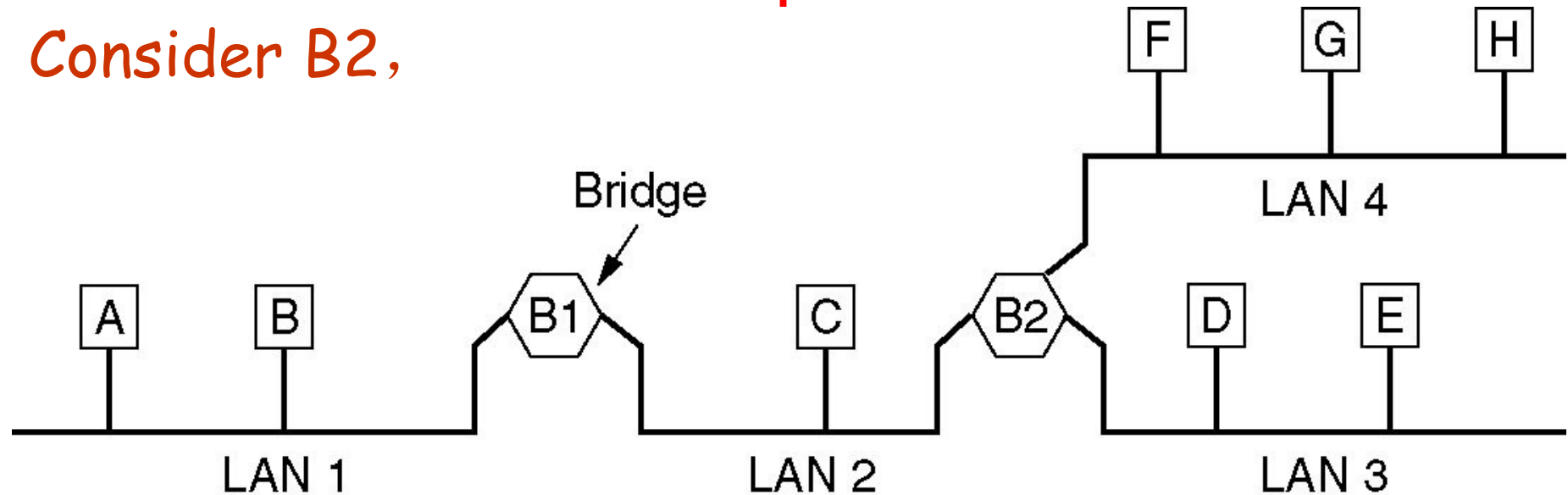


- 1 A send frame to C
- 2 E send frame to A
- 3 D send frame to E
- 4 E move to LAN2
- 5 E send frame to A

	B1	B2
1	<u>A LAN1</u>	<u>A LAN2</u>
2	<u>E LAN2</u>	<u>E LAN3</u>
3		<u>D LAN3</u>
4		
5	<u>E LAN2</u>	<u>E LAN2</u>

Routing and Building Station Table: Example2

Consider B2,

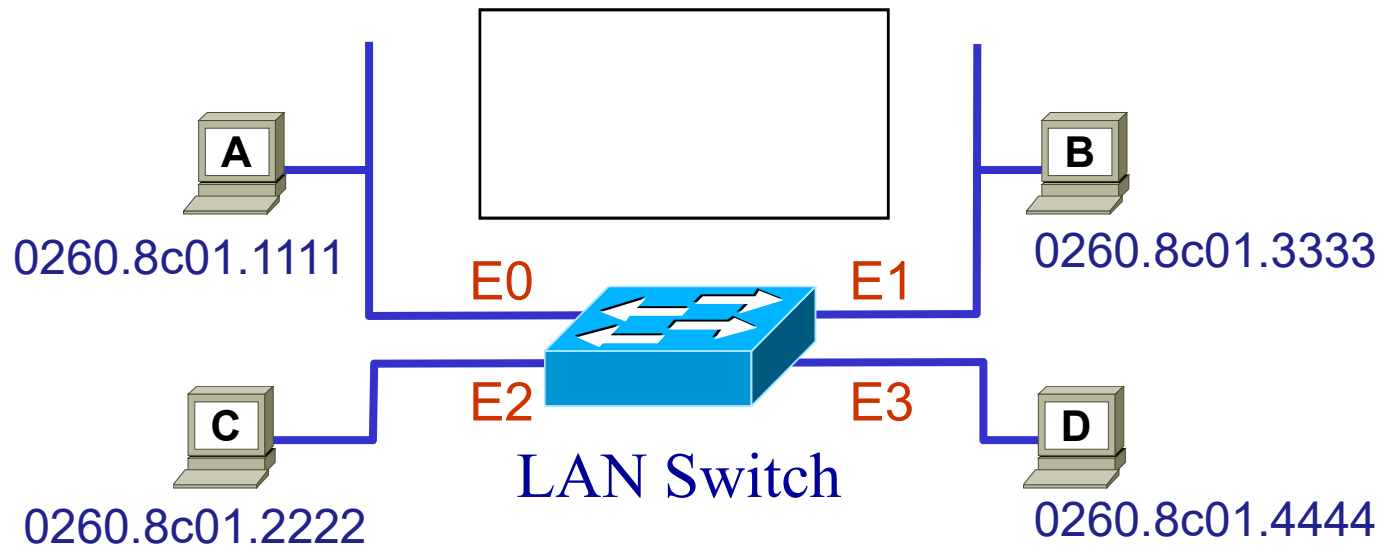


- Step 1: A send frame to C, flooding
- Step 2: E send frame to A, backward learning
- Step 3: D send frame to E, bridge discarding
- Step 4: E move to LAN2
- Step 5: E send frame to A, backward learning

How LAN Switches Learn Host Addresses and Locations(1)

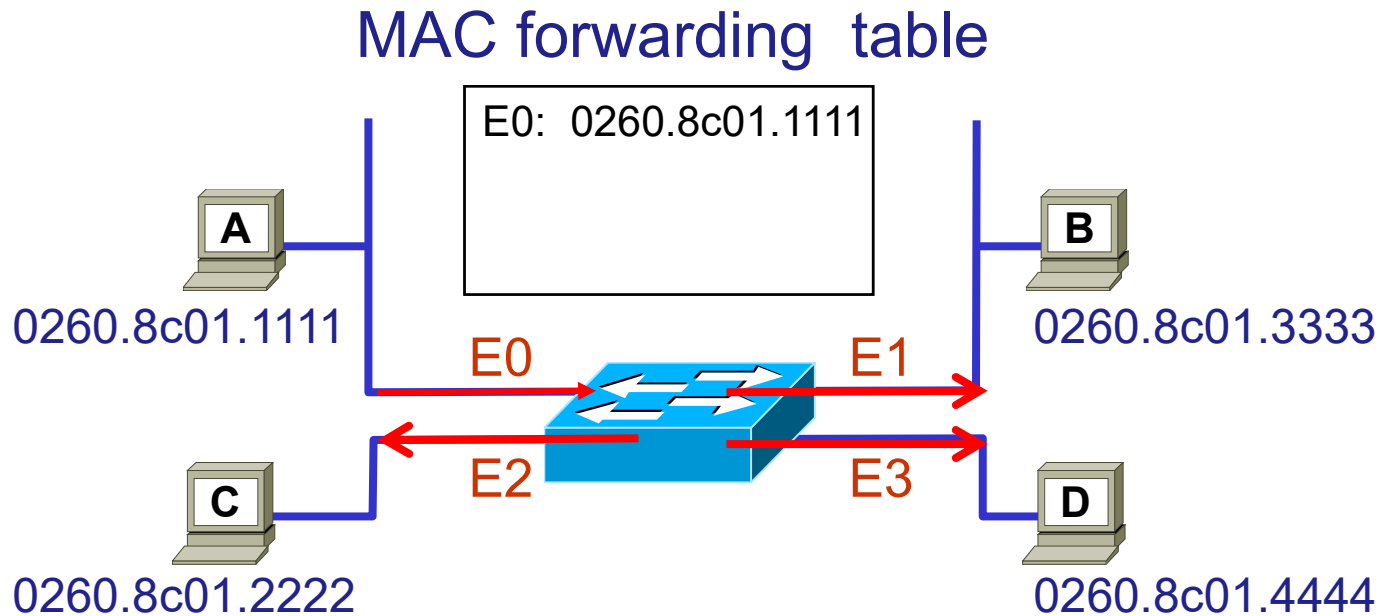
交换机根据转发表将收到的帧转发到相应端口

MAC forwarding table(转发表)



Initial MAC forwarding table is empty

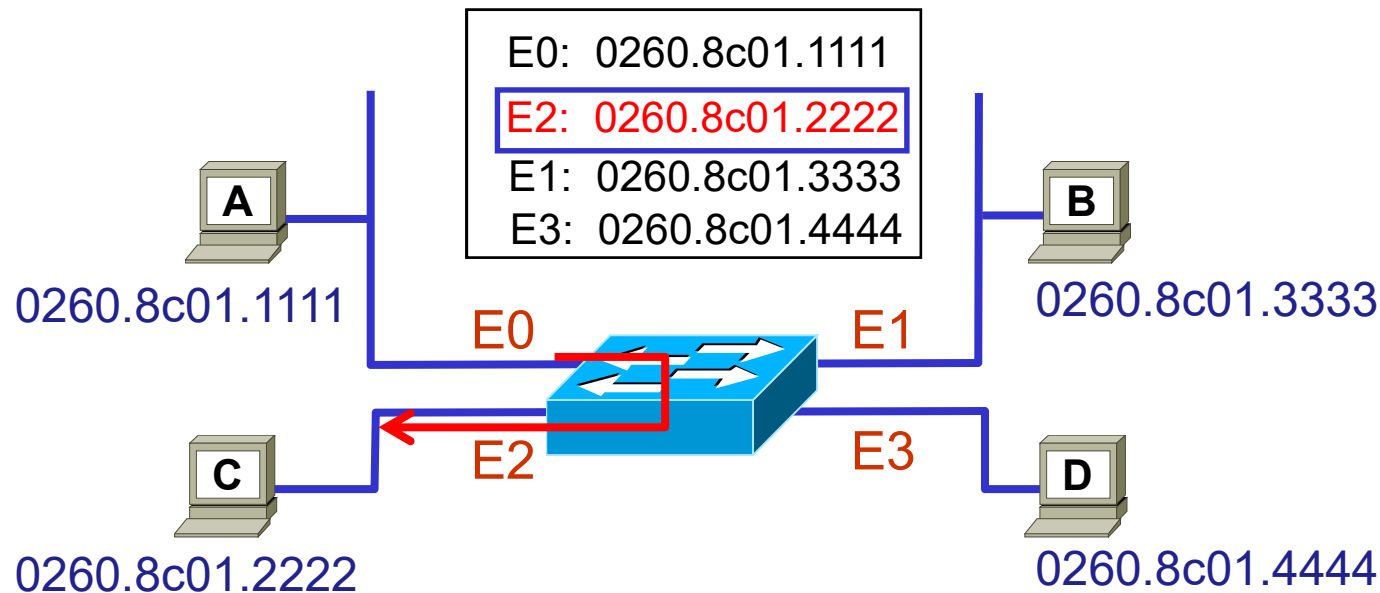
How LAN Switches Learn Host Addresses and Locations(2)



- Station **A** sends a frame to Station **C**
- The frame from station A to station C is **flooded** (洪泛转发) out to all ports except port E0 (unknown unicasts are flooded)
- Switch caches MAC address of station **A** to port **E0** by learning the source address of data frames (逆向学习)

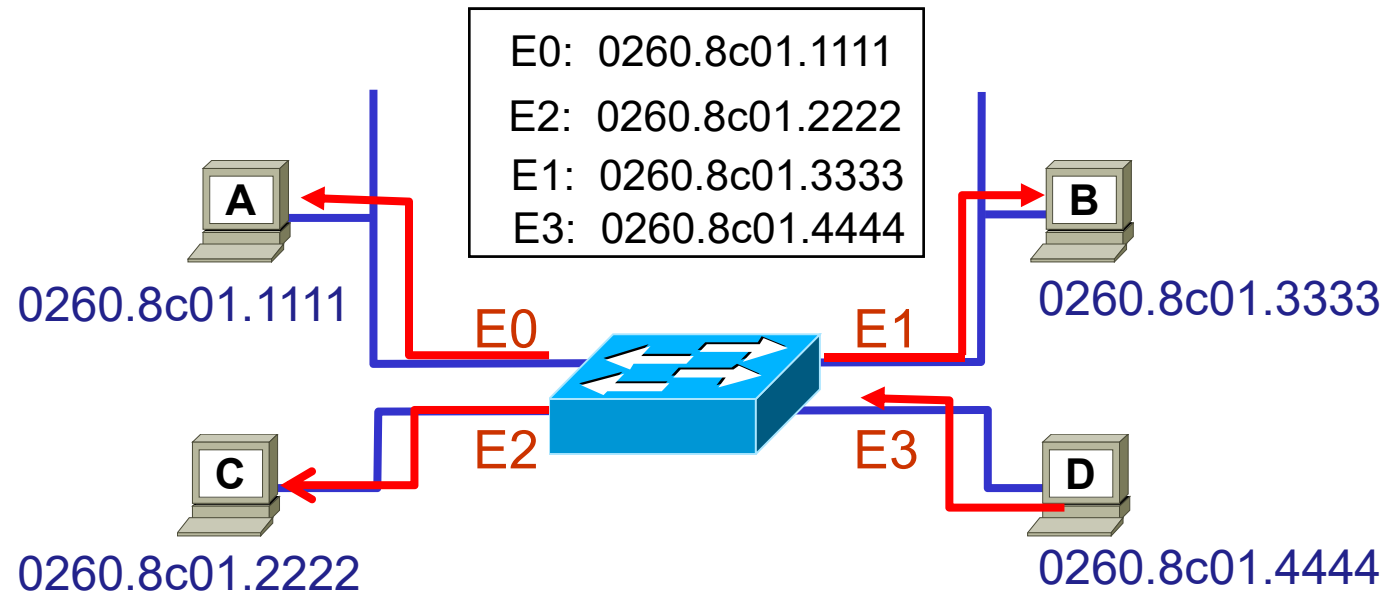
How Switches Filter Frames

MAC forwarding table



- Station **A** sends a frame to station **C**
- Destination is known, frame is **not flooded**

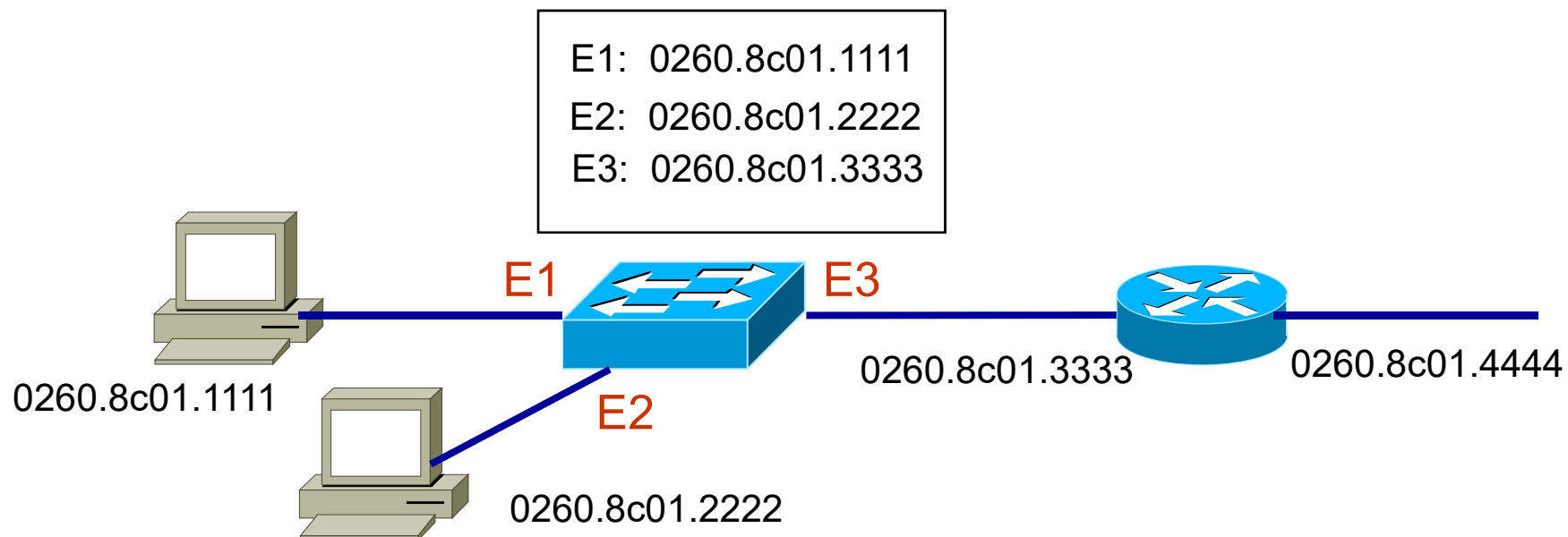
Broadcast Frames



- Station **D** sends a **broadcast** frame (Dest MAC Address: FF-FF-FF-FF-FF-FF)
- Broadcast frames are **flooded** to all ports other than the originating port

MAC Forwarding Table

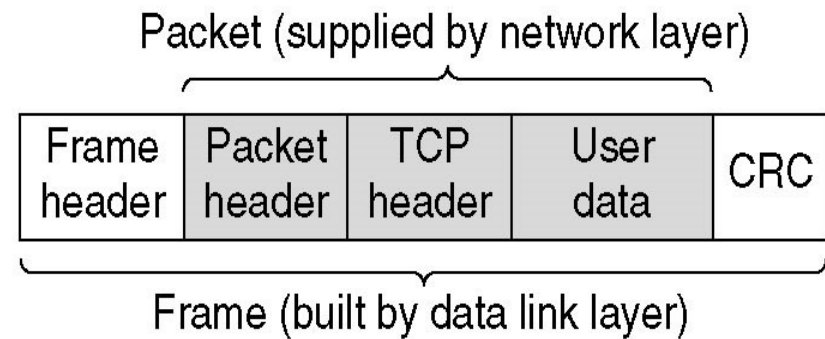
- Unlike IP address, MAC address is *local*.
- MAC forwarding table is also *local*, within the *single broadcast domain* (which is an *IP subnet*).
- MAC address learning stops at the router, and a switch does not learn the MAC addresses at the other side of the router.
- ◆ The switch learns one MAC address of the router.



Interconnection Devices

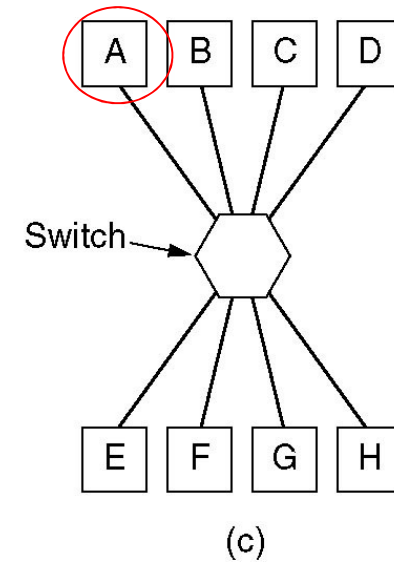
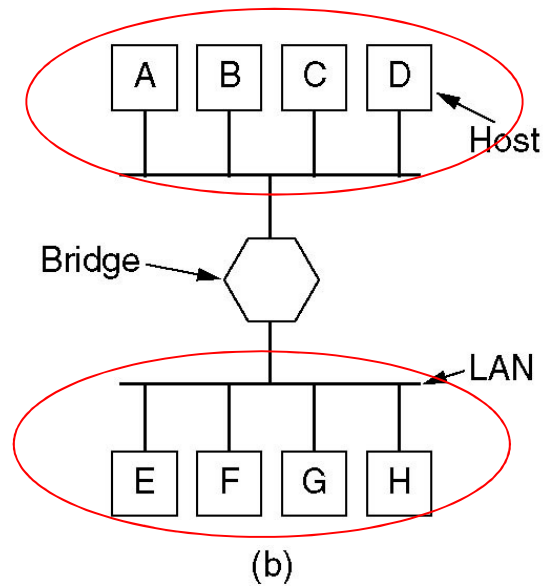
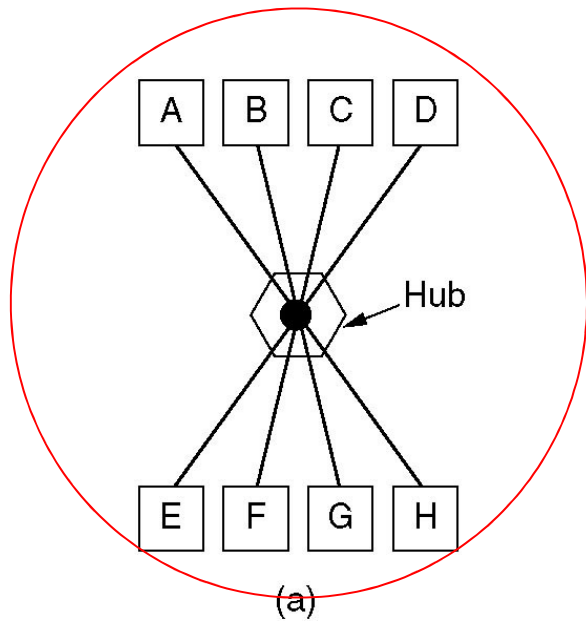
Application layer	Application gateway
Transport layer	Transport gateway
Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub

(a)



(b)

Hubs, Bridges, Switches



 Collision Domain

Bridge(网桥) vs. Switch(交换机)

- Bridge: frame handling done in software
- Switch performs address recognition and frame forwarding in hardware
- Bridge only analyzes and forwards one frame at a time
- Switch has multiple parallel data paths
 - ◆ Can handle multiple frames at a time
- Bridge uses store-and-forward(存储-转发) operation
- Switch can have cut-through(直通) operation
- Bridge suffered commercially
 - ◆ New installations typically include layer 2 switches with bridge functionality rather than bridges

What are VLANs

- VLAN: Virtual LAN(虚拟局域网)
- Decoupling the logical topology from the physical topology.
- Label each port with a "color", and prohibit multicast and broadcast across colors.
- Prohibit packets from traveling to destinations on different VLANs....at least without IP routing
- Based on VLAN-aware bridges/switches
- Modifies Ethernet frame to implement VLAN

Reason for using Virtual LANs

■ Security

- ◆ Many departments have information that they do not want passed outside their department

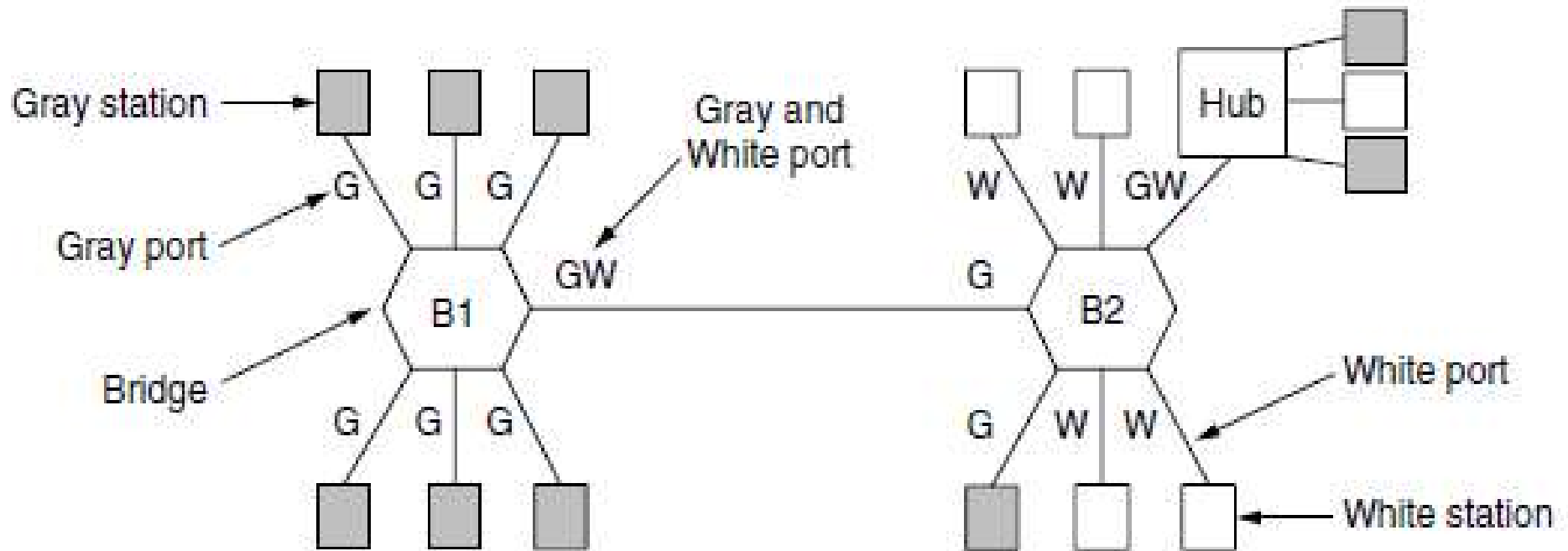
■ Broadcast storm(广播风暴)

- ◆ LAN capacity is occupied by these frames
- ◆ Machines on all the interconnected LANs are crippled just processing and discarding all the frames being broadcast

■ Lower cost

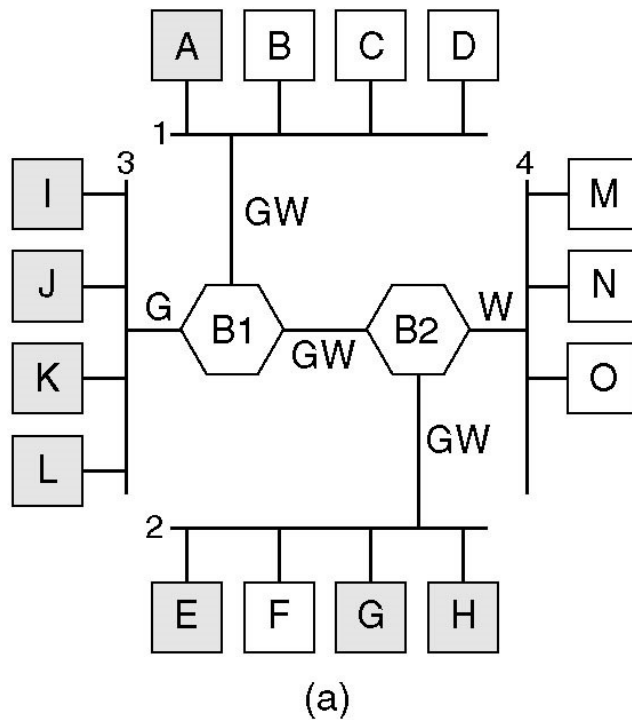
- ◆ A VLAN saves company time and money because reconfiguration is done through **software**. Physical reconfiguration is not necessary.

VLAN Example

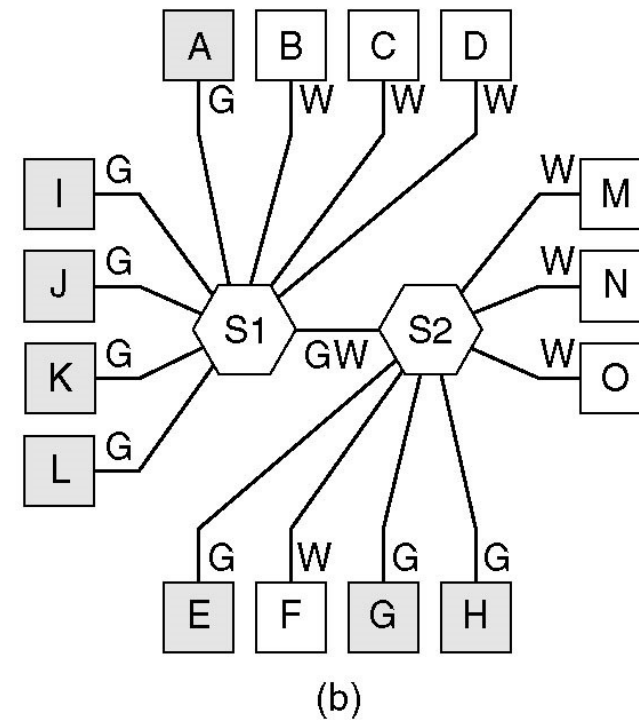


A grey VLAN and a white VLAN

VLAN: Using bridges or LAN switches



VLAN-aware
bridges



VLAN-aware
switches

VLAN ID

- In the bridges or switches, configuration tables tell which **VLANs** are accessible via which **ports**
- The bridges and switches know what **VLAN ID** an incoming frame is
 - ◆ Every **port** is assigned a VLAN ID
 - ◆ This method only works if all machines on a port belong to the same VLAN
 - ◆ Every **MAC address** is assigned a VLAN ID
 - ◆ When a frame arrives, the bridge or switch extracts the MAC address and look it up in a table to see which VLAN the frame came from
 - ◆ Every **layer 3 protocol or IP address** is assigned a VLAN ID

IEEE 802.1Q Frame Tagging

■ IEEE802.1Q Frame

Length 1522 Bytes

6	6	2	2	2	1500	4
Destination Address	Source Address	802.1Q Tag		Type/Length	Data Max of 1500 Bytes	New FCS
		8100	Tag			

Ethernet Type

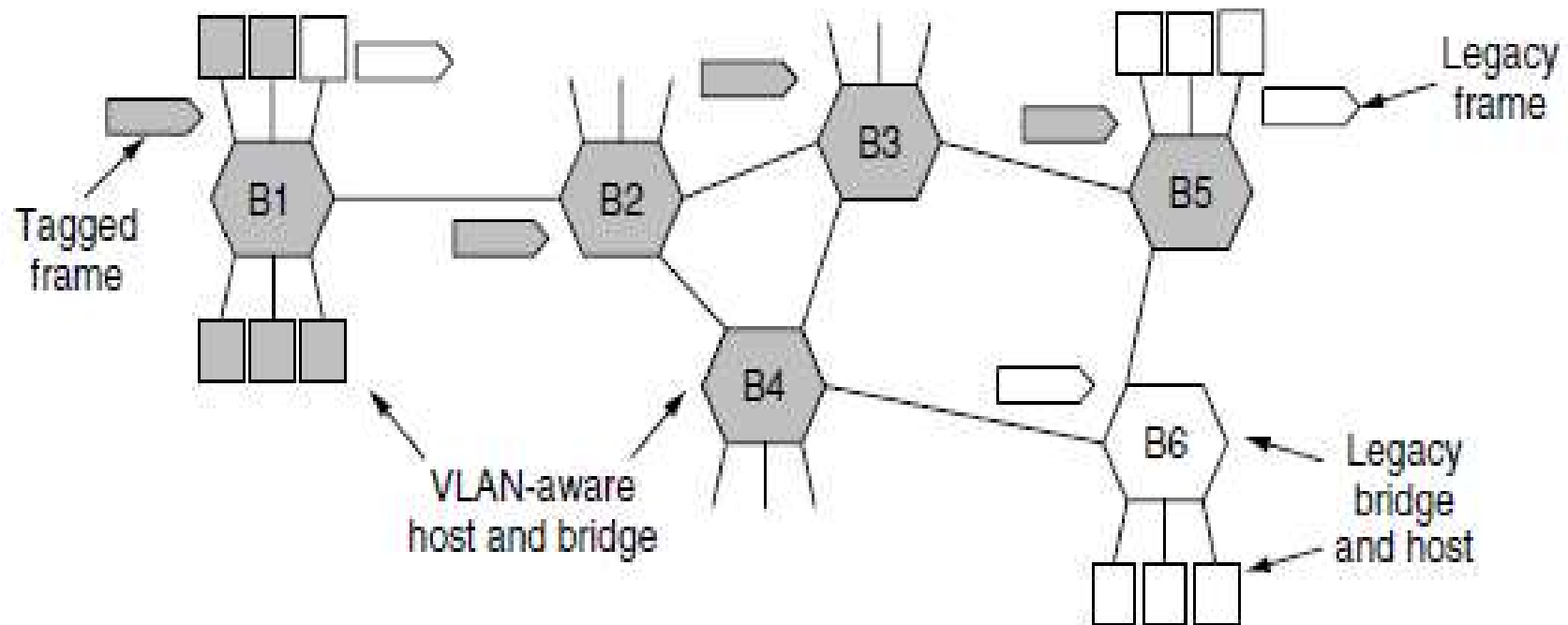
3 Bits	1 Bit	12 Bits
User Priority	CFI	VLAN ID

Canonical Format Identifier

CFI=1, VLAN ID有效

The IEEE 802.1Q Standard

Transition from legacy Ethernet to VLAN-aware Ethernet.
The shaded symbols are VLAN aware. The empty ones are not.



First VLAN-aware bridge(switch) to touch a frame adds VLAN fields and the last one down the road removes them.

Key points of MAC sublayer (1)

- Static & Dynamic Channel Allocation
- Multiple Access Protocols
 - ◆ Carrier Sense Multiple Access (**CSMA/CD**)
 - ◆ carrier sense, collision detection, jam, binary exponential backoff
 - ◆ Wireless LAN Protocols (**MACA**)
- IEEE 802 Reference Model
 - ◆ Functions of LLC, MAC and physical layer
- Ethernet
 - ◆ Ethernet Cabling
 - ◆ **Manchester Encoding**
 - ◆ **Operations of Ethernet**
 - ◆ Fast Ethernet / Gigabit Ethernet

Key points of MAC sublayer (2)

■ WLAN

- ◆ IEEE802.11: Usage, **MAC protocols**
- ◆ IEEE802.15: Usage

■ Data Link Layer Switching

- ◆ **Learning Bridges and LAN Switches**
- ◆ **Repeaters, Hubs, Bridges, Switches, Routers, Gateways**
 - ◆ **Layering and collision domain**
- ◆ **Virtual LANs**

Terms

缩写	英文全称	中文
	1-persisten CSMA	1-坚持CSMA
AP	Access Point	接入点
AIFS	Arbitration InterFrame Spacing	仲裁帧间隔
	auto-negotiation	自动协商
	backward learning	逆向学习
	beacon frame	信标帧
	binary exponential backoff	二进制指数退避
	binary countdown	二进制倒数法
	bit-map protocol	位图协议
	bluetooth	蓝牙
	broadcast storm	广播风暴
	bridge	网桥
	broadcasting	广播
	carrier extension	载波扩展
Spring 2020	checksum Computer Networks, MAC Sublayer	校验和/检查和

缩写	英文全称	中文
CRC	Cyclic Redundancy Check	循环冗余校验
	Collision Domain	冲突域
	collision free protocol	无冲突协议
	carrier sense	载波监听
CSMA	Carrier Sense Multiple Access	载波监听多点接入
	classic Ethernet	传统以太网
CAMA/CA	CSMA with Collision Avoidance	带冲突避免的载波监听多点接入
CSMA/CD	Carrier Sense Multiple Access/Collision Detection	带冲突检测的载波监听多点接入
CTS	Clear To Send	允许发送
	cut-through switching	直通交换
DCF	Distributed Coordination Function	分布协调功能
DIFS	DCF InterFrame Spacing	DCF帧间隔
EIFS	Extended InterFrame Spacing	扩展帧间隔
Spring 2020	Fast Ethernet	快速以太网

Computer Networks, MAC Sublayer

缩写	英文全称	中文
	Ethernet	以太网
	Exposed Terminal Problem	暴露站问题
	fragment	(帧)片段
	flow control	流量控制
FCS	Frame Check Sequence	帧校验序列
FDDI	Fiber Distributed Data Interface	光纤分布数据接口
	frame	帧
	frame bursting	帧突发
	gateway	网关
	Gigabit Ethernet	千兆以太网
	hidden terminal problem	隐蔽站问题
	handoff	切换 (AP)
	HUB	集线器
IEEE	Institute of Electrical and Electronics Engineers	国际电气电子工程师协会
	jam	强化信号

缩写	英文全称	中文
LAN	Local Area Network	局域网
	Learning Bridge	学习网桥
	limited contention protocol	有限竞争协议
LLC	Logical Link Control	逻辑链路控制
MAC	Media Access Control	媒体(介质)访问控制
MACA	Multiple Access with Collision Avoidance	带冲突避免的多点接入
	multi-access channel	多接入信道
	Manchester Encoding	曼彻斯特编码
	multicasting	多播/组播
NAV	Network Allocation Vector	网络分配向量
NIC	Network Interface Card	网络接口卡
	non-persistent CSMA	非坚持CSMA
OUI	Organization Unique Identifier	组织唯一标识符
	packet	分组/包
Spring 2020	pad Computer Networks, MAC Sublayer	填充

缩写	英文全称	中文
PCF	Point Coordination Function	点协调功能
	p-persistent CSMA	p坚持CSMA
PPP	Point-to-Point Protocol	点对点协议
	preamble	前导码
	promiscuous mode	混杂模式
	pure ALOHA	纯ALOHA
	random access channel	随机接入信道
	repeater	中继器
	router	路由器
	rate anomaly	速率异常
RTS	Request To Send	请求发送
SIFS	Short InterFrame Spacing	短帧间隔
	store-and-forward	存储-转发
	slot	时隙/时槽
	slotted ALOHA	时隙ALOHA
	Spanning Tree Bridge	生成树网桥
Spring 2020	station	站点

缩写	英文全称	中文
	subnet	子网
	switch	交换机
	switching	交换
	switched Ethernet	交换式以太网
	thick Ethernet	粗缆以太网
	thin Ethernet	细缆以太网
	token	令牌
	token bus	令牌总线
	token ring	令牌环
TXOP	Transmission Opportunity	发送机会
VLAN	Virtual LAN	虚拟局域网
	Vulnerable Period	脆弱期
WAN	Wide Area Network	广域网
WiFi	Wireless Fidelity	无线保真
WiMAX	Worldwide Interoperability for Microwave Access	全球微波互联接入
Spring 2020 WLAN	Computer Networks, MAC Sublayer Wireless Local Area Network	无线局域网

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 - ◆ "Data communications and networking", B.A.Forouzan, which are marked with [Forouzan]
 - ◆ "Data and Computer Communications", William Stallings, which are marked with [Stallings]
 - ◆ "计算机网络", 谢希仁, 第五版, 电子工业出版社, 2008年, which are marked with [谢]
 - ◆ Nen-Fu Huang, Lecture notes "IEEE 802.3 Carrier Sense Multiple Access with Collision Detection (CSMA/CD)", Natinal Tsinghua University, which are marked with [Huang]

Self-Study

Static Channel Allocation (静态信道分配)

- Channel is divided into N static sub-channels

- ◆ FDM: user number is small and constant, with heavy load of traffic of each user

$$T_{FDM} = \frac{1}{\mu C / N - \lambda / N} = NT$$

WHY?

Using M/M/1 model



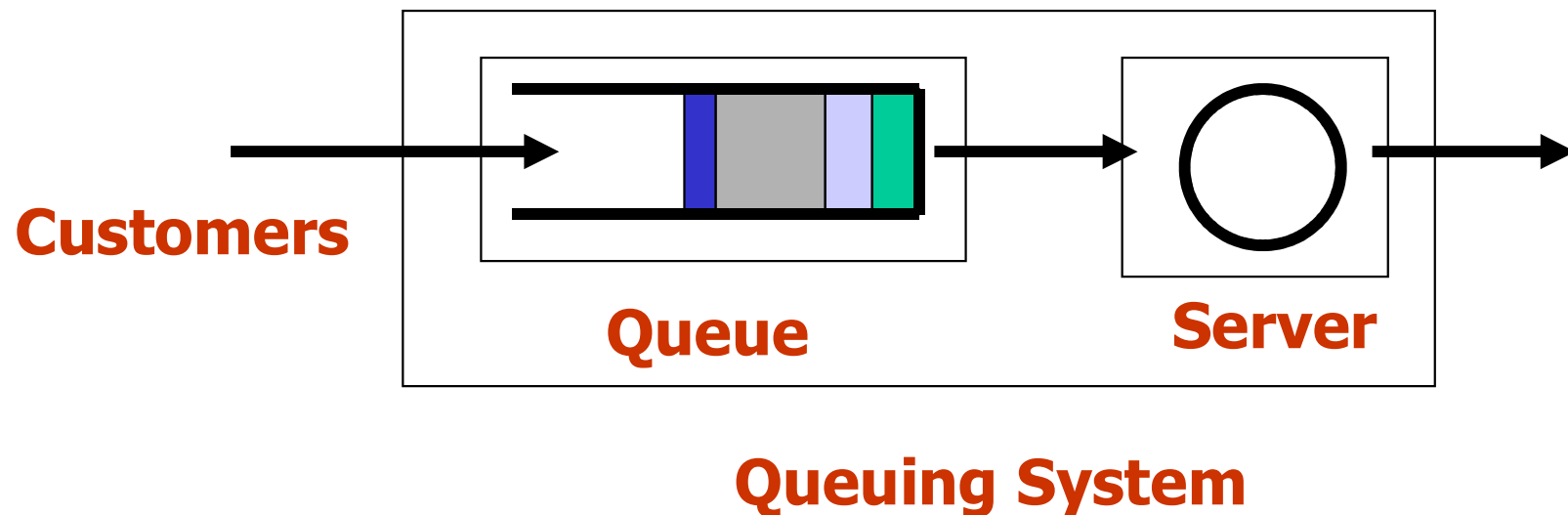
- ◆ TDM: same problem. Poor performance.

- None of the static channel allocation methods work well with bursty traffic
- Dynamic Channel Allocation are adopted



M/M/1 Model

- First M: Arrivals are a “Markovian / Poisson” process (泊松过程)
- Second M: **exponential** distribution for service time (指数分布)
- 1: **1** server



M/M/1 Model

- Probability that k customers arrived during period T

$$P[k] = \frac{(\lambda T)^k e^{-\lambda T}}{k!}$$

- We can get

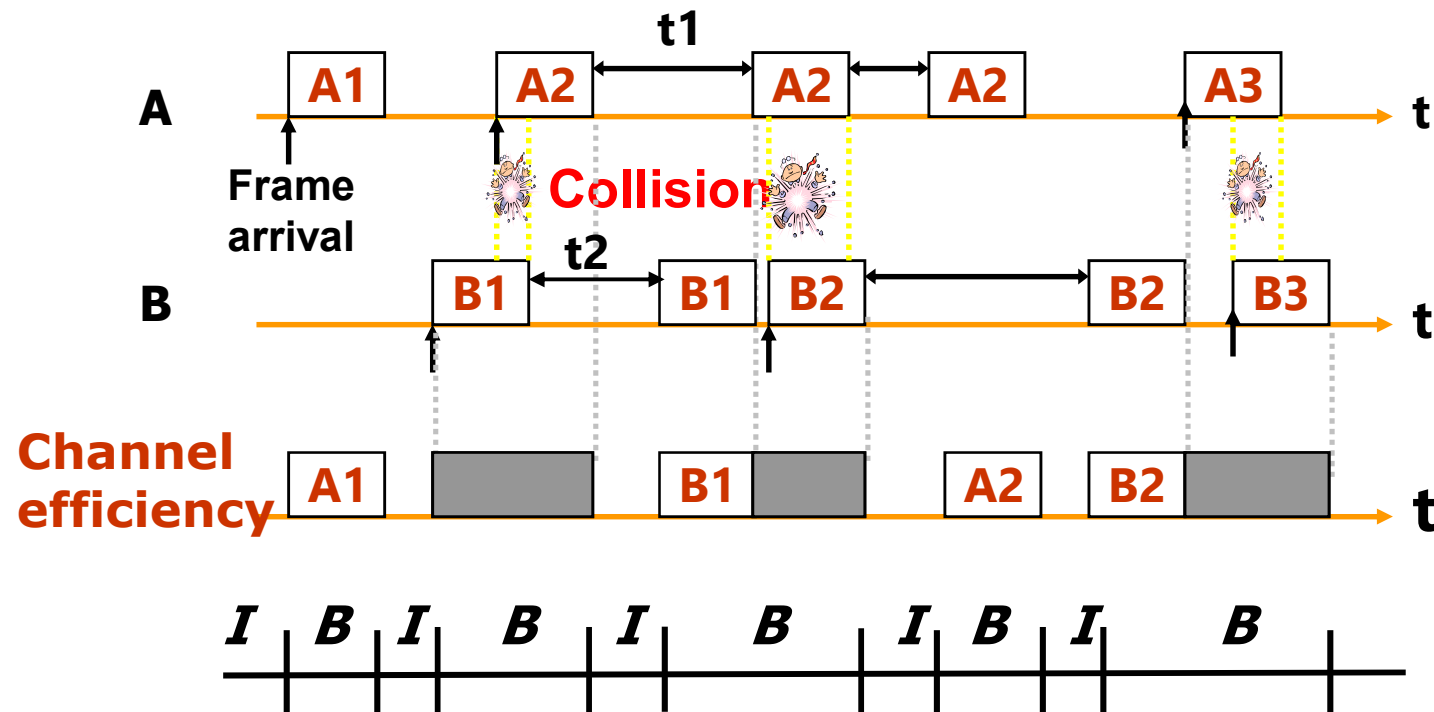
$$T = \frac{1}{\mu - \lambda}$$

μ : *AverageServiceRate*

λ : *AverageArrivalRate*

T : *AverageWaitingTime(QueuingTime + ServiceTime)*

Illustration of Pure ALOHA



Frames are transmitted at completely arbitrary times.

Pure ALOHA: Performance Analysis (1)

- Performance parameters: Assumed each frame has same duration T_0

- ◆ Throughput(吞吐量) S --- Frames successfully transmitted in T_0

- ◆ Network Load(网络负载) G --- Frames arrived in T_0

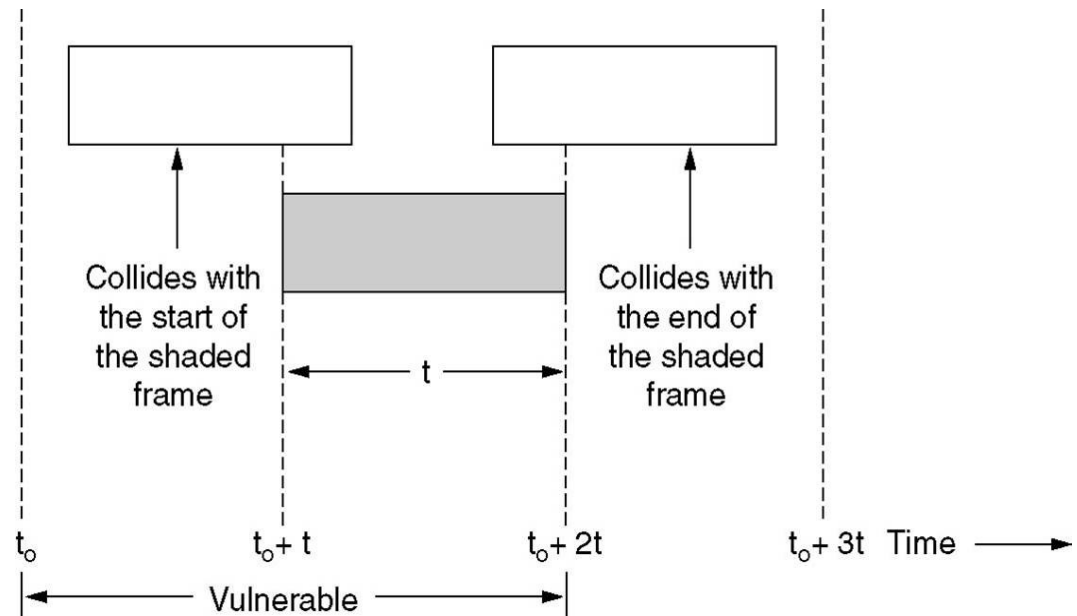
$$S = G \times P \text{ [successful transmission]}$$

- A frame is successful if

- ◆ no packets arrive within T_0 seconds before it starts or while it is being transmitted

Vulnerable Period: $2T_0$

脆弱期（易冲突期）



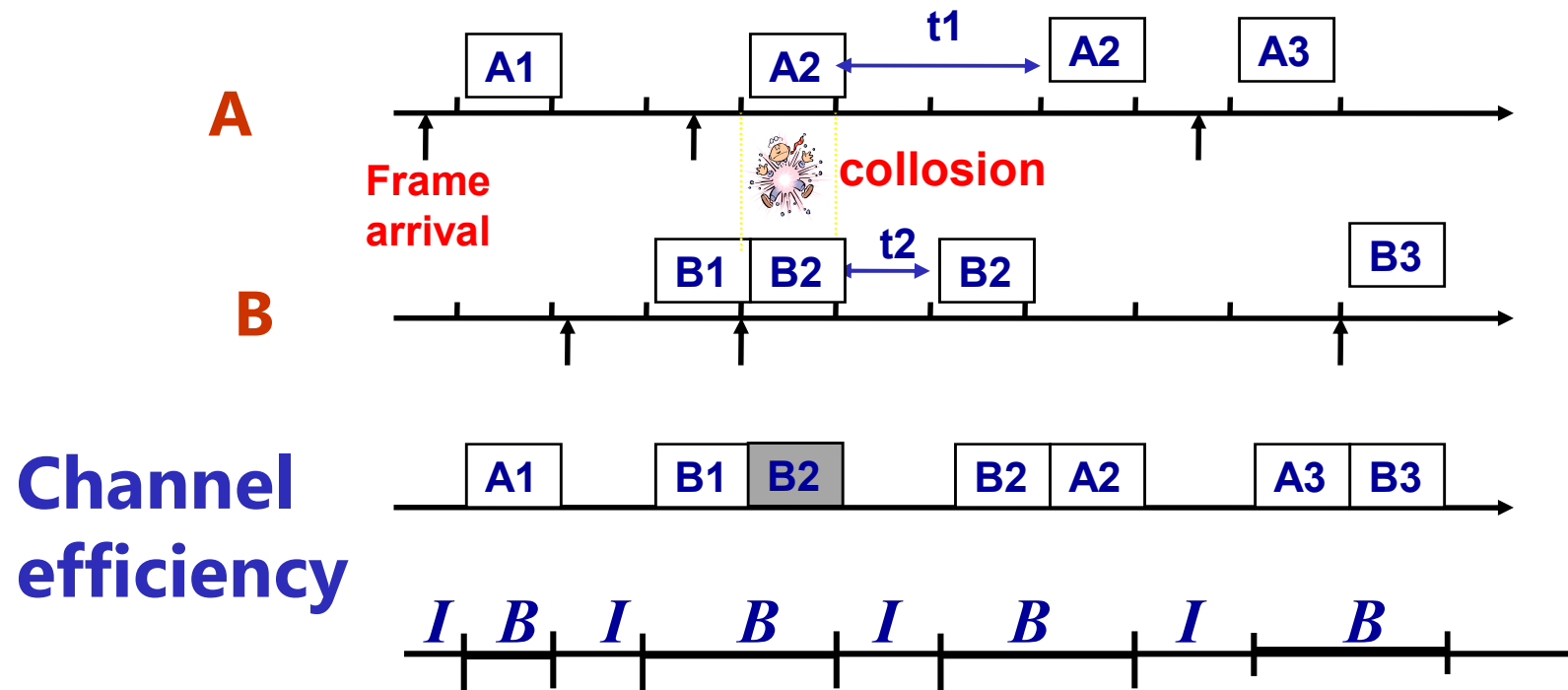
Pure ALOHA: Performance Analysis(2)

$$P_r(k) = \frac{G^k e^{-G}}{k!}$$

$$P_r(0) = P_0 = e^{-G} \text{ (within frame time } t\text{)}$$

- $P_0 = P$ [no other traffic being initiated during the entire vulnerable period] = e^{-2G}
- Throughput: $S = G \times P_0 = G e^{-2G}$
- Maximum throughput occurs when $G=0.5$,
 $S=1/2e \approx 0.184$

Illustration of Slotted ALOHA



Vulnerable Period: T_0

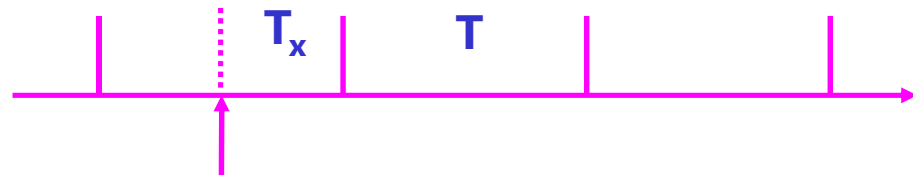
Slotted ALOHA: Performance Analysis

■ Collision probability

- ◆ No frame sent between t_0 and t_0+t

$$P_r(0) = P_0 = e^{-G} \quad (\text{within frame time } t_0)$$

$$S = GP_0 = Ge^{-G}$$



- ◆ Maximum $S=0.37$ at $G=1$

37%: Slots empty

37%: Successes

26%: Collisions

Performance of IEEE 802.3

- Premise: heavy and constant load, i.e. k stations always ready to transmit, and retransmission probability is also constant
- Assumed each station emits during a contention slot with probability p

$$A = P[\text{some station emits successfully}] = k p (1-p)^{k-1}$$

Then probability of j slots in a contention interval is $A(1-A)^{j-1}$,

$$\text{so mean number of slots per contention} = \sum_{j=1}^{\infty} j A(1-A)^{j-1} = \underline{1/A}$$

When $p=1/k$, $A_{\max} = (1-1/k)^{k-1}$, $A_{\max} = 1/e$ as $k \rightarrow \infty$

Performance of IEEE 802.3 (cont.)

$A = P[\text{Some station emits successfully}]$ and

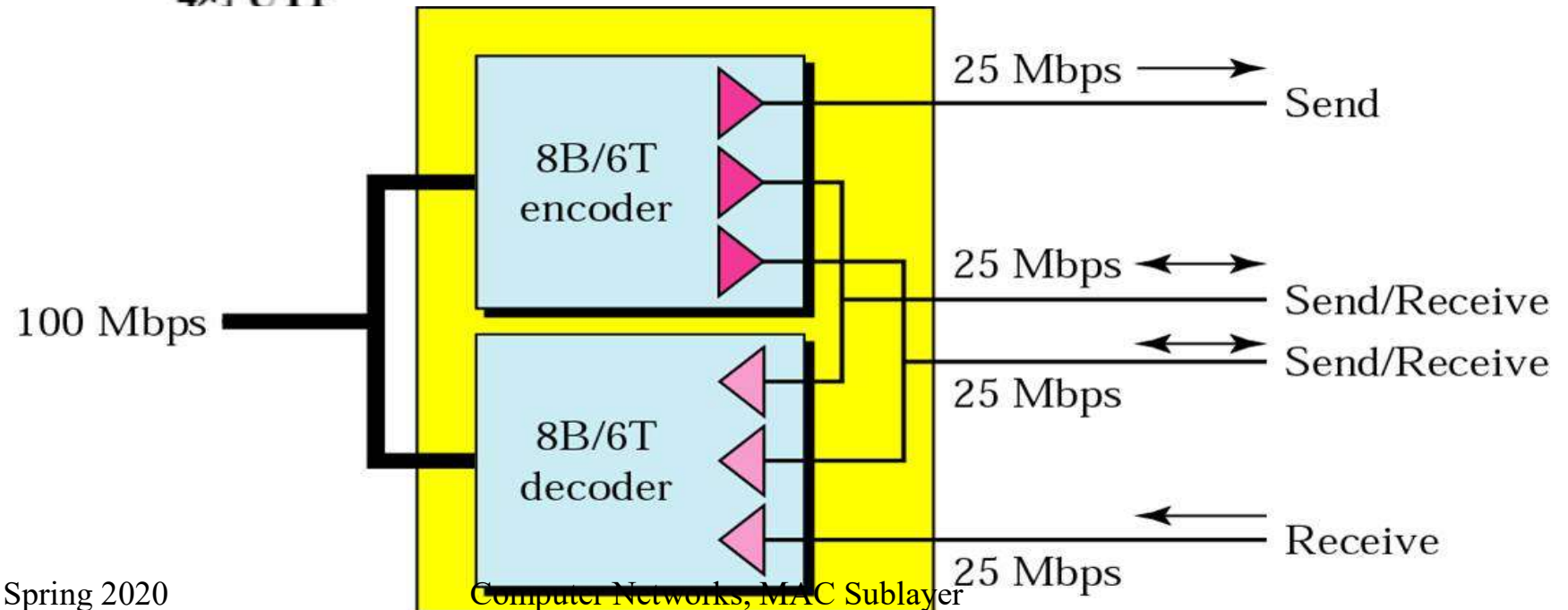
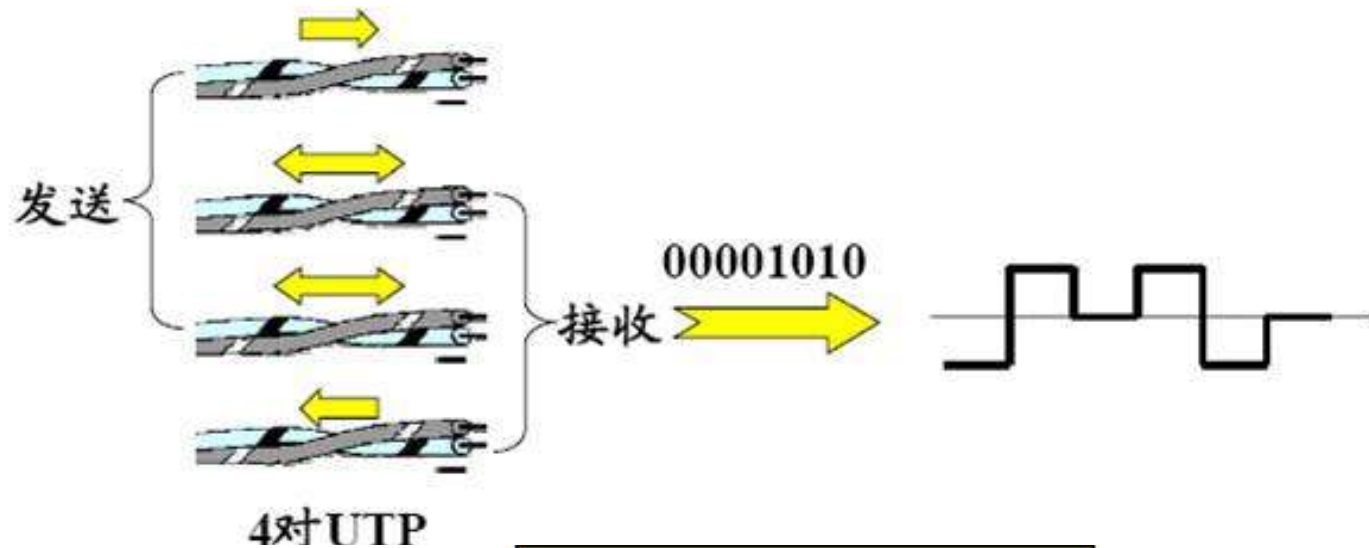
mean number of slots per contention = $\frac{1}{A}$

- Slot duration is 2τ , then mean contention interval: $w = 2\tau / A$
- If mean frame time is P sec, Channel efficiency = $P / (P + 2\tau / A)$
- Assumed frame length F , network bandwidth B , cable length L , speed of signal propagation c , for optimal case $A = 1/e$

$$\text{Channel efficiency} = \frac{1}{1 + 2BLE/cF}$$

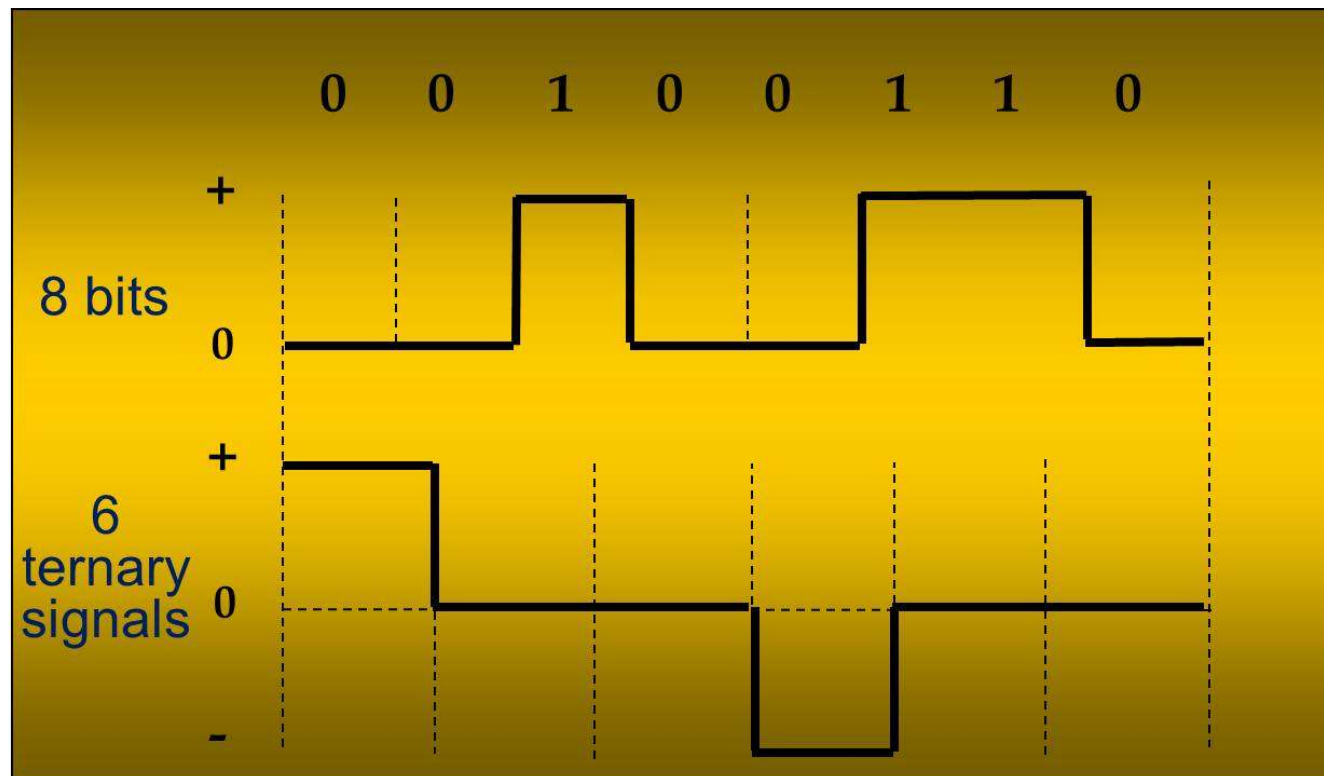
Increasing network bandwidth or distance reduces efficiency for a given frame size

100Base-T4 Encoding



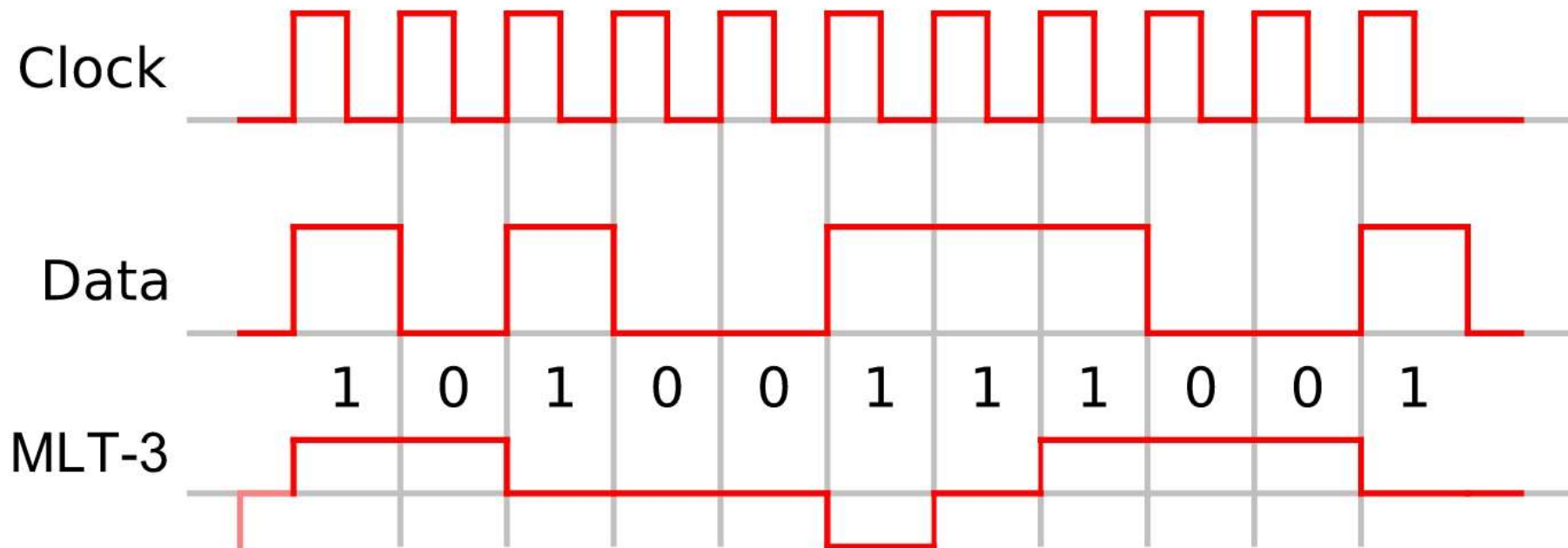
8B6T Encoding Example

- 8 bits are encoded as 6 ternary signals
- $3^6=729$ possible combinations, 256 combinations are used for data

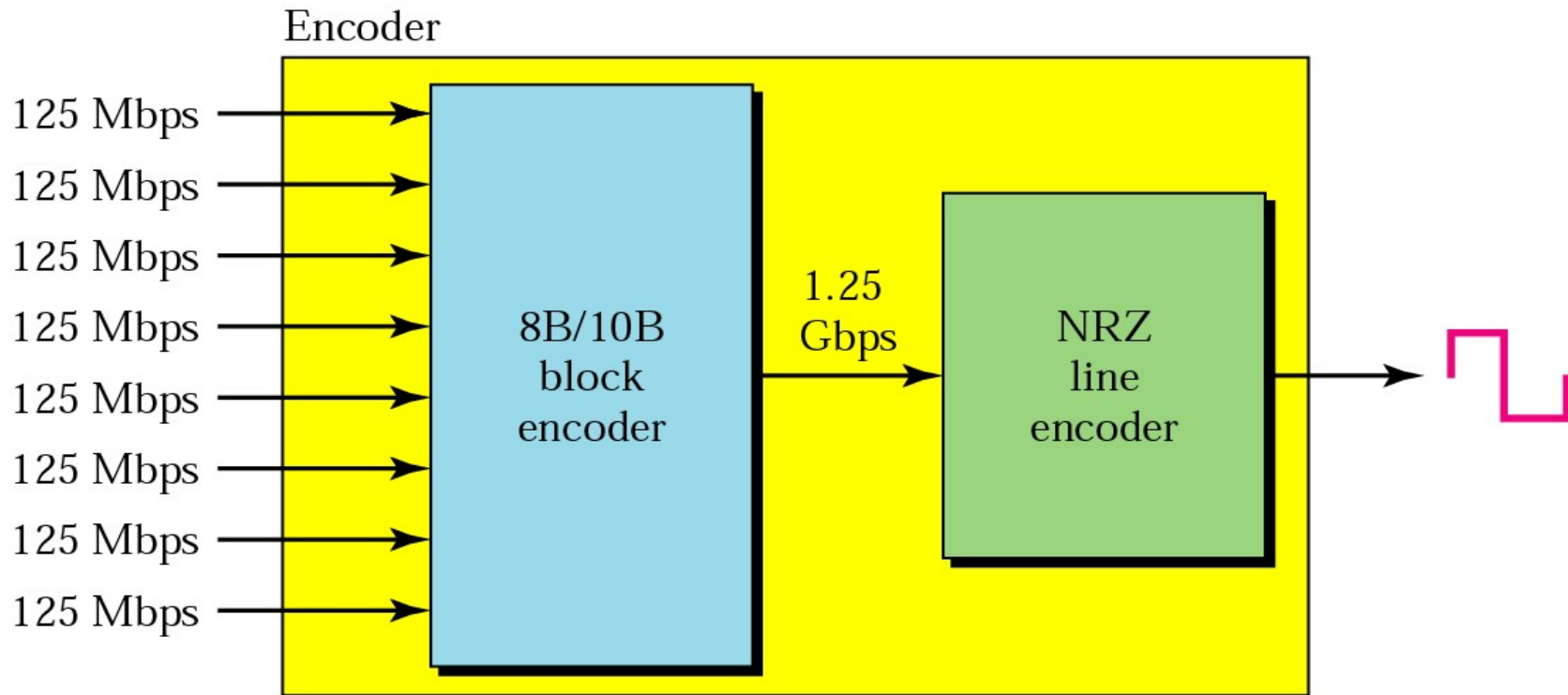


MLT Encoding Example

- MLT-3 cycles sequentially through the voltage levels -1, 0, and +1.
- It moves to the next state to transmit a 1 bit, and stays in the same state to transmit a 0 bit.



Encoding in 1000Base-X



10-Gigabit Ethernet cabling (自阅)

Name	Cable	Max. segment	Advantages
10GBase-SR	Fiber optics	Up to 300 m	Multimode fiber (0.85 μ)
10GBase-LR	Fiber optics	10 km	Single-mode fiber (1.3 μ)
10GBase-ER	Fiber optics	40 km	Single-mode fiber (1.5 μ)
10GBase-CX4	4 Pairs of twinax	15 m	Twinaxial copper
10GBase-T	4 Pairs of UTP	100 m	Category 6a UTP

Outline

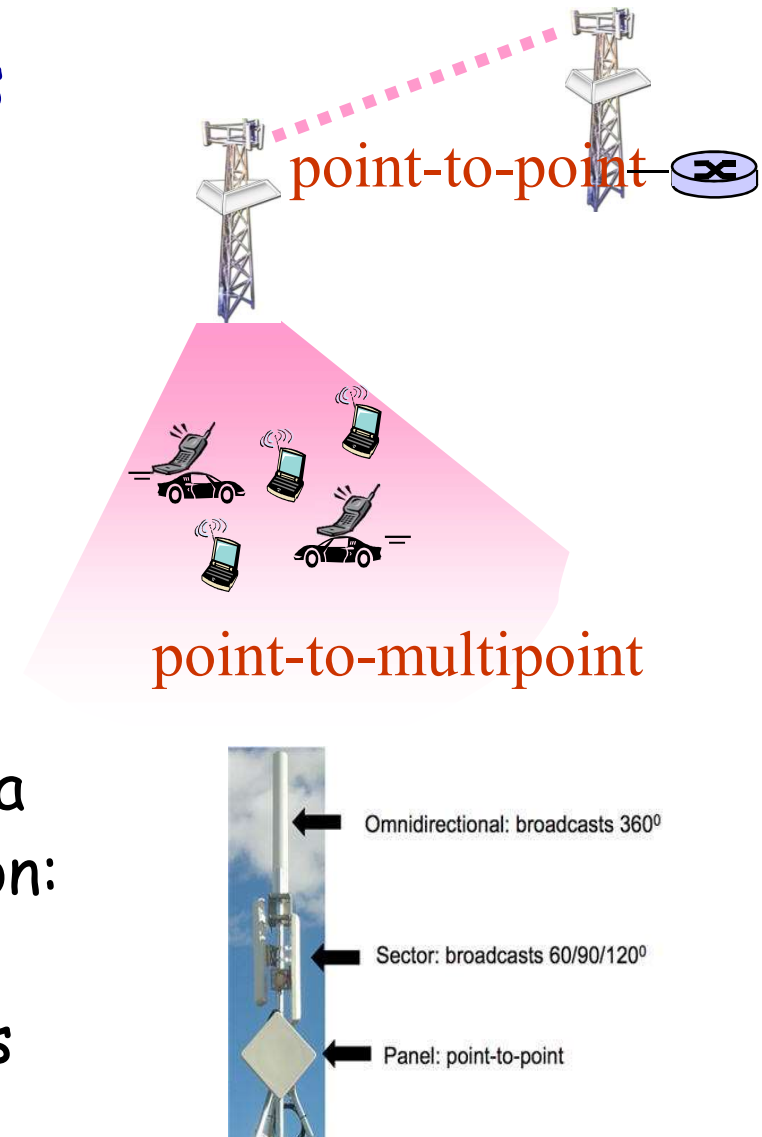
- Channel Allocation Problem
- Multiple Access Protocols
- IEEE 802 Reference Model
- Ethernet
- Wireless LANs
 - ◆ IEEE 802.11 (WiFi)
- Broadband wireless
 - ◆ IEEE 802.16 (WiMax)
- Bluetooth
 - ◆ IEEE 802.15
- Data Link Layer Switching

IEEE802.16: Broadband Wireless

- Wireless MAN or Wireless local loop
- 802.16 vs. 802.11

Similarity

- ◆ Designed to provide high-bandwidth wireless commun.
- ◆ Using base station model
- ◆ Host $\leftrightarrow$ Base Station:
omnidirectional antenna
- ◆ Base station $\leftrightarrow$ Base station:
point-to-point antenna
- ◆ Designed to carry IP packets
- ◆ Using OFDM and MIMO



了解WiMAX 与WiFi的相似点与区别

[Kurose]

802.16 vs 802.11

■ Differences

- ◆ More than 1 computers are connected to one node
- ◆ Better radios maybe used and it can have **full duplex** connection
- ◆ The distance to base station is much longer
- ◆ **Base stations** are **more powerful**, using more power and better antennas
- ◆ **Security and privacy** is mandatory
- ◆ **More bandwidth** is needed, ISM band can't provide, licensed spectrum is used.
- ◆ Absorption by water and rain is big problem
- ◆ Supporting real time traffic like phone, movie and music

■ **WiMax:** Worldwide Interoperability for Microwave Access

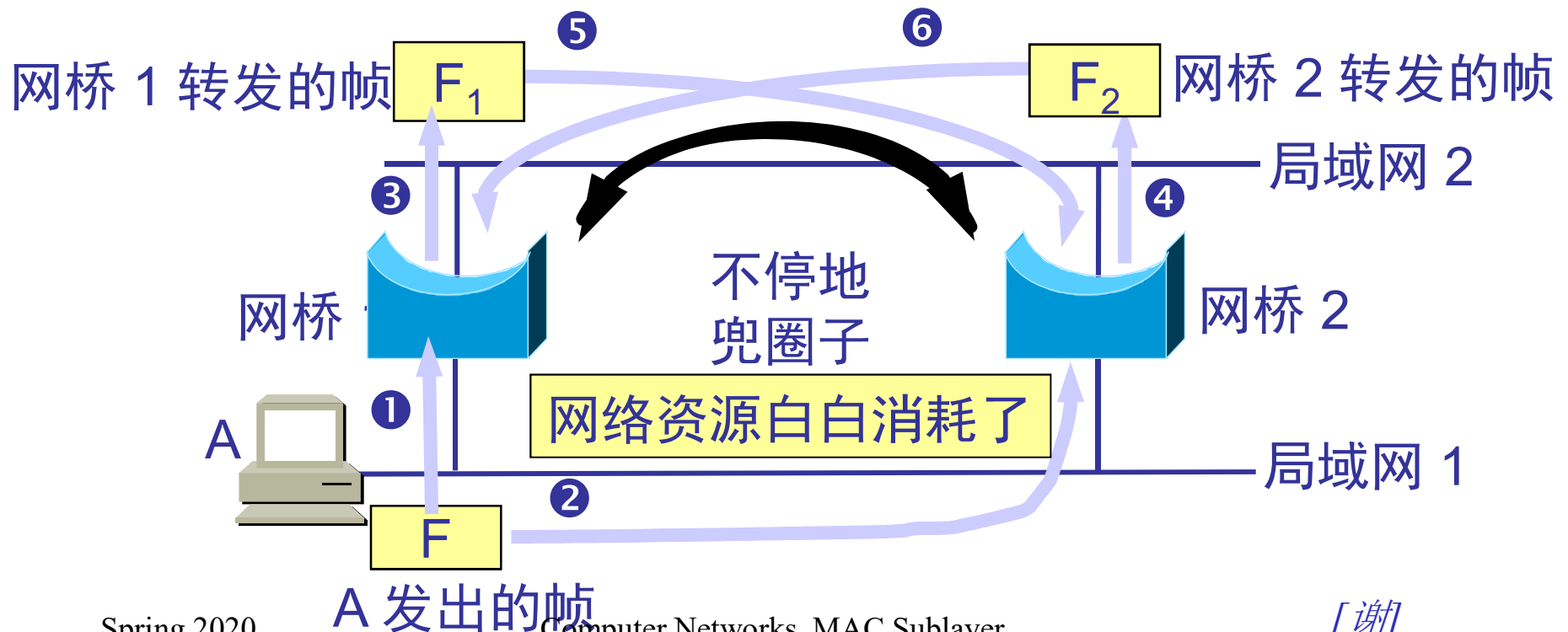
Data Link Layer Switching

- Bridges from 802.x to 802.y
- Learning Bridges and LAN Switches
- Spanning Tree Bridges
- Summary of connection devices
- Virtual LANs

Spanning Tree Bridges(生成树网桥): Motivation

■ Problem

- ◆ To increase reliability, some sites use two or more bridges **in parallel** between pairs of LANs,
- ◆ This arrangement introduces problems because it creates **loops in the topology**



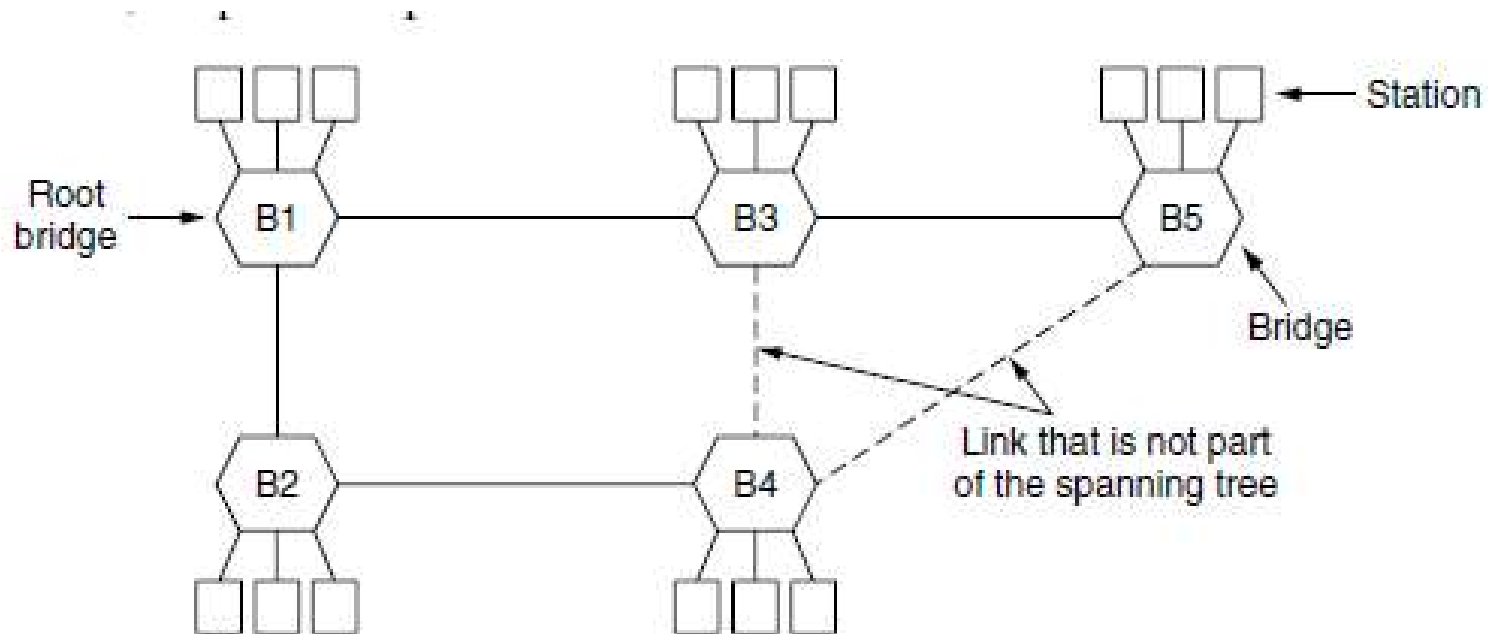
Spanning Tree Bridges: Ideas

■ Solution

- ◆ Bridges communicate with each other and overlay the actual topology with a **spanning tree** that reaches every LAN
- ◆ Using this spanning tree, there is exactly **one path** from every LAN to every other LAN, loops are impossible
- ◆ Some potential connections between LANs are ignored in the interest of constructing a fictitious **loop-free** topology

断环，构成树形拓扑

Spanning Tree Bridges: Examples



The dashed lines are links that are not part of the spanning tree.

IEEE 802.1d: Distributed algorithm for constructing the spanning tree

- Each bridge periodically **broadcasts its identifier** (based on its *MAC* address) out all of its ports to its neighbors
- The bridge with the **lowest identifier** becomes the **root**
- A tree of **shortest paths from the root** to every bridge and LAN is constructed
- **Automatically detect** topology changes, if a bridge or LAN fails, a new one is computed

生成树网桥：思路

- 解决方案(Radia Perlman, 1983)
 - ◆ 网桥之间相互通信，用一棵可以到达每个网桥的**生成树**覆盖实际的拓扑结构
 - ◆ 在构造一个**虚拟的无环拓扑结构**时，网桥之间一些潜在的连接被忽略了，无环拓扑结构是**实际拓扑结构**的一个子集
 - ◆ 利用生成树，从每个站到每个其他站实际上只有一条路径，不存在环路



Radia Perlman 2009
(born 1951)

断环，构成树形拓扑

About ALOHA



■ Norman Abramson先后毕业于

Harvard大学、UCLA（加州大学
Los Angeles分校）、Stanford

大学，分别获学士、硕士、博士学位，毕业后在Stanford大学电子工程系任教。1967年开始任Hawaii大学信息与计算机科学系教授，并从事通信理论研究，领导ALOHA系统研究项目，提出了著名的ALOHA多址通信协议和CRC（循环冗余校验）专利技术，并成功建立了世界上第一个数据通信网络ALOHANET，开创了计算机局域网和现代数据通信网，同时也首先实现了无线分组数据接入方法。

About ETHERNET

- 1973年，为了实现几台电脑之间的简单联接和信息交互，施乐公司Palo Alto研究中心(PARC)的研究员Bob Metcalfe描绘出了大致的网络构想，并将这项技术命名为“Ethernet”。当时的数据传输速率为2.94Mbps，以太网就此诞生。

